Research report after round 2. The run is still in progress: this is the report text as it stands after this round, text only, with no figures. The results and the writing are not final.

How an LLM picks a solution idea, and whether internal edits can make it pick new ones: research report

Framing

This run studies one question from the user's request: when an instruction-tuned open-weight model is asked an open design task such as writing a path search algorithm, how is the choice of solution idea (BFS, DFS, Dijkstra, A*, ...) made inside the model, and can that choice be pushed toward ideas the model does not usually produce? Practitioners treat sampling temperature as the creativity knob and add requests for originality, or prohibitions of named algorithms, to the prompt. Behavioural work finds that temperature is only weakly correlated with novelty in narrative generation, and moderately with incoherence [2], and that incidental prompt cues move algorithm-family choice in code generation without an identified mechanism [3]. The public repository that inspired this run (github.com/DrejcPesjak/llm-creativity) reported, for a weighted shortest-path task on a closed model, that random context seeds changed nothing, that a ban list pushed most programs to a new attractor (SPFA), and that a two-stage pipeline that first lists approaches and then picks one at random (generate-then-select) gave the widest spread.

The run's working hypothesis treats idea choice as a discrete choice made at a few early tokens and governed by a readable vector of internal idea utilities. Under that view each knob acts on a different parameter of a random-utility model: temperature rescales the utilities (it flattens the idea distribution but cannot reorder it), random seeds are utility shocks, bans restrict the choice set with substitution toward internally similar ideas, and novelty instructions shift utilities along a typicality axis. The hypothesis further claims that editing the utilities directly, at the layer and position where the choice forms, is the knob that produces new ideas. Four alternatives were written down with it, and we refer to them by these names throughout: the *gradual-commitment alternative* (commitment spread over many tokens, so heat can reorder ideas); the *far-side ban alternative* (bans push probability to the far side of the idea manifold); the *menu-erasure alternative* (erasing the subspace of the usual ideas is the route to novelty); and the *unusualness-dial alternative* (a single task-general direction controls atypicality).

The model throughout is Qwen2.5-3B-Instruct [1]. The report is chronological, one section per iteration. Each section records what was planned, what ran, every table the artifacts produced, and what the evidence did to the hypothesis.

Iteration 1: a pre-registered three-arm screen, of which one arm delivered

1.1 Strategy and why it was chosen

The hypothesis bundles three separable claims, one per knob family, and each has a rival that predicts the opposite observable on the same data. The iteration therefore planned a wide screen rather than a deep test of one claim: three experiments run in parallel on one model and one shared protocol, each a head-to-head between candidate accounts with a selection rule fixed before any data existed.

- **Temperature arm.** A temperature by top-k grid (T in {0.3, 0.5, 0.7, 1.0, 1.3, 1.6}, top-k in {10, none}, n = 100 per cell, three tasks) with forking-path resampling [4] to locate the commitment token (defined in the hypothesis as the first generated position after which resampling no longer changes the idea class), Arrhenius fits of idea log-odds against 1/T, an ideas-versus-phrasings diversity partition with Hill numbers [11] to answer the user's questions (a), (b) and (c), and a seed-string side condition.
- **Ban arm.** Bans of the top-1, top-3 and top-5 ideas plus a rare-idea priming control, with a geometry-weighted nested logit, a far-side repulsion model and IIA renormalisation as competing predictors of where banned probability goes, scored by held-out Jensen-Shannon divergence. This arm was to draw the low-dimensional map of path-search ideas in activation space, with before/after-ban arrows, that the user asked for.
- **Influence arm.** Three internal knobs (flattening the idea utilities, erasing the menu subspace, a cross-task novelty direction) against tuned temperature, novelty prompts, a top-3 ban and generate-then-select, at matched correctness and against matched-norm random controls.

The shared protocol fixed the model checkpoint, chat template (Qwen's default system message, nothing else), task prompts, per-sample seed equal to the sample index (the user asked for a fixed seed throughout), max 768 new tokens, the idea taxonomy, a comment-stripping AST classifier, and an execution oracle. Two things were reserved and touched by no artifact this iteration: a second model family (gemma-2-2b-it, which also has public SAEs for naming features) and a fourth task (k-th smallest element), both for confirming any survivor in iteration 2. The strategy states plainly that nothing is claimed from the screen alone.

The choice of one model was deliberate: a screen has to compare candidates on identical evidence, and a second model would have halved the depth of each arm. SAE features and circuit discovery were left out on purpose; the field's own evidence rates raw SAE latents as a weak default for steering and single-draw circuits as unstable, so causal weight was to come from steering and erasure with random controls.

1.2 What ran and what did not

Only the influence arm produced an artifact. The other two arms are recorded here as not delivered.

- **Temperature arm: not delivered.** Its executor froze and committed a pre-registration before any data existed and built an idea classifier that passed its 41 fixtures. Setup then consumed most of the time budget: the container had no C compiler, so vLLM's compiled path failed and the executor moved to eager mode and a compiler workaround. The arm ended with an interim status file that explicitly reports no results. Its raw generations were not analysed and are not part of the evidence base; no number from that arm appears in this report.
- **Ban arm: not delivered.** Its executor failed with a model-API streaming error before producing results.

> **Correction (iteration 2).** Both statements above understate what the two arms left on disk, and this was established by the iteration-2 review and confirmed by the iteration-2 artifacts. The ban arm produced 4,360 generations on the bare prompt (no output contract), 4,210 of them labelled by its own rule classifier: baseline n = 300 and ten ban or placebo conditions at n = 100 on each

of three tasks, forced-idea cells at n = 10 and co-mention lists at n = 50. Only its execution oracle, its activation capture (a missing `accelerate` dependency) and its analysis stage did not run. The temperature arm completed its full grid (3 tasks x 6 temperatures x 2 top-k settings x n = 100, plus greedy and repeat rows: 3,623 generations) and its 360 seed-string generations; only the labelling step was killed. 51% of the grid rows stopped at the 768-token cap, and only 2 of 20 repeated grid rows reproduced token for token under vLLM. Both data sets are labelled, executed and written up in Sections 2.4 to 2.6.

As a consequence, the questions that depended on those arms remain unanswered after iteration 1: the Arrhenius test and the ideas-versus-phrasings partition (the user's questions a, b, c), the forking-path localisation of the commitment token, the seed-shock test, and the nested-logit ban prediction with the manifold figure. The influence arm does contain a temperature sweep, a top-k cell, a top-3 ban and two novelty prompts as baselines, so partial behavioural evidence on those knobs exists and is reported below; it was not designed to test the temperature law or the ban substitution model.

1.3 Influence arm: design

Question. Does any internal intervention make Qwen2.5-3B-Instruct write more diverse, and genuinely novel, correct path-search programs than the knobs practitioners use?

Tasks. Two of the three planned tasks were used; the sorting task served only to extract the cross-task direction. Each prompt is the task sentence followed by an output contract that fixes the function signature, so that every program can be executed:

- *Open path-search task*, or the open task for short: the sentence `Implement a path search algorithm in Python.` plus a contract for `find_path(graph, start, goal)` on an unweighted directed adjacency dict, returning any valid path or None.
- *Weighted shortest-path task*, or the weighted task for short: the sentence `Implement a function that finds a shortest path in a weighted graph.` plus a contract for `shortest_path(graph, source, target)` with non-negative weights, returning `(distance, path)` or `(inf, None)`.

Engine. All conditions ran through one HuggingFace engine (bf16) with a custom sampler: sample *i* draws its token uniforms from a random generator seeded with *i* and samples by inverse CDF, so the same index sees the same random numbers in every condition (common random numbers). Generation stops at the closing fence of the first code block; this rule applied to every condition. Hardware was one 32 GB RTX PRO 4500 GPU.

Correctness. A sandboxed oracle runs only the requested code (imports, definitions and safe assignments; module-level demo code is removed because it crashed 22% of weighted-task programs for reasons unrelated to the algorithm) on 50 fixed random directed graphs (seed 12345; 10 to 60 nodes; edge density 0.05 to 0.3; start equal to goal, isolated goals, self-loops, unreachable targets and float weights included), with 5 s CPU and 1 GB RAM per program. A program passes only if all 50 graphs are correct: a valid path for the open task, the exact optimal distance with a consistent path for the weighted task.

Idea labels. An AST classifier assigns one of 13 classes (Floyd-Warshall, Bellman-Ford, SPFA, bidirectional search, A* search, greedy best-first search, Dijkstra, Dial, iterative deepening, BFS, DFS, random walk, and a catch-all class for anything else) after stripping comments and docstrings; syntax errors and missing code get non-idea labels. A blind LLM judge (gpt-4.1-mini, temperature

0) audited the classifier and also decided whether passing programs in the catch-all class were genuinely distinct algorithms.

Definitions used throughout this section. *Idea entropy* is the Shannon entropy, in bits, of the idea-class distribution over the samples of a setting, with syntax-error and no-code outputs excluded. We call the set of idea classes with at least 3% baseline share a task's *menu*. The *off-menu correct share* is the fraction of samples that pass and carry an idea label outside the menu. The *judge-confirmed novel-and-correct share* is the fraction that pass, fall in the catch-all class of the classifier, and are judged a distinct novel algorithm. *Atypicality* is the mean of $-\log p0(\text{label})$, with p0 the baseline idea frequency (add-0.5 smoothing). A setting is *eligible* when its pass rate is at least 0.8 times the baseline pass rate (0.56 on the open task, 0.299 on the weighted task, 0.429 pooled); eligibility at the pooled level decided selection.

Interventions. All internal knobs act on the residual stream at a band of three layers (29, 30, 31 of 36), at the prompt-final token and the first 64 generated tokens. An additive knob adds α times the median residual norm of the layer (122.0, 135.3 and 170.6 at layers 29 to 31), divided by three, along a unit direction at each band layer. We call the two utility-flattening steering vectors the *probe-weight direction* and the *difference-in-means direction*, the two erasure variants *menu erasure* by rank-1 projection and by LEACE, and the transferred vector the *cross-task novelty direction*; the table below gives the construction of each and the setting names used in later tables.

setting name in tables	what it is
baseline $T = 0.7$	temperature 0.7, plain prompt, seeds 0..149 (the 300-sample draw is used only for menus, probes and vectors)
temperature 1.0 to 2.0; temperature 1.6, top-k 10	temperature 1.0, 1.3, 1.6, 2.0; temperature 1.6 with top-k = 10
new-algorithm prompt	the sentence `Make a new algorithm.` appended to the task sentence
2000-word novelty prompt	the sentence `Make a fully novel algorithm never seen before. Explain the idea in detail (about 2000 words) and implement it.` appended
top-3 ban	a prohibition sentence of the form `Do not use any of the following algorithms: ...` naming the three most frequent baseline ideas (open task: BFS, DFS, Dijkstra; weighted task: Dijkstra, BFS, SPFA)
generate-then-select	list 10 approaches, pick one with an external seeded RNG, implement it
probe-weight direction $\alpha = \ldots$	mean of the probe weight vectors of the menu classes minus the vector of the modal class
difference-in-means direction $\alpha = \ldots$	minus (mean activation of modal-idea samples minus mean of the rest), in the style of contrastive activation addition [6, 7, 12]
... (post hoc)	lower-strength settings of the same two directions, added after the sweep and excluded from the survival rules
rank-1 projection $a = \ldots$	menu erasure by partial orthogonal projection off the centred menu-centroid subspace (rank 1), strength a in [0, 1]
LEACE erasure $a = \ldots$	menu erasure with LEACE [8] fitted on one-hot menu labels, applied at strength a
novelty direction $\alpha = \ldots$	cross-task novelty direction: difference in means of prompt-final states between plain prompts and the same prompts followed by `Make a fully novel algorithm never seen before.`, over 5 non-path tasks x 40 phrasings

random direction α = ..., draw k	matched-norm random direction (3 draws, seed 777) at the same strength
random projection a = ..., draw k	random rank-1 projection (3 draws) at the same strength
baseline rerun (noise floor)	the baseline regenerated in differently composed batches
..., held-out	re-draw on fresh seeds 1000..1099

Pre-registered rules (frozen at 15:39:37 on 2026-10-15, before the sweep). The flattening knobs survive if, on the held-out draw, their idea entropy is at least 2x that of the best eligible temperature and the bootstrap 95% CI of their entropy difference from the matched random direction excludes 0. Menu erasure survives if its judge-confirmed novel-and-correct share is at least 3x the best of temperature, the novelty prompts and the ban, at least 0.05 absolute, and above random projection. The cross-task novelty direction survives if atypicality rises monotonically with strength (Spearman at least 0.8 over α in {0, five strengths}) on both tasks and the top strength beats random. The best setting per family was picked on indices 0..99 and judged on the fresh held-out seeds, to remove winner's-curse bias. We define the *prompting ceiling* as the outcome in which no internal knob survives these rules; the plan fixed it as the headline in that case, naming the prompt-level method on top of each diversity front.

1.4 Baseline: two ideas per task, and a weighted task that the model often gets wrong

The 300-sample baseline at T = 0.7 establishes each task's menu and how often each idea is correct. On the open task the model writes BFS about two times in three and DFS about one time in four; on the weighted task it writes Dijkstra three times in four and a BFS-style queue relaxation almost one time in four. The baseline menus are therefore two ideas each, {BFS, DFS} and {Dijkstra, BFS}, which made the planned rank-k menu erasure a rank-1 operation.

Table 1. Baseline idea distribution and per-class pass rate, T = 0.7, seeds 0..299.

task	idea class	count	share	passing	pass rate within class
open path search	BFS	205	0.683	150	0.732
open path search	DFS	80	0.267	54	0.675
open path search	other	12	0.040	5	0.417
open path search	Dijkstra	2	0.007	0	0.000
open path search	A*	1	0.003	1	1.000
open path search	**all**	300	1.000	210	0.700
weighted shortest path	Dijkstra	225	0.750	108	0.480
weighted shortest path	BFS	71	0.237	3	0.042
weighted shortest path	SPFA	2	0.007	0	0.000
weighted shortest path	other	1	0.003	1	1.000
weighted shortest path	bidirectional	1	0.003	0	0.000
weighted shortest path	**all**	300	1.000	112	0.373

Correctness differs sharply by idea on the weighted task: BFS-style programs pass on 3 of 71 samples because hop-order expansion is wrong under weights, and even Dijkstra passes only about half the time, failing on edge cases (failing Dijkstra programs are correct on a mean of 33% of the

50 graphs). Any knob that moves mass from Dijkstra toward BFS on this task will raise idea entropy and lower the pass rate at the same time. The planned screening assumption that at least 3 idea classes reach 3% on at least 2 tasks fails here: both tasks have exactly two classes above 3%. The plan's narrow-menu fallback (keep the task, report the narrow menu) applied.

> **Correction (iteration 2).** The two-idea menus above are a property of the influence arm's prompt, not of the model. That arm appended an output contract fixing the function signature; the user's request and the ban arm used the bare task sentence. On the bare prompt the same model writes BFS 179, DFS 44, A* 39, Dijkstra 22, other 14 and SPFA 2 of 300 on the open task (idea entropy 1.764 bits, four ideas above 3%), and Dijkstra 300 of 300 on the weighted task (Table 29 in Section 2.4). Every influence-arm conclusion in Sections 1.4 to 1.17 is therefore conditional on the contract prompt. The two arms also used different classifiers, so part of the weighted-task gap may be labelling (Section 2.3).

1.5 Classifier audit

A stratified sample of 200 programs was labelled blind by the judge. Agreement is 0.715 and Cohen's kappa [10] is 0.684, above the pre-registered 0.6 threshold, so the rule labels stand. The main confusions are a rule label of SPFA against a judge label of BFS (12 programs: queue relaxation without in-queue tracking) and a rule label of greedy best-first search against a judge label of Dijkstra (6). The menu classes themselves are labelled consistently: BFS 20/23, DFS 22/22 and Dijkstra 25/26 rule labels match the judge.

Table 2. Classifier audit: rule label (rows) against blind judge label (columns), n = 200, agreement = 0.715, Cohen's kappa = 0.684.

rule \ judge	A*	Bellman-Ford	BFS	bidirectional	DFS	Dijkstra	Floyd-Warshall	greedy best-first search	iterative deepening	other	random walk	SPFA	row total
A*	15	·	·	·	·	·	·	·	·	1	·	·	16
Bellman-Ford	·	15	1	·	·	·	·	·	·	·	·	·	16
BFS	·	·	20	·	·	·	·	·	·	3	·	·	23
bidirectional	·	·	3	7	·	3	·	·	·	3	·	·	16
DFS	·	·	·	·	22	·	·	·	·	·	·	·	22
Dijkstra	·	·	·	·	·	25	·	·	·	1	·	·	26
Floyd-Warshall	·	·	·	·	·	·	17	·	·	·	·	·	17
greedy best-first search	3	·	1	·	·	6	·	1	·	5	·	·	16
iterative deepening	·	·	1	·	·	·	·	·	8	·	·	·	9
other	·	2	·	·	·	8	·	·	·	6	·	·	16
random walk	·	·	·	·	·	·	·	·	·	·	7	·	7
SPFA	·	·	12	·	·	·	·	·	·	4	·	·	16

1.6 Locating the choice: probes at the first tokens see nothing, so the late-window fallback was used

The plan chose the intervention band by cross-validated multinomial probes for the final idea at the prompt-final token and the first 8 generated tokens. Two facts defeated that plan. First, prefill is deterministic, so the prompt-final residual is identical for all samples of a task; a within-task probe there sits at the majority baseline by construction. Second, the first 8 generated tokens are

boilerplate (an opening of the form `Certainly! Below is ...`). Probes on generated tokens 1 to 6 return a pooled balanced accuracy of 0.429 to 0.433 at every layer from 1 to 36, and tokens 7 and 8 (alone or averaged with the earlier ones) reach only 0.45 to 0.47. The value 0.433 is exactly the balanced accuracy of predicting each task's majority idea (BFS recall 205/276 = 0.743, Dijkstra recall 225/227 = 0.991, DFS and the rare class 0, mean 0.433): the early tokens carry task identity and nothing about which idea the sample will implement.

The algorithm name, where it appears at all, comes much later. Only 136 of 300 open-task and 152 of 300 weighted-task generations contain an explicit algorithm name anywhere in the text; among those, the first mention never falls within 8 tokens and has a median position of 23.5 tokens on the open task and 46.5 on the weighted task.

Table 3. Position of the first explicit algorithm name in baseline generations (token index after the assistant turn starts).

task	generations	with an algorithm name	median token of first mention	quartiles (25/50/75)	share within 8 tokens	within 32	within 64
open path search	300	136	23.5	20.0 / 23.5 / 40.0	0.000	0.654	0.904
weighted shortest path	300	152	46.5	24.75 / 46.5 / 72.25	0.000	0.447	0.645

The plan's pre-registered fallback for chance-level early probes, which we call the late-window fallback, therefore applied: probes were trained on the mean residual over gen1..32, and the hook window was widened to the prompt-final token plus gen1..64. On that feature the probe reaches a pooled balanced accuracy of 0.638 to 0.713 across layers, against 0.392 to 0.429 for within-task shuffled labels (a control-task baseline in the sense of [9]). The curve is flat across depth: layer 0 already gives 0.648. A flat curve that is high from the embedding layer onward is what a readout of name and data-structure tokens inside the window would produce; it is not evidence of an early plan. The band rule picked layers 29 to 31 (mean gap over shuffled 0.282).

[Figure fig_probe: generated in the report phase, not in this draft]

Table 4. Per-layer idea probes (5-fold CV, logistic regression with C chosen per layer). The gen1-only, gen8-only and mean(gen1..8) columns are the original plan's features; the mean(gen1..32) columns are the late-window feature actually used.

layer	C	bal. acc. mean(gen1..32), pooled	open path search	weighted shortest path	shuffled-label pooled (mean ± sd, 5 perms)	gen1 only	gen8 only	mean(gen1..8)
0	0.1	0.648	0.424	0.589	0.410 ± 0.026	0.247	0.359	0.450
1	1.0	0.664	0.440	0.602	0.402 ± 0.022	0.433	0.468	0.466
2	1.0	0.672	0.455	0.599	0.403 ± 0.015	0.433	0.466	0.466
3	1.0	0.647	0.420	0.597	0.409 ± 0.022	0.433	0.463	0.467
4	1.0	0.663	0.444	0.597	0.424 ± 0.018	0.433	0.463	0.466
5	1.0	0.661	0.438	0.591	0.420 ± 0.020	0.433	0.463	0.467
6	1.0	0.663	0.437	0.605	0.421 ± 0.018	0.433	0.467	0.466
7	0.1	0.661	0.440	0.594	0.413 ± 0.009	0.433	0.468	0.467
8	1.0	0.644	0.418	0.592	0.416 ± 0.014	0.433	0.467	0.466

9	0.01	0.681	0.464	0.589	0.421 ± 0.011	0.433	0.466	0.465
10	0.01	0.684	0.467	0.589	0.420 ± 0.019	0.433	0.465	0.467
11	1.0	0.713	0.500	0.606	0.422 ± 0.035	0.433	0.463	0.467
12	0.1	0.668	0.443	0.598	0.427 ± 0.025	0.433	0.467	0.463
13	1.0	0.666	0.443	0.600	0.407 ± 0.017	0.433	0.468	0.467
14	1.0	0.699	0.485	0.598	0.423 ± 0.023	0.433	0.468	0.467
15	0.01	0.692	0.475	0.594	0.424 ± 0.018	0.433	0.468	0.467
16	0.1	0.690	0.470	0.606	0.416 ± 0.015	0.433	0.465	0.463
17	0.01	0.663	0.439	0.600	0.423 ± 0.005	0.433	0.467	0.467
18	1.0	0.664	0.438	0.605	0.419 ± 0.023	0.433	0.466	0.467
19	1.0	0.675	0.458	0.602	0.411 ± 0.019	0.433	0.467	0.467
20	1.0	0.658	0.434	0.600	0.405 ± 0.009	0.433	0.467	0.467
21	0.01	0.667	0.446	0.589	0.421 ± 0.007	0.433	0.465	0.467
22	1.0	0.665	0.440	0.600	0.410 ± 0.015	0.433	0.467	0.467
23	0.01	0.659	0.439	0.595	0.429 ± 0.014	0.433	0.468	0.465
24	0.1	0.668	0.445	0.603	0.415 ± 0.030	0.433	0.467	0.466
25	1.0	0.661	0.437	0.605	0.421 ± 0.012	0.433	0.468	0.463
26	1.0	0.676	0.455	0.605	0.417 ± 0.009	0.433	0.464	0.466
27	0.1	0.669	0.447	0.600	0.416 ± 0.010	0.433	0.463	0.466
28	1.0	0.656	0.434	0.594	0.408 ± 0.020	0.433	0.465	0.467
29	1.0	0.691	0.477	0.598	0.416 ± 0.013	0.433	0.464	0.464
30	1.0	0.694	0.481	0.595	0.399 ± 0.021	0.433	0.468	0.467
31	1.0	0.680	0.460	0.601	0.401 ± 0.011	0.433	0.466	0.465
32	1.0	0.660	0.437	0.600	0.422 ± 0.021	0.433	0.467	0.468
33	0.1	0.665	0.442	0.603	0.424 ± 0.007	0.433	0.468	0.467
34	1.0	0.638	0.409	0.598	0.416 ± 0.015	0.433	0.467	0.467
35	0.1	0.665	0.442	0.603	0.411 ± 0.015	0.433	0.468	0.465
36	1.0	0.670	0.450	0.597	0.392 ± 0.015	0.433	0.468	0.463

What this settles for the hypothesis is limited but real. In this model and protocol, no linear readout at the prompt-final token or the first 8 generated tokens predicts the idea. The idea becomes decodable only once the 32-token window includes the explanatory preamble, which places the commitment token, if there is a single one, somewhere after token 8. Whether it precedes the first mention of the algorithm's name cannot be read from these data. Forking-path resampling, which would locate it per sample, belonged to the temperature arm and did not run.

1.7 Building the intervention vectors, with offline checks

All vectors were built from the baseline only, at layers 29 to 31. Three checks were run before any steered generation.

The two flattening directions are nearly orthogonal. The probe-weight direction and the difference-in-means direction have cosine 0.16 to 0.23, so the two flattening variants are different interventions, not two estimates of one.

Table 5. Cosine between the probe-weight and difference-in-means flattening directions.

layer	cos(probe-weight dir., diff-in-means dir.) open path search	weighted shortest path
layer 29	0.207	0.159
layer 30	0.225	0.163
layer 31	0.213	0.170

LEACE [8] erases idea information offline; rank-1 projection nearly does. On cached activations, a fresh probe for the menu label reaches a balanced accuracy of 0.58 to 0.61 on the open task and 0.88 to 0.89 on the weighted task before erasure. After LEACE it reaches 0.20 and 0.33 to 0.35 (chance is 0.33); after rank-1 projection, 0.32 to 0.34. The menu-centroid subspace is effectively one-dimensional: the second singular value of the centred centroid matrix is about 6 times smaller than the first on the open task and 56 to 61 times smaller on the weighted task.

Table 6. Offline erasure check: balanced accuracy of a fresh 5-fold probe for the menu label (menu classes plus a pooled class for everything else) before and after erasure.

task, layer	probe bal. acc. before	after LEACE	after rank-1 projection	chance	majority share
open path search, layer 29	0.607	0.198	0.334	0.333	0.683
weighted shortest path, layer 29	0.888	0.334	0.325	0.333	0.750
open path search, layer 30	0.582	0.199	0.337	0.333	0.683
weighted shortest path, layer 30	0.884	0.352	0.323	0.333	0.750
open path search, layer 31	0.577	0.198	0.339	0.333	0.683
weighted shortest path, layer 31	0.893	0.332	0.323	0.333	0.750

Table 7. Singular values of the centred menu-centroid matrix ($k = 2$ menu classes, so the rank is at most 1).

task, layer	singular value 1	singular value 2
open path search, layer 29	3.632	0.576
weighted shortest path, layer 29	7.930	0.134
open path search, layer 30	4.023	0.626
weighted shortest path, layer 30	8.834	0.145
open path search, layer 31	4.848	0.811
weighted shortest path, layer 31	10.148	0.180

The cross-task novelty direction transfers as a readout. Extracted from sorting, substring search, primality testing, string compression and matrix multiplication prompts only, the direction separates novel-request from plain prompts on held-out path-search and weighted-path prompts with AUROC 1.0 at all three layers. Leave-one-task-out directions have cosine 0.92 to 0.98 with each held-out task's own difference vector. The direction is close to orthogonal to both flattening directions and to the erasure subspace (absolute cosines at most 0.095).

Table 8. Cross-task novelty direction: transfer and overlap checks.

layer	leave-one-task-out cosine: sorting	substring	primality	compression	matmul	held-out AUROC open path search	weighted shortest path	cos with probe-weight dir. (path / weighted)	cos with diff-in-means dir. (path / weighted)	cos with erasure subspace (path / weighted)
layer 29	0.941	0.978	0.956	0.934	0.966	1.0	1.0	0.037 / -0.027	0.008 / 0.093	0.009 / 0.095
layer 30	0.943	0.977	0.956	0.925	0.966	1.0	1.0	0.027 / -0.013	-0.021 / 0.082	0.024 / 0.088
layer 31	0.946	0.978	0.952	0.919	0.964	1.0	1.0	0.021 / -0.028	0.011 / 0.038	0.039 / 0.039

1.8 Strength calibration

Strengths for the three additive knobs were calibrated on the open task with n = 20 per point over a grid of 10 values. The lower bound is the largest grid value with pass at least 0.95x baseline, the upper bound the smallest larger value with pass at most 0.40x baseline, and five strengths are spaced geometrically between them. The calibration was frozen before the full run and reused for the weighted task and the random controls. The calibration rows already show what the full sweep confirms: both flattening directions convert the open task to DFS well before they break code (difference-in-means at $\alpha = 0.2$ gives DFS 19 of 20), and the probe-weight direction keeps pass near baseline up to $\alpha = 0.75$ only because its samples are almost all DFS.

Table 9. Strength calibration on the open task (n = 20 per row).

knob	strength α	pass (n=20)	degenerate share	idea labels
probe-weight direction	0.02	0.75	0.00	DFS 4, BFS 15, Dijkstra 1
probe-weight direction	0.05	0.55	0.00	DFS 8, BFS 11, A* 1
probe-weight direction	0.1	0.60	0.00	DFS 12, A* 1, BFS 6, other 1
probe-weight direction	0.2	0.90	0.00	DFS 14, BFS 5, other 1
probe-weight direction	0.35	0.85	0.00	DFS 19, other 1
probe-weight direction	0.5	0.65	0.00	DFS 20
probe-weight direction	0.75	0.70	0.00	DFS 19, BFS 1
probe-weight direction	1.0	0.45	0.25	DFS 14, syntax error 5, BFS 1
probe-weight direction	1.5	0.05	0.70	other 3, syntax error 14, DFS 3
probe-weight direction	2.0	0.20	0.85	no code 14, DFS 4, syntax error 1, BFS 1
probe-weight direction	**chosen**	α_lo = 0.75, α_hi = 1.5		0.75, 0.892, 1.06, 1.26, 1.5
difference-in-means direction	0.02	0.65	0.00	DFS 9, BFS 10, A* 1
difference-in-means direction	0.05	0.75	0.00	DFS 13, BFS 6, other 1
difference-in-means direction	0.1	0.65	0.00	DFS 15, BFS 5

difference-in-means direction	0.2	0.60	0.00	DFS 19, BFS 1
difference-in-means direction	0.35	0.40	0.00	DFS 17, BFS 2, other 1
difference-in-means direction	0.5	0.65	0.00	DFS 19, BFS 1
difference-in-means direction	0.75	0.30	0.15	syntax error 1, other 1, DFS 16, BFS 2
difference-in-means direction	1.0	0.00	0.95	no code 10, syntax error 9, DFS 1
difference-in-means direction	1.5	0.10	0.80	DFS 1, no code 16, BFS 2, other 1
difference-in-means direction	2.0	0.00	0.95	no code 19, other 1
difference-in-means direction	**chosen**	α_lo = 0.05, α_hi = 1.0		0.05, 0.106, 0.224, 0.473, 1.0
novelty direction	0.02	0.55	0.00	DFS 3, BFS 16, A* 1
novelty direction	0.05	0.65	0.00	DFS 5, BFS 14, A* 1
novelty direction	0.1	0.80	0.00	DFS 7, BFS 12, A* 1
novelty direction	0.2	0.85	0.00	DFS 10, BFS 10
novelty direction	0.35	0.60	0.00	A* 2, BFS 9, DFS 7, other 1, Dijkstra 1
novelty direction	0.5	0.45	0.00	Dijkstra 2, DFS 11, BFS 4, A* 3
novelty direction	0.75	0.60	0.00	BFS 11, Dijkstra 1, DFS 7, greedy best-first search 1
novelty direction	1.0	0.40	0.00	DFS 9, BFS 8, A* 1, Dijkstra 2
novelty direction	1.5	0.60	0.05	BFS 12, DFS 4, Dijkstra 2, no code 1, A* 1
novelty direction	2.0	0.65	0.05	DFS 6, BFS 12, no code 1, Dijkstra 1
novelty direction	**chosen**	α_lo = 0.2, α_hi = 2.0		0.2, 0.356, 0.632, 1.12, 2.0

1.9 Main sweep: practitioner knobs

The sweep used seeds 0..149 for every method setting and seeds 0..74 for random controls (n = 150 and 75 per task). Table 10 holds the knobs a practitioner would reach for. Three results stand out.

First, temperature buys almost no idea diversity before it breaks the code. Between T = 0.7 and T = 1.0 idea entropy rises from 1.175 to 1.278 bits on the open task and from 0.916 to 1.165 on the weighted task, while the weighted-task pass rate falls from 0.380 to 0.233. T = 1.0 misses pooled eligibility by 0.006 (0.423 against 0.429), so the best eligible temperature is T = 0.7 itself. At T = 1.3 a quarter to a third of outputs are syntax errors, at T = 1.6 almost all are, and at T = 2.0 one output in 300 contains a parseable program. Top-k = 10 at T = 1.6 keeps most outputs parseable but the pass rate is still 0.227 and 0.040. Across all temperatures the same two menu ideas

dominate the outputs that parse; no temperature introduced a third idea class above 5% of samples.

[Figure fig_temp: generated in the report phase, not in this draft]

Second, the top-3 ban is largely ignored. On the open task, 142 of 150 programs (0.947) still implement a banned idea, and the ban only evens BFS and DFS out (70/70), which raises entropy from 1.175 to 1.404 bits. On the weighted task 131 of 150 (0.873) still implement Dijkstra, BFS or SPFA; the ban shifts mass from Dijkstra to BFS (from 111/37 to 62/66), which raises entropy to 1.779 bits and drops the pass rate from 0.380 to 0.153, because BFS-style programs fail on weighted graphs. Small amounts of Bellman-Ford (7) and Floyd-Warshall (3) appear. The entropy gain from the ban on the weighted task is mostly a move toward the incorrect menu idea. Non-compliance with negative instructions is a known failure: naming a forbidden item can prime it [5]. The artifact also rendered a PCA view of layer-30 window activations with per-idea centroids for the baseline, the ban and the best projection setting. It is not reproduced here, because its coordinates were not written to the results files, and it is not the planned manifold analysis, which belonged to the ban arm.

> **Correction (iteration 2).** The statement that the ban is largely ignored holds only for this prompt: the contract prompt with a list-style prohibition (`Do not use any of the following algorithms: ...`). On the bare prompt with the short wording `Do not use X, Y or Z.`, the same model complies far more often: 1 of 100 programs still uses BFS when BFS is banned on the open task, 29 of 100 still use a banned idea under the top-3 ban and 38 of 100 under the top-5 ban; on the sorting task 0 of 100 under the top-3 ban; on the weighted task 12, 46 and 61 of 100 for the top-1, top-3 and top-5 bans (Section 2.4). The contract and the wording are confounded, so the difference is unexplained.

Third, the novelty prompts split. The new-algorithm prompt leaves the pooled numbers almost unchanged (pass 0.553, entropy 1.053) but moves the two tasks in opposite directions: more DFS, A* and Dijkstra on the open task (entropy 1.533), and more Dijkstra on the weighted task (137 of 150; entropy 0.573, pass 0.547, the highest weighted-task pass rate in the sweep; the held-out re-draw gives 0.580). The 2000-word novelty prompt raises entropy to 2.205 bits pooled but collapses the pass rate to 0.083, with 39 and 64 syntax errors. Generate-then-select gives the highest entropy of all (2.561 pooled; 9 idea classes on each task) at pass 0.263: listing approaches first does put A*, bidirectional, IDDFS, Bellman-Ford and Floyd-Warshall into the outputs, but these implementations fail the oracle more often.

Table 10. Practitioner knobs and noise-floor rerun (sweep seeds 0..149).

setting	n per task	path: pass	path: idea entropy	path: idea labels	weighted: pass	weighted: idea entropy	weighted: idea labels	pooled pass	pooled idea entropy	eligible (path / weighted / pooled)
baseline T = 0.7	150	0.673	1.175	BFS 98, DFS 44, other 7, Dijkstra 1	0.380	0.916	Dijkstra 111, BFS 37, other 1, SPFA 1	0.527	1.045	y / y / y
baseline rerun (noise floor)	150	0.673	1.133	BFS 95, DFS 50, other 4, Dijkstra 1	0.413	0.891	Dijkstra 109, BFS 40, SPFA 1	0.543	1.012	y / y / y

temperature 1.0	150	0.613	1.278	BFS 82, DFS 61, other 3, Dijkstra 3, A* 1	0.233	1.165	Dijkstra 105, BFS 35, other 6, DFS 2, greedy best-first search 1, syntax error 1	0.423	1.222	y / n / n
temperature 1.3	150	0.320	1.288	BFS 52, DFS 51, syntax error 41, other 5, Dijkstra 1	0.067	1.377	Dijkstra 73, syntax error 48, BFS 15, other 7, DFS 4, greedy best-first search 2, A* 1	0.193	1.333	n / n / n
temperature 1.6	150	0.007	0.811	syntax error 139, no code 7, BFS 3, other 1	0.000	1.000	syntax error 134, no code 14, Dijkstra 1, other 1	0.003	0.906	n / n / n
temperature 1.6, top-k 10	150	0.227	1.366	BFS 71, DFS 56, syntax error 13, other 6, Dijkstra 4	0.040	1.301	Dijkstra 84, syntax error 27, BFS 26, other 7, DFS 4, greedy best-first search 1, no code 1	0.133	1.333	n / n / n
temperature 2.0	150	0.000	0.000	syntax error 78, no code 72	0.000	0.000	no code 78, syntax error 71, other 1	0.000	0.000	n / n / n
new-algorithm prompt	150	0.560	1.533	BFS 75, DFS 61, A* 6, Dijkstra 4, other 2, bidirectional 1, iterative deepening 1	0.547	0.573	Dijkstra 137, BFS 8, Bellman-Ford 2, Floyd-Warshall 1, SPFA 1, other 1	0.553	1.053	y / y / y
2000-word novelty prompt	150	0.113	1.881	DFS 65, syntax error 39, BFS 24, other 6, bidirectional 4, Dijkstra 4, greedy best-first search 2, random walk 2, iterative deepening 2, no code 1, A* 1	0.053	2.528	syntax error 64, Dijkstra 37, BFS 17, other 7, greedy best-first search 5, DFS 5, bidirectional 4, random walk 3, Bellman-Ford 3, no code 2, Floyd-Warshall 2, A* 1	0.083	2.205	n / n / n
top-3 ban	150	0.707	1.404	DFS 70, BFS 70, other 5, Dijkstra 2, A* 2, greedy best-first search 1	0.153	1.779	BFS 66, Dijkstra 62, Bellman-Ford 7, other 6, DFS 3, Floyd-Warshall 3, SPFA 3	0.430	1.591	y / n / y

generate-then-select	150	0.353	2.549	Dijkstra 45, DFS 40, BFS 26, bidirectional 10, A* 10, iterative deepening 6, other 6, syntax error 3, random walk 2, greedy best-first search 2	0.173	2.574	Dijkstra 58, A* 21, BFS 19, Floyd-Warshall 13, Bellman-Ford 13, bidirectional 9, syntax error 7, greedy best-first search 5, DFS 3, other 2	0.263	2.561	n / n / n

Table 11 lists what the generate-then-select pipeline chose in stage 1. The picked names are far more varied than what the model writes unprompted; on the weighted task they include the Johnson, Kruskal and Prim algorithms (the last two are minimum-spanning-tree methods, not shortest-path methods), A* search and Bellman-Ford.

Table 11. Generate-then-select, stage 1: items picked by the seeded RNG from the 10-item lists the model wrote itself (possessive apostrophes dropped from item names).

task	stage-1 lists parsed	mean items listed	15 most frequent picked items (count)
open path search	150/150	9.99	Bidirectional Search (15); Depth-First Search (14); Dijkstra Algorithm (11); Uniform Cost Search (11); Breadth-First Search (10); Depth-First Search (DFS) (9); A* Search Algorithm (8); Greedy Best-First Search (8); Hill Climbing Search (6); Best-First Search (3); A* Search (3); Depth-First Search with Iterative Deepening (3); Wavefront Search (2); Bi-directional Search (2); Iterative Deepening Search (2)
weighted shortest path	150/150	9.97	A* Search Algorithm (19); Bellman-Ford Algorithm (14); Dijkstra Algorithm (11); Floyd-Warshall Algorithm (11); Bidirectional Search (9); Johnson Algorithm (9); Breadth-First Search (BFS) with Relaxation (4); Kruskal Algorithm (3); Breadth-First Search (BFS) with relaxation (3); A* Search (2); Breadth-First Search (with relaxation) (2); Bi-directional Search (2); Johnson algorithm (2); Prim Minimum Spanning Tree (MST) Algorithm (2); Kruskal MST Approach for Shortest Path (via MST) (1)

1.10 Main sweep: internal knobs

Flattening directions. Neither flattening direction flattens. The difference-in-means direction at $\alpha = 0.224$ turns the open task into DFS on 145 of 150 samples (baseline BFS 98, DFS 44) at an unchanged pass rate (0.680 against 0.673); at $\alpha = 0.106$ it already gives DFS 126. Idea entropy therefore falls, to 0.235 bits. The probe-weight direction does the same at $\alpha = 0.75$ (DFS 148 of 150) at a lower pass rate (0.580). On the weighted task the same directions push Dijkstra toward BFS: at $\alpha = 0.224$ the split is 77/60 against 111/37 at baseline, entropy rises to 1.444 bits, and the pass rate falls to 0.287 because more programs are BFS. With only two menu ideas, a direction that points from the mode toward the mean of the menu overshoots the uniform point and selects the minority idea. At the top strengths both directions destroy the output: the difference-in-means direction at $\alpha = 1.0$ leaves 148 of 150 open-task samples with no code or a syntax error, the probe-weight direction at $\alpha = 1.5$ leaves 125.

Table 12. Flattening directions (rows marked post hoc are excluded from survival).

setting	n per task	path: pass	path: idea entropy	path: idea labels	weighted: pass	weighted: idea entropy	weighted: idea labels	pooled pass	pooled idea entropy	eligible (path / weighted / pooled)
probe-weight direction $\alpha = 0.75$	150	0.580	0.102	DFS 148, BFS 2	0.173	1.543	BFS 88, Dijkstra 42, DFS 11, SPFA 4, other 3, syntax error 1, bidirectional 1	0.377	0.823	y / n / n
probe-weight direction $\alpha = 0.892$	150	0.387	0.323	DFS 129, syntax error 14, BFS 6, other 1	0.087	1.522	BFS 76, Dijkstra 44, DFS 18, syntax error 8, other 2, no code 1, SPFA 1	0.237	0.922	n / n / n
probe-weight direction $\alpha = 1.06$	150	0.247	0.806	DFS 81, syntax error 54, BFS 10, other 4, Dijkstra 1	0.127	1.603	BFS 60, Dijkstra 45, syntax error 19, DFS 16, no code 6, other 3, Floyd-Warshall 1	0.187	1.205	n / n / n
probe-weight direction $\alpha = 1.26$	150	0.020	0.994	syntax error 117, DFS 18, other 15	0.200	1.410	Dijkstra 71, BFS 28, no code 23, syntax error 17, DFS 5, other 4, bidirectional 1, Bellman-Ford 1	0.110	1.202	n / n / n
probe-weight direction $\alpha = 1.5$	150	0.007	0.943	syntax error 121, other 16, DFS 9, no code 4	0.140	0.808	no code 64, Dijkstra 62, syntax error 12, BFS 9, other 2, DFS 1	0.073	0.875	n / n / n
difference-in-means direction $\alpha = 0.05$	150	0.727	1.149	DFS 87, BFS 58, other 5	0.420	0.972	Dijkstra 106, BFS 40, other 3, syntax error 1	0.573	1.061	y / y / y

difference-in-means direction α = 0.106	150	0.727	0.714	DFS 126, BFS 22, other 1, A* 1	0.360	1.192	Dijkstra 94, BFS 49, other 3, Bellman-Ford 2, syntax error 1, SPFA 1	0.543	0.953	y / y / y
difference-in-means direction α = 0.224	150	0.680	0.235	DFS 145, BFS 4, other 1	0.287	1.444	Dijkstra 77, BFS 60, SPFA 6, other 4, bidirectional 1, syntax error 1, DFS 1	0.483	0.840	y / n / y
difference-in-means direction α = 0.473	150	0.580	0.450	DFS 138, BFS 7, other 4, syntax error 1	0.313	1.459	Dijkstra 80, BFS 52, other 6, syntax error 4, DFS 3, no code 2, Floyd-Warshall 1, Bellman-Ford 1, SPFA 1	0.447	0.954	y / y / y
difference-in-means direction α = 1	150	0.000	1.000	no code 79, syntax error 69, DFS 1, other 1	0.033	1.141	no code 110, syntax error 20, Dijkstra 13, BFS 6, other 1	0.017	1.071	n / n / n
probe-weight direction α = 0.1 (post hoc)	150	0.633	1.158	DFS 83, BFS 62, other 3, A* 1, syntax error 1	0.247	1.152	Dijkstra 86, BFS 58, other 5, syntax error 1	0.440	1.155	y / n / y
probe-weight direction α = 0.2 (post hoc)	150	0.733	1.046	DFS 100, BFS 47, other 2, Dijkstra 1	0.287	1.223	Dijkstra 78, BFS 66, other 5, SPFA 1	0.510	1.135	y / n / y
probe-weight direction α = 0.35 (post hoc)	150	0.807	0.415	DFS 140, BFS 8, other 1, Dijkstra 1	0.313	1.340	BFS 73, Dijkstra 69, SPFA 4, other 2, bidirectional 1, Bellman-Ford 1	0.560	0.877	y / y / y

probe-weight direction α = 0.5 (post hoc)	150	0.753	0.160	DFS 147, BFS 2, Dijkstra 1	0.220	1.352	BFS 74, Dijkstra 67, SPFA 5, DFS 2, other 2	0.487	0.756	y / n / y
difference-in-means direction α = 0.075 (post hoc)	150	0.740	0.978	DFS 106, BFS 41, other 3	0.380	1.100	Dijkstra 102, BFS 42, other 3, syntax error 1, DFS 1, bidirectional 1	0.560	1.039	y / y / y

[Figure fig_select: generated in the report phase, not in this draft]

Menu erasure. Rank-1 projection and LEACE leave the open-task distribution close to baseline and the weighted task more Dijkstra-heavy, at pass rates near baseline. LEACE at a = 1.0 gives open-task BFS 117 and weighted-task Dijkstra 127; rank-1 projection at a = 0.25 evens the open task (DFS 75, BFS 73) and raises weighted-task BFS to 47. No erasure setting produced a single passing program outside the known classes that the judge accepted as novel. Rank-1 projection at a = 0.5 and above produced 17 to 23 syntax errors on the open task.

Table 13. Menu erasure.

setting	**n per task**	**path: pass**	**path: idea entropy**	**path: idea labels**	**weighted: pass**	**weighted: idea entropy**	**weighted: idea labels**	**pooled pass**	**pooled idea entropy**	**eligible (path / weighted / pooled)**
rank-1 projection a = 0.25	150	0.720	1.089	DFS 75, BFS 73, other 2	0.380	1.027	Dijkstra 100, BFS 47, other 3	0.550	1.058	y / y / y
rank-1 projection a = 0.5	150	0.640	0.983	DFS 76, BFS 56, syntax error 18	0.347	0.958	Dijkstra 101, BFS 48, DFS 1	0.493	0.971	y / y / y
rank-1 projection a = 0.75	150	0.627	1.086	DFS 74, BFS 57, syntax error 17, other 2	0.233	1.117	Dijkstra 96, BFS 49, other 2, SPFA 1, Bellman-Ford 1, syntax error 1	0.430	1.101	y / n / y
rank-1 projection a = 1	150	0.607	0.976	DFS 75, BFS 52, syntax error 23	0.340	1.113	Dijkstra 101, BFS 43, other 2, Bellman-Ford 2, syntax error 1, DFS 1	0.473	1.045	y / y / y

LEACE erasure a = 0.5	150	0.740	1.079	BFS 100, DFS 46, other 2, Dijkstra 2	0.293	0.792	Dijkstra 122, BFS 24, bidirectional 2, DFS 1, syntax error 1	0.517	0.936	y / n / y
LEACE erasure a = 1	150	0.673	1.072	BFS 117, DFS 23, A* 5, other 2, Dijkstra 2, greedy best-first search 1	0.387	0.635	Dijkstra 127, BFS 17, syntax error 4, Floyd-Warshall 1, DFS 1	0.530	0.853	y / y / y

Cross-task novelty direction. This is the internal knob with the clearest effect on diversity. At $\alpha = 0.356$ the open task shows 7 idea classes (DFS 76, BFS 51, A* 9, Dijkstra 6, other 5, greedy best-first 2, bidirectional 1) and entropy 1.750 bits, at pass 0.560, just at the per-task eligibility line. The weighted task barely moves (entropy 0.984). At $\alpha = 2.0$ about 15% of outputs contain no code.

Table 14. Cross-task novelty direction.

setting	n per task	path: pass	path: idea entropy	path: idea labels	weighted: pass	weighted: idea entropy	weighted: idea labels	pooled pass	pooled idea entropy	eligible (path / weighted / pooled)
novelty direction α = 0.2	150	0.647	1.354	BFS 77, DFS 63, Dijkstra 4, other 3, A* 2, syntax error 1	0.440	1.100	Dijkstra 114, BFS 27, other 4, DFS 2, SPFA 2, Bellman-Ford 1	0.543	1.227	y / y / y
novelty direction α = 0.356	150	0.560	1.750	DFS 76, BFS 51, A* 9, Dijkstra 6, other 5, greedy best-first search 2, bidirectional 1	0.373	0.984	Dijkstra 115, BFS 26, other 7, Bellman-Ford 1, syntax error 1	0.467	1.367	y / y / y

novelty direction α = 0.632	150	0.493	1.769	DFS 75, BFS 48, A* 13, Dijkstra 8, other 3, greedy best-first search 1, syntax error 1, Floyd-Warshall 1	0.340	1.270	Dijkstra 108, BFS 26, DFS 9, other 5, Bellman-Ford 2	0.417	1.519	n / y / n
novelty direction α = 1.12	150	0.493	1.520	BFS 75, DFS 59, Dijkstra 11, A* 2, greedy best-first search 2, other 1	0.360	0.950	Dijkstra 119, BFS 22, other 6, greedy best-first search 1, DFS 1, no code 1	0.427	1.235	n / y / n
novelty direction α = 2	150	0.567	1.330	BFS 76, DFS 42, no code 22, Dijkstra 7, A* 3	0.327	0.585	Dijkstra 112, no code 24, BFS 12, SPFA 1, DFS 1	0.447	0.958	y / y / y

Random controls. Matched-norm random directions and random projections produce entropy changes of the same size as the targeted knobs. Several random settings reach higher pooled entropy than any targeted internal setting at comparable pass rate (draw 2 at $\alpha = 1.12$: 1.407 bits at pass 0.467; draw 2 at $\alpha = 1.0$: 1.370 at 0.487), and the random draws disagree with each other in direction: draw 1 tends toward DFS on the open task and draw 0 toward BFS on the weighted task.

> **Correction (iteration 2).** The parity claim in this paragraph, and in Table 19 and conclusion 3 of Section 1.17, compares single targeted settings at $n = 150$ with individual random draws at $n = 75$, and ranks the best of 27 random cells against 10 targeted ones. Re-analysed against the draw-averaged random entropy (Miller-Madow, n-matched seeds 0..74, hierarchical bootstrap over draws and samples), 3 of 18 targeted settings beat the random draw mean, 5 are significantly below it and 10 are undetermined (Table 67 in Section 2.9). Parity therefore does not hold as stated.

Table 15. Random controls ($n = 75$ per task).

setting	n per task	path: pass	path: idea entropy	path: idea labels	weighted: pass	weighted: idea entropy	weighted: idea labels	pooled pass	pooled idea entropy	eligible (path / weighted / pooled)
random direction α = 0.224, draw 0	75	0.707	0.978	BFS 44, DFS 31	0.307	1.088	Dijkstra 38, BFS 36, other 1	0.507	1.033	y / y / y

random direction $\alpha = 0.224$, draw 1	75	0.693	1.183	BFS 50, DFS 21, Dijkstra 2, other 2	0.373	0.837	Dijkstra 55, BFS 20	0.533	1.010	y / y / y
random direction $\alpha = 0.224$, draw 2	75	0.680	1.121	BFS 44, DFS 29, other 2	0.440	1.117	Dijkstra 53, BFS 18, other 3, Bellman-Ford 1	0.560	1.119	y / y / y
random direction $\alpha = 0.473$, draw 0	75	0.680	1.009	BFS 54, DFS 19, Dijkstra 1, other 1	0.253	0.990	BFS 42, Dijkstra 33	0.467	0.999	y / n / y
random direction $\alpha = 0.473$, draw 1	75	0.533	1.320	BFS 39, DFS 32, Dijkstra 2, other 1, A* 1	0.253	1.025	Dijkstra 48, BFS 26, other 1	0.393	1.173	n / n / n
random direction $\alpha = 0.473$, draw 2	75	0.733	1.000	BFS 38, DFS 37	0.360	1.250	Dijkstra 50, BFS 20, other 3, DFS 1, SPFA 1	0.547	1.125	y / y / y
random direction $\alpha = 0.632$, draw 0	75	0.640	1.093	BFS 49, DFS 24, other 1, Dijkstra 1	0.280	0.953	BFS 47, Dijkstra 28	0.460	1.023	y / n / y
random direction $\alpha = 0.632$, draw 1	75	0.560	1.492	DFS 37, BFS 29, Dijkstra 7, other 2	0.373	0.917	Dijkstra 58, BFS 15, SPFA 1, bidirectional 1	0.467	1.204	y / y / y
random direction $\alpha = 0.632$, draw 2	75	0.787	0.994	DFS 41, BFS 34	0.387	1.067	Dijkstra 59, BFS 11, other 2, A* 1, SPFA 1, bidirectional 1	0.587	1.030	y / y / y
random direction $\alpha = 1$, draw 0	75	0.613	1.092	BFS 48, DFS 23, syntax error 2, other 1, Dijkstra 1	0.093	0.986	BFS 49, Dijkstra 22, syntax error 3, other 1	0.353	1.039	y / n / n
random direction $\alpha = 1$, draw 1	75	0.453	1.191	DFS 50, BFS 19, syntax error 2, Dijkstra 2, A* 1, other 1	0.347	0.743	Dijkstra 64, BFS 5, syntax error 2, DFS 2, bidirectional 1, other 1	0.400	0.967	n / y / n
random direction $\alpha = 1$, draw 2	75	0.707	1.284	BFS 41, DFS 30, A* 2, Dijkstra 2	0.267	1.456	Dijkstra 50, BFS 15, other 3, syntax error 2, SPFA 1, Bellman-Ford 1, A* 1, Floyd-Warshall 1, greedy best-first search 1	0.487	1.370	y / n / y

random direction α = 1.06, draw 0	75	0.547	1.095	BFS 50, DFS 22, other 3	0.187	1.218	BFS 47, Dijkstra 19, syntax error 5, other 2, bidirectional 1, Floyd-Warshall 1	0.367	1.156	n / n / n
random direction α = 1.06, draw 1	75	0.400	1.075	DFS 54, BFS 15, syntax error 2, A* 2, Dijkstra 2	0.373	0.871	Dijkstra 61, BFS 11, DFS 2, other 1	0.387	0.973	n / y / n
random direction α = 1.06, draw 2	75	0.640	1.284	BFS 39, DFS 32, other 3, A* 1	0.293	1.241	Dijkstra 54, BFS 14, other 4, Floyd-Warshall 2, DFS 1	0.467	1.262	y / n / y
random direction α = 1.12, draw 0	75	0.440	1.167	BFS 45, DFS 21, syntax error 5, other 4	0.120	1.526	BFS 38, Dijkstra 24, syntax error 6, other 3, bidirectional 2, DFS 1, SPFA 1	0.280	1.346	n / n / n
random direction α = 1.12, draw 1	75	0.573	1.144	DFS 51, BFS 18, A* 2, Dijkstra 2, syntax error 2	0.347	0.697	Dijkstra 63, BFS 7, DFS 3, syntax error 2	0.460	0.920	y / y / y
random direction α = 1.12, draw 2	75	0.520	1.497	BFS 40, DFS 27, A* 5, other 2, greedy best-first search 1	0.413	1.316	Dijkstra 53, BFS 12, other 6, bidirectional 1, greedy best-first search 1, Bellman-Ford 1, syntax error 1	0.467	1.407	n / y / y
random direction α = 1.26, draw 0	75	0.373	0.776	BFS 54, DFS 16, syntax error 5	0.080	1.623	BFS 44, Dijkstra 20, bidirectional 3, greedy best-first search 2, syntax error 2, SPFA 1, other 1, A* 1, Bellman-Ford 1	0.227	1.199	n / n / n
random direction α = 1.26, draw 1	75	0.400	1.219	DFS 52, BFS 14, other 3, syntax error 3, Dijkstra 2, A* 1	0.307	1.110	Dijkstra 55, BFS 13, DFS 3, syntax error 2, A* 1, SPFA 1	0.353	1.164	n / y / n
random direction α = 1.26, draw 2	75	0.453	1.613	BFS 36, DFS 24, A* 10, other 3, syntax error 2	0.280	1.409	Dijkstra 51, BFS 13, other 5, syntax error 2, greedy best-first search 1, DFS 1, Bellman-Ford 1, A* 1	0.367	1.511	n / n / n

random direction α = 1.5, draw 0	75	0.560	0.703	BFS 63, DFS 8, syntax error 2, Dijkstra 1, other 1	0.120	1.264	BFS 42, Dijkstra 27, other 3, syntax error 2, bidirectional 1	0.340	0.983	y / n / n
random direction α = 1.5, draw 1	75	0.387	1.224	DFS 45, BFS 12, Dijkstra 6, syntax error 6, no code 5, other 1	0.213	1.341	Dijkstra 47, BFS 15, other 6, syntax error 4, DFS 1, A* 1, no code 1	0.300	1.283	n / n / n
random direction α = 1.5, draw 2	75	0.360	1.849	BFS 29, DFS 26, A* 9, Dijkstra 4, syntax error 3, no code 2, greedy best-first search 1, other 1	0.280	1.550	Dijkstra 50, other 9, BFS 8, syntax error 2, DFS 2, Bellman-Ford 2, SPFA 1, Floyd-Warshall 1	0.320	1.699	n / n / n
random direction α = 2, draw 0	75	0.187	1.481	no code 48, BFS 16, DFS 7, Dijkstra 3, bidirectional 1	0.093	0.889	no code 48, Dijkstra 19, BFS 4, syntax error 3, other 1	0.140	1.185	n / n / n
random direction α = 2, draw 1	75	0.267	1.420	no code 25, BFS 24, DFS 11, syntax error 10, Dijkstra 4, other 1	0.173	0.965	Dijkstra 33, no code 22, syntax error 12, BFS 5, DFS 2, other 1	0.220	1.193	n / n / n
random direction α = 2, draw 2	75	0.147	1.564	syntax error 37, BFS 13, no code 11, DFS 11, other 1, A* 1, Dijkstra 1	0.120	1.095	syntax error 35, Dijkstra 25, BFS 7, no code 6, SPFA 1, other 1	0.133	1.329	n / n / n
random projection a = 0.5, draw 0	75	0.560	1.115	BFS 51, DFS 21, Dijkstra 2, other 1	0.293	1.063	Dijkstra 51, BFS 22, other 1, Bellman-Ford 1	0.427	1.089	y / n / n
random projection a = 0.5, draw 1	75	0.613	1.226	BFS 49, DFS 22, other 2, A* 1, Dijkstra 1	0.387	1.131	Dijkstra 50, BFS 22, other 2, Bellman-Ford 1	0.500	1.179	y / y / y
random projection a = 0.5, draw 2	75	0.680	1.092	BFS 47, DFS 26, other 2	0.480	0.864	Dijkstra 59, BFS 14, other 2	0.580	0.978	y / y / y
random projection a = 1, draw 0	75	0.600	1.021	BFS 54, DFS 18, other 3	0.347	1.020	Dijkstra 52, BFS 21, Bellman-Ford 2	0.473	1.021	y / y / y
random projection a = 1, draw 1	75	0.720	0.892	BFS 56, DFS 18, Dijkstra 1	0.480	1.037	Dijkstra 51, BFS 22, other 2	0.600	0.964	y / y / y

random projection a = 1, draw 2	75	0.667	0.999	BFS 50, DFS 24, Dijkstra 1	0.427	0.748	Dijkstra 59, BFS 16	0.547	0.873	y / y / y

Table 16 carries the secondary metrics for every method setting: off-menu correct share, atypicality, degeneration, flip rate against the same-seed baseline sample, the order-1 Hill number, idea entropy among passing programs only, and phrasing diversity (mean normalised edit distance among up to 25 programs of the modal idea). Two regularities are worth stating. Phrasing diversity within the modal idea stays between 0.39 and 0.67 in every setting and rises with temperature (0.503 at T = 0.7 to 0.599 at T = 1.3 on the open task), while idea entropy rises little (1.175 to 1.288 bits). Read against the user's cases, this is closer to (a), more phrasings of the same two ideas, than to (b). It rests on two temperature points that still produce code and one phrasing measure; the dedicated ideas-versus-phrasings partition belonged to the temperature arm. And on the weighted task the entropy among passing programs is 0 in many settings: whatever the knob does to the idea mix, the programs that pass are nearly all Dijkstra.

Table 16. Secondary metrics for every method setting (sweep).

setting | path: off-menu correct | path: atypicality | path: degeneration | path: flip rate vs baseline | path: Hill-1 | path: entropy among passing | path: phrasing diversity (modal idea) | weighted: off-menu correct | weighted: atypicality | weighted: degeneration | weighted: flip rate | weighted: Hill-1 | weighted: entropy among passing | weighted: phrasing diversity

baseline T = 0.7 | 0.027 | 0.835 | 0.000 | n/a | 2.258 | 1.097 | 0.503 | 0.007 | 0.654 | 0.000 | n/a | 1.887 | 0.254 | 0.467

baseline rerun (noise floor) | 0.007 | 0.816 | 0.000 | 0.167 | 2.194 | 1.013 | 0.513 | 0.007 | 0.643 | 0.000 | 0.107 | 1.855 | 0.324 | 0.495

temperature 1.0 | 0.013 | 0.958 | 0.000 | 0.387 | 2.425 | 1.087 | 0.557 | 0.007 | 0.902 | 0.007 | 0.300 | 2.243 | 0.373 | 0.557

temperature 1.3 | 0.013 | 1.007 | 0.280 | 0.633 | 2.442 | 1.217 | 0.599 | 0.000 | 1.239 | 0.360 | 0.580 | 2.598 | 0.469 | 0.635

temperature 1.6 | 0.000 | 1.100 | 0.980 | 1.000 | 1.755 | 0.000 | 0.570 | 0.000 | 2.813 | 1.000 | 1.000 | 2.000 | 0.000 | n/a

temperature 1.6, top-k 10 | 0.000 | 1.034 | 0.107 | 0.560 | 2.577 | 0.834 | 0.610 | 0.000 | 1.090 | 0.213 | 0.500 | 2.464 | 0.000 | 0.663

temperature 2.0 | 0.000 | n/a | 1.000 | 1.000 | 1.000 | 0.000 | n/a | 0.000 | 5.320 | 1.000 | 1.000 | 1.000 | 0.000 | n/a

new-algorithm prompt | 0.027 | 1.213 | 0.020 | 0.520 | 2.893 | 1.213 | 0.485 | 0.007 | 0.554 | 0.007 | 0.293 | 1.487 | 0.190 | 0.396

2000-word novelty prompt | 0.013 | 1.858 | 0.447 | 0.747 | 3.684 | 1.277 | 0.603 | 0.000 | 2.578 | 0.693 | 0.787 | 5.767 | 0.000 | 0.624

top-3 ban | 0.013 | 1.095 | 0.000 | 0.467 | 2.647 | 1.126 | 0.500 | 0.033 | 1.633 | 0.000 | 0.580 | 3.431 | 1.328 | 0.465

generate-then-select | 0.113 | 3.272 | 0.040 | 0.787 | 5.851 | 1.751 | 0.530 | 0.047 | 3.196 | 0.113 | 0.673 | 5.953 | 1.000 | 0.605

probe-weight direction α = 0.75 | 0.000 | 1.324 | 0.000 | 0.707 | 1.073 | 0.091 | 0.516 | 0.013 | 1.692 | 0.007 | 0.627 | 2.914 | 1.217 | 0.477

probe-weight direction α = 0.892 | 0.007 | 1.309 | 0.093 | 0.720 | 1.251 | 0.418 | 0.537 | 0.013 | 1.809 | 0.060 | 0.673 | 2.871 | 1.352 | 0.526

probe-weight direction $\alpha = 1.06$ | 0.000 | 1.353 | 0.380 | 0.773 | 1.749 | 0.800 | 0.598 | 0.000 | 1.810 | 0.180 | 0.673 | 3.039 | 0.629 | 0.559

probe-weight direction $\alpha = 1.26$ | 0.000 | 2.184 | 0.867 | 0.980 | 1.992 | 0.000 | 0.620 | 0.007 | 1.160 | 0.287 | 0.647 | 2.657 | 0.561 | 0.514

probe-weight direction $\alpha = 1.5$ | 0.000 | 2.529 | 0.947 | 0.980 | 1.922 | 0.000 | n/a | 0.000 | 0.665 | 0.533 | 0.693 | 1.751 | 0.276 | 0.513

difference-in-means direction $\alpha = 0.05$ | 0.007 | 1.037 | 0.000 | 0.420 | 2.218 | 1.050 | 0.505 | 0.013 | 0.716 | 0.007 | 0.313 | 1.962 | 0.320 | 0.491

difference-in-means direction $\alpha = 0.106$ | 0.000 | 1.238 | 0.000 | 0.647 | 1.640 | 0.688 | 0.475 | 0.033 | 0.898 | 0.007 | 0.407 | 2.285 | 0.795 | 0.484

difference-in-means direction $\alpha = 0.224$ | 0.000 | 1.324 | 0.000 | 0.700 | 1.177 | 0.191 | 0.477 | 0.020 | 1.160 | 0.007 | 0.480 | 2.721 | 0.935 | 0.506

difference-in-means direction $\alpha = 0.473$ | 0.000 | 1.343 | 0.007 | 0.720 | 1.366 | 0.269 | 0.521 | 0.013 | 1.174 | 0.040 | 0.480 | 2.748 | 0.893 | 0.501

difference-in-means direction $\alpha = 1$ | 0.000 | 2.268 | 0.993 | 0.993 | 2.000 | 0.000 | n/a | 0.000 | 0.902 | 0.880 | 0.940 | 2.206 | 0.000 | 0.508

probe-weight direction $\alpha = 0.1$ (post hoc) | 0.000 | 1.011 | 0.007 | 0.467 | 2.232 | 0.982 | 0.501 | 0.007 | 0.922 | 0.007 | 0.313 | 2.222 | 0.357 | 0.473

probe-weight direction $\alpha = 0.2$ (post hoc) | 0.007 | 1.091 | 0.000 | 0.560 | 2.065 | 0.982 | 0.479 | 0.007 | 1.009 | 0.000 | 0.427 | 2.335 | 0.603 | 0.420

probe-weight direction $\alpha = 0.35$ (post hoc) | 0.007 | 1.323 | 0.000 | 0.700 | 1.333 | 0.419 | 0.466 | 0.033 | 1.127 | 0.000 | 0.513 | 2.531 | 1.232 | 0.511

probe-weight direction $\alpha = 0.5$ (post hoc) | 0.000 | 1.348 | 0.000 | 0.700 | 1.117 | 0.000 | 0.445 | 0.013 | 1.172 | 0.007 | 0.533 | 2.552 | 0.851 | 0.499

difference-in-means direction $\alpha = 0.075$ (post hoc) | 0.007 | 1.118 | 0.000 | 0.533 | 1.970 | 0.958 | 0.470 | 0.013 | 0.806 | 0.007 | 0.333 | 2.143 | 0.381 | 0.475

rank-1 projection $a = 0.25$ | 0.000 | 0.906 | 0.000 | 0.340 | 2.127 | 0.996 | 0.541 | 0.013 | 0.767 | 0.000 | 0.180 | 2.038 | 0.219 | 0.453

rank-1 projection $a = 0.5$ | 0.000 | 0.939 | 0.120 | 0.547 | 1.977 | 0.992 | 0.489 | 0.000 | 0.715 | 0.000 | 0.207 | 1.943 | 0.000 | 0.434

rank-1 projection $a = 0.75$ | 0.000 | 0.963 | 0.113 | 0.547 | 2.122 | 0.997 | 0.464 | 0.007 | 0.823 | 0.007 | 0.227 | 2.168 | 0.501 | 0.459

rank-1 projection $a = 1$ | 0.000 | 0.953 | 0.153 | 0.567 | 1.967 | 0.975 | 0.470 | 0.007 | 0.829 | 0.007 | 0.240 | 2.163 | 0.460 | 0.395

LEACE erasure $a = 0.5$ | 0.013 | 0.783 | 0.000 | 0.480 | 2.113 | 1.009 | 0.484 | 0.000 | 0.600 | 0.007 | 0.373 | 1.732 | 0.156 | 0.486

LEACE erasure $a = 1$ | 0.013 | 0.844 | 0.000 | 0.473 | 2.103 | 0.682 | 0.428 | 0.000 | 0.524 | 0.027 | 0.360 | 1.553 | 0.126 | 0.485

novelty direction $\alpha = 0.2$ | 0.013 | 1.037 | 0.007 | 0.427 | 2.557 | 1.082 | 0.539 | 0.020 | 0.830 | 0.000 | 0.327 | 2.143 | 0.502 | 0.445

novelty direction $\alpha = 0.356$ | 0.033 | 1.560 | 0.007 | 0.633 | 3.364 | 1.345 | 0.508 | 0.013 | 0.784 | 0.007 | 0.367 | 1.977 | 0.386 | 0.420

novelty direction $\alpha = 0.632$ | 0.033 | 1.675 | 0.047 | 0.647 | 3.408 | 1.337 | 0.537 | 0.007 | 1.121 | 0.000 | 0.467 | 2.411 | 0.139 | 0.452

novelty direction $\alpha = 1.12$ | 0.007 | 1.256 | 0.007 | 0.627 | 2.868 | 1.030 | 0.495 | 0.013 | 0.760 | 0.020 | 0.360 | 1.932 | 0.398 | 0.469

novelty direction $\alpha = 2$ | 0.013 | 1.064 | 0.167 | 0.540 | 2.514 | 1.016 | 0.474 | 0.007 | 0.501 | 0.173 | 0.487 |

1.500 | 0.144 | 0.414

Table 17. Secondary metrics for random controls and the held-out draw.

setting	path: off-menu correct	path: atypicality	path: degeneration	path: flip rate vs baseline	path: Hill-1	path: entropy among passing	path: phrasing diversity (modal idea)	weighted: off-menu correct	weighted: atypicality	weighted: degeneration	weighted: flip rate	weighted: Hill-1	weighted: entropy among passing	weighted: phrasing diversity
random direction α = 0.224, draw 0	0.000	0.787	0.000	0.293	1.970	0.924	0.524	0.000	0.925	0.000	0.400	2.126	0.258	0.459
random direction α = 0.224, draw 1	0.027	0.854	0.000	0.320	2.271	0.966	0.520	0.000	0.613	0.000	0.213	1.786	0.000	0.456
random direction α = 0.224, draw 2	0.000	0.837	0.000	0.400	2.175	0.897	0.514	0.013	0.865	0.000	0.333	2.169	0.196	0.445
random direction α = 0.473, draw 0	0.000	0.733	0.000	0.347	2.013	0.787	0.521	0.000	0.950	0.000	0.493	1.986	0.297	0.408
random direction α = 0.473, draw 1	0.000	1.020	0.000	0.440	2.497	0.934	0.516	0.000	0.772	0.000	0.307	2.035	0.000	0.478
random direction α = 0.473, draw 2	0.000	0.862	0.000	0.533	2.000	1.000	0.493	0.027	0.955	0.000	0.373	2.379	0.605	0.482
random direction α = 0.632, draw 0	0.000	0.796	0.000	0.493	2.134	0.871	0.512	0.000	1.027	0.000	0.507	1.936	0.702	0.422
random direction α = 0.632, draw 1	0.067	1.348	0.000	0.533	2.812	1.456	0.544	0.000	0.663	0.000	0.267	1.889	0.000	0.514
random direction α = 0.632, draw 2	0.000	0.912	0.000	0.533	1.991	1.000	0.501	0.000	0.817	0.000	0.267	2.095	0.362	0.476
random direction α = 1, draw 0	0.000	0.794	0.027	0.467	2.132	0.828	0.522	0.000	1.158	0.040	0.667	1.981	0.592	0.512
random direction α = 1, draw 1	0.000	1.268	0.027	0.627	2.283	0.960	0.528	0.000	0.690	0.027	0.320	1.674	0.000	0.502
random direction α = 1, draw 2	0.013	1.023	0.000	0.480	2.435	1.090	0.473	0.040	1.145	0.027	0.547	2.744	0.848	0.490
random direction α = 1.06, draw 0	0.000	0.787	0.013	0.493	2.136	0.712	0.504	0.013	1.380	0.067	0.693	2.327	0.946	0.506
random direction α = 1.06, draw 1	0.027	1.349	0.027	0.627	2.107	1.111	0.506	0.013	0.705	0.000	0.387	1.829	0.222	0.515
random direction α = 1.06, draw 2	0.013	0.977	0.000	0.533	2.435	1.031	0.520	0.013	1.033	0.000	0.493	2.364	0.530	0.556
random direction α = 1.12, draw 0	0.000	0.841	0.067	0.560	2.245	0.533	0.524	0.013	1.457	0.080	0.640	2.879	0.503	0.489
random direction α = 1.12, draw 1	0.027	1.310	0.027	0.653	2.210	1.118	0.513	0.000	0.668	0.027	0.333	1.621	0.391	0.504
random direction α = 1.12, draw 2	0.013	1.220	0.000	0.547	2.823	1.129	0.483	0.027	1.133	0.013	0.400	2.490	0.612	0.449
random direction α = 1.26, draw 0	0.000	0.614	0.067	0.440	1.712	0.371	0.551	0.013	1.670	0.027	0.680	3.079	1.252	0.505
random direction α = 1.26, draw 1	0.040	1.384	0.053	0.600	2.328	1.249	0.505	0.000	0.908	0.027	0.453	2.158	0.258	0.536
random direction α = 1.26, draw 2	0.000	1.497	0.027	0.640	3.058	0.960	0.511	0.040	1.190					

0.027 | 0.453 | 2.655 | 0.857 | 0.505

random direction $\alpha = 1.5$, draw 0 | 0.000 | 0.601 | 0.027 | 0.453 | 1.627 | 0.371 | 0.518 | 0.000 | 1.242 | 0.040 | 0.613 | 2.401 | 0.000 | 0.522

random direction $\alpha = 1.5$, draw 1 | 0.013 | 1.516 | 0.147 | 0.747 | 2.336 | 1.091 | 0.540 | 0.013 | 1.157 | 0.080 | 0.493 | 2.533 | 0.669 | 0.502

random direction $\alpha = 1.5$, draw 2 | 0.000 | 1.758 | 0.080 | 0.680 | 3.602 | 0.918 | 0.538 | 0.027 | 1.531 | 0.027 | 0.467 | 2.928 | 0.454 | 0.508

random direction $\alpha = 2$, draw 0 | 0.013 | 1.356 | 0.653 | 0.827 | 2.791 | 0.946 | 0.475 | 0.013 | 0.707 | 0.680 | 0.787 | 1.851 | 0.592 | 0.520

random direction $\alpha = 2$, draw 1 | 0.000 | 1.168 | 0.520 | 0.747 | 2.675 | 0.722 | 0.541 | 0.000 | 0.867 | 0.467 | 0.707 | 1.953 | 0.000 | 0.461

random direction $\alpha = 2$, draw 2 | 0.000 | 1.231 | 0.693 | 0.893 | 2.956 | 0.845 | 0.521 | 0.013 | 0.823 | 0.560 | 0.747 | 2.136 | 0.986 | 0.462

random projection $a = 0.5$, draw 0 | 0.013 | 0.817 | 0.000 | 0.200 | 2.166 | 0.945 | 0.515 | 0.000 | 0.792 | 0.000 | 0.147 | 2.090 | 0.000 | 0.435

random projection $a = 0.5$, draw 1 | 0.013 | 0.874 | 0.000 | 0.173 | 2.339 | 1.049 | 0.524 | 0.013 | 0.859 | 0.000 | 0.187 | 2.191 | 0.216 | 0.422

random projection $a = 0.5$, draw 2 | 0.013 | 0.799 | 0.000 | 0.200 | 2.131 | 1.046 | 0.519 | 0.027 | 0.655 | 0.000 | 0.160 | 1.820 | 0.310 | 0.435

random projection $a = 1$, draw 0 | 0.000 | 0.737 | 0.000 | 0.160 | 2.029 | 0.867 | 0.506 | 0.000 | 0.791 | 0.000 | 0.160 | 2.028 | 0.235 | 0.434

random projection $a = 1$, draw 1 | 0.013 | 0.683 | 0.000 | 0.227 | 1.856 | 0.890 | 0.514 | 0.013 | 0.778 | 0.000 | 0.240 | 2.052 | 0.593 | 0.465

random projection $a = 1$, draw 2 | 0.013 | 0.758 | 0.000 | 0.213 | 1.999 | 1.012 | 0.492 | 0.000 | 0.552 | 0.000 | 0.147 | 1.679 | 0.000 | 0.439

LEACE erasure $a = 0.5$, held-out | 0.000 | 0.796 | 0.000 | n/a | 2.035 | 0.966 | 0.466 | 0.000 | 0.552 | 0.010 | n/a | 1.639 | 0.191 | 0.465

rank-1 projection $a = 0.25$, held-out | 0.000 | 0.887 | 0.000 | n/a | 1.998 | 0.998 | 0.515 | 0.000 | 0.640 | 0.000 | n/a | 1.826 | 0.191 | 0.469

difference-in-means direction $\alpha = 0.05$, held-out | 0.010 | 0.962 | 0.000 | n/a | 2.077 | 1.070 | 0.443 | 0.000 | 0.852 | 0.000 | n/a | 2.147 | 0.316 | 0.449

random direction $\alpha = 0.05$, draw 0, held-out | 0.000 | 0.765 | 0.000 | n/a | 1.952 | 0.928 | 0.529 | 0.000 | 0.690 | 0.000 | n/a | 1.925 | 0.000 | 0.452

random projection $a = 0.25$, draw 0, held-out | 0.010 | 0.756 | 0.000 | n/a | 2.012 | 0.986 | 0.507 | 0.010 | 0.656 | 0.000 | n/a | 1.870 | 0.575 | 0.450

random projection $a = 0.5$, draw 0, held-out | 0.000 | 0.728 | 0.000 | n/a | 1.911 | 0.887 | 0.513 | 0.010 | 0.728 | 0.000 | n/a | 2.007 | 0.201 | 0.467

top-3 ban, held-out | 0.010 | 1.072 | 0.000 | n/a | 2.598 | 1.087 | 0.499 | 0.020 | 1.480 | 0.000 | n/a | 3.203 | 1.472 | 0.479

new-algorithm prompt, held-out | 0.070 | 1.603 | 0.020 | n/a | 3.175 | 1.332 | 0.495 | 0.000 | 0.460 | 0.000 | n/a | 1.430 | 0.000 | 0.411

2000-word novelty prompt, held-out | 0.000 | 1.771 | 0.520 | n/a | 4.004 | 0.971 | 0.639 | 0.000 | 1.882 | 0.730 | n/a | 4.136 | 0.000 | 0.639

temperature 0.7, held-out | 0.000 | 0.840 | 0.000 | n/a | 2.191 | 0.928 | 0.509 | 0.010 | 0.675 | 0.010 | n/a | 1.888 | 0.420 | 0.488

1.11 Pareto fronts across all settings

The figure below places every setting on the pooled pass-rate and idea-entropy plane. Above the eligibility line the front, in order of rising entropy, runs through the difference-in-means direction at $\alpha = 0.05$, the cross-task novelty direction at $\alpha = 0.2$ and 0.356, and the top-3 ban; generate-then-select extends it below the line. The ban is the eligible setting with the highest entropy (1.591 bits at pass 0.430), and the best internal setting (the novelty direction at $\alpha = 0.356$: 1.367 at 0.467) sits below it. Seven random-control settings would join the entropy front if controls were allowed on it.

[Figure fig_pareto: generated in the report phase, not in this draft]

Table 18. Pareto fronts (pass, metric) for each objective, with the random controls that would join them.

objective (pooled, vs pass rate)	Pareto front (methods only)	random controls that would join the front	best prompt or temperature	best internal knob
idea entropy	novelty direction $\alpha = 0.2$ (0.543, 1.227); novelty direction $\alpha = 0.356$ (0.467, 1.367); difference-in-means direction $\alpha = 0.05$ (0.573, 1.061); generate-then-select (0.263, 2.561); top-3 ban (0.430, 1.591)	random direction $\alpha = 0.224$, draw 2, random direction $\alpha = 0.473$, draw 2, random direction $\alpha = 0.632$, draw 2, random direction $\alpha = 1.12$, draw 2, random direction $\alpha = 1.5$, draw 2, random direction $\alpha = 1$, draw 2, random projection $a = 1$, draw 1	top-3 ban (1.591)	novelty direction $\alpha = 0.356$ (1.367)
off-menu correct share	novelty direction $\alpha = 0.356$ (0.467, 0.023); difference-in-means direction $\alpha = 0.05$ (0.573, 0.010); probe-weight direction $\alpha = 0.35$ (post hoc) (0.560, 0.020); generate-then-select (0.263, 0.080)	random direction $\alpha = 0.632$, draw 1, random direction $\alpha = 1$, draw 2, random projection $a = 0.5$, draw 2, random projection $a = 1$, draw 1	top-3 ban (0.023)	novelty direction $\alpha = 0.356$ (0.023)
judge-confirmed novel-and-correct share	difference-in-means direction $\alpha = 0.05$ (0.573, 0.000)	random projection $a = 1$, draw 1	baseline $T = 0.7$ (0.000)	LEACE erasure $a = 0.5$ (0.000)
idea entropy among passing programs	novelty direction $\alpha = 0.356$ (0.467, 0.866); difference-in-means direction $\alpha = 0.05$ (0.573, 0.685); probe-weight direction $\alpha = 0.35$ (post hoc) (0.560, 0.826); generate-then-select (0.263, 1.375); top-3 ban (0.430, 1.227)	random direction $\alpha = 1$, draw 2, random projection $a = 1$, draw 1	top-3 ban (1.227)	novelty direction $\alpha = 0.356$ (0.866)

Table 19. Ten highest-entropy eligible settings (pooled). Five of the ten are random controls.

rank	setting	pooled idea entropy

1	top-3 ban	1.591
2	random direction $\alpha = 1.12$, draw 2	1.407
3	random direction $\alpha = 1$, draw 2	1.370
4	novelty direction $\alpha = 0.356$	1.367
5	random direction $\alpha = 1.06$, draw 2	1.262
6	novelty direction $\alpha = 0.2$	1.227
7	random direction $\alpha = 0.632$, draw 1	1.204
8	random projection $a = 0.5$, draw 1	1.179
9	probe-weight direction $\alpha = 0.1$ (post hoc)	1.155
10	probe-weight direction $\alpha = 0.2$ (post hoc)	1.135

1.12 Selection, held-out draw and survival verdicts

The selection step picked, among eligible settings on indices 0..99, the difference-in-means direction at $\alpha = 0.05$ (highest entropy), rank-1 projection at $a = 0.25$ and LEACE at $a = 0.5$ (on the novel-and-correct share, all tied at 0, broken by off-menu share and entropy). The probe-weight direction had no eligible setting. The best eligible temperature was $T = 0.7$. These were re-drawn on fresh seeds 1000..1099 with their matched controls.

Table 20. Held-out draw, seeds 1000..1099 ($n = 100$ per task).

setting	**n per task**	**path: pass**	**path: idea entropy**	**path: idea labels**	**weighted: pass**	**weighted: idea entropy**	**weighted: idea labels**	**pooled pass**	**pooled idea entropy**	**eligible (path / weighted / pooled)**
temperature 0.7, held-out	100	0.700	1.131	BFS 59, DFS 38, other 3	0.300	0.917	Dijkstra 74, BFS 23, other 2, syntax error 1	0.500	1.024	y / y / y
new-algorithm prompt, held-out	100	0.670	1.667	BFS 44, DFS 38, Dijkstra 14, bidirectional 3, A* 1	0.580	0.516	Dijkstra 90, BFS 9, bidirectional 1	0.625	1.091	y / y / y
2000-word novelty prompt, held-out	100	0.100	2.002	DFS 34, syntax error 27, BFS 23, other 4, random walk 3, Dijkstra 3, no code 2, greedy best-first search 1, iterative deepening 1, bidirectional 1, A* 1	0.060	2.048	syntax error 55, Dijkstra 25, BFS 8, other 4, DFS 2, A* 1, SPFA 1, greedy best-first search 1, no code 1, bidirectional 1, Bellman-Ford 1	0.080	2.025	n / n / n

top-3 ban, held-out	100	0.670	1.378	BFS 51, DFS 43, other 2, greedy best-first search 2, A* 1, Dijkstra 1	0.150	1.679	Dijkstra 45, BFS 43, Bellman-Ford 4, SPFA 2, DFS 2, Floyd-Warshall 2, other 2	0.410	1.528	y / n / n
difference-in-means direction $\alpha = 0.05$, held-out	100	0.750	1.054	DFS 57, BFS 42, other 1	0.350	1.102	Dijkstra 66, BFS 30, other 4	0.550	1.078	y / y / y
random direction $\alpha = 0.05$, draw 0, held-out	100	0.700	0.965	BFS 61, DFS 39	0.270	0.945	Dijkstra 70, BFS 29, other 1	0.485	0.955	y / n / y
rank-1 projection $a = 0.25$, held-out	100	0.700	0.999	DFS 52, BFS 48	0.340	0.869	Dijkstra 71, BFS 29	0.520	0.934	y / y / y
random projection $a = 0.25$, draw 0, held-out	100	0.720	1.009	BFS 64, DFS 35, other 1	0.290	0.903	Dijkstra 73, BFS 26, other 1	0.505	0.956	y / n / y
LEACE erasure $a = 0.5$, held-out	100	0.790	1.025	BFS 62, DFS 37, A* 1	0.340	0.713	Dijkstra 83, BFS 16, DFS 1	0.565	0.869	y / y / y
random projection $a = 0.5$, draw 0, held-out	100	0.690	0.934	BFS 65, DFS 35	0.320	1.005	Dijkstra 73, BFS 24, other 2, SPFA 1	0.505	0.969	y / y / y

On the held-out draw the difference-in-means direction gives 0.93x the entropy of the held-out $T = 0.7$ draw on the open task and 1.20x on the weighted task, against a required 2x, and its CI against the random direction includes 0 on both tasks. Nothing survives.

Table 21. Pre-registered survival verdicts.

knob family	selected setting (indices 0..99)	rule	held-out numbers	verdict
probe-weight flattening	none	entropy $\geq 2\times$ best eligible temperature and CI vs random direction excludes 0	no eligible setting at matched pass rate	fails

difference-in-means flattening	difference-in-means direction $\alpha = 0.05$	same	path: entropy 1.054 vs 1.131 (ratio 0.932), diff vs random 0.090 [-0.054, 0.234], pass 0.75; weighted: 1.102 vs 0.917 (ratio 1.203), diff 0.158 [-0.069, 0.367], pass 0.35; pooled 1.078 vs 1.024, diff 0.124 [-0.019, 0.251]	fails
menu erasure, rank-1 projection	rank-1 projection $a = 0.25$	judge-confirmed novel-and-correct share $\geq 3\times$ best of temperature/prompts/ban, ≥ 0.05 absolute, CI vs random projection excludes 0	share 0.000 (comparators all 0.000); rule-only upper bound 0.000; pass 0.520	fails
menu erasure, LEACE	LEACE erasure $a = 0.5$	judge-confirmed novel-and-correct share $\geq 3\times$ best of temperature/prompts/ban, ≥ 0.05 absolute, CI vs random projection excludes 0	share 0.000 (comparators all 0.000); rule-only upper bound 0.000; pass 0.565	fails
cross-task novelty direction	all 5 strengths (no selection)	Spearman(α, atypicality) ≥ 0.8 on both tasks and top strength beats random	Spearman path 0.486 [-0.029, 0.543], weighted -0.257 [-0.486, 0.143]; top vs random path -0.176 [-0.529, 0.187], weighted -0.313 [-0.612, -0.041]	fails

Table 22. Targeted additive settings against the pooled random directions at the same strength (unpaired bootstrap; * = CI excludes 0).

setting	**path: Δ entropy vs random [95% CI]**	**path: Δ pass vs random [95% CI]**	**weighted: Δ entropy [95% CI]**	**weighted: Δ pass [95% CI]**
novelty direction $\alpha = 0.632$	+0.496 [+0.252, +0.719] *	-0.169 [-0.267, -0.064] *	+0.118 [-0.154, +0.383]	-0.007 [-0.104, +0.096]
novelty direction $\alpha = 1.12$	+0.092 [-0.166, +0.319]	-0.018 [-0.120, +0.096]	-0.454 [-0.748, -0.156] *	+0.067 [-0.033, +0.160]
novelty direction $\alpha = 2$	-0.219 [-0.514, +0.101]	+0.367 [+0.278, +0.462] *	-0.456 [-0.788, -0.075] *	+0.198 [+0.113, +0.282] *
difference-in-means direction $\alpha = 0.224$	-0.892 [-1.092, -0.678] *	-0.013 [-0.113, +0.078]	+0.374 [+0.130, +0.584] *	-0.087 [-0.182, +0.007]
difference-in-means direction $\alpha = 0.473$	-0.713 [-0.946, -0.475] *	-0.069 [-0.171, +0.029]	+0.302 [+0.058, +0.533] *	+0.024 [-0.071, +0.124]
difference-in-means direction $\alpha = 1$	-0.287 [-1.391, -0.196] *	-0.591 [-0.653, -0.524] *	-0.210 [-0.762, +0.149]	-0.202 [-0.262, -0.142] *
probe-weight direction $\alpha = 1.06$	-0.498 [-0.805, -0.173] *	-0.282 [-0.380, -0.187] *	+0.278 [+0.028, +0.497] *	-0.158 [-0.233, -0.076] *
probe-weight direction $\alpha = 1.26$	-0.463 [-0.637, -0.313] *	-0.389 [-0.453, -0.324] *	-0.201 [-0.546, +0.102]	-0.022 [-0.107, +0.058]
probe-weight direction $\alpha = 1.5$	-0.634 [-0.923, -0.434] *	-0.429 [-0.493, -0.367] *	-0.763 [-1.124, -0.427] *	-0.064 [-0.140, +0.004]

The cross-task novelty direction fails its own test in an instructive way. Atypicality rises from α =

0 to 0.632 on the open task (0.835 to 1.675) and then falls back (1.064 at $\alpha = 2.0$); on the weighted task it never rises consistently. The rule required Spearman at least 0.8 on both tasks; the values are 0.486 and -0.257. At the top strength the direction is less atypical than random directions on the weighted task (CI excludes 0 in the wrong direction).

[Figure fig_a4: generated in the report phase, not in this draft]

Table 23. Cross-task novelty direction: dose-response used by the survival rule ($\alpha = 0$ is the baseline).

strength α	path: atypicality	path: pass	weighted: atypicality	weighted: pass
0.0	0.835	0.673	0.654	0.380
0.2	1.037	0.647	0.830	0.440
0.356	1.560	0.560	0.784	0.373
0.632	1.675	0.493	1.121	0.340
1.12	1.256	0.493	0.760	0.360
2.0	1.064	0.567	0.501	0.327
slope of Δlog p vs −log p0 at top strength	0.242		0.230	

> **Correction (iteration 2).** The Spearman confidence intervals in Table 21 came from resampling samples within each strength cell with the baseline frequencies p0 held fixed, which is why the open-task point estimate 0.486 sat at the edge of its interval $[-0.029, 0.543]$. With six strength points Spearman's rho takes only multiples of 1/35. The exact permutation p-values (720 permutations) are 0.36 on the open task and 0.66 on the weighted task, two-sided; a bootstrap that also resamples the baseline gives $[-0.43, 0.60]$ and $[-1.00, 0.09]$. The verdict is unchanged.

1.13 Manipulation checks: the steering does what it was built to do internally, and the output follows only partly

Baseline probes (trained on unsteered mean(gen1..32) residuals) were applied to the residuals captured during steered generation, at layer 30 (band middle) and layer 33 (downstream). Two quantities are reported: the probe menu entropy, defined as the entropy of the probe softmax restricted to the menu, and the probe decodability, defined as the balanced accuracy with which the probe predicts the realised idea of each steered sample. For the flattening directions the probe menu entropy goes to 0.000 bits at every strength from $\alpha = 0.224$ (difference-in-means; from $\alpha = 0.106$ on the open task) and at every probe-weight strength: the steered state reads as one idea with full confidence, the opposite of flattening, and consistent with the DFS takeover in the outputs. For erasure, probe decodability drops to 0.27 to 0.44 at full strength (unsteered rerun: 0.48 and 0.57), yet the outputs remain BFS, DFS and Dijkstra. The internal readout can be erased without the behaviour changing.

Table 24. Manipulation checks with baseline probes on steered residuals (unsteered reference rows first).

setting	path, layer 30: probe menu entropy (bits)	path, layer 30: probe decodability	path, layer 33: menu entropy	path, layer 33: decodability	weighted, layer 30: menu entropy	weighted, layer 30: decodability	weighted, layer 33: menu entropy	weighted, layer 33: decodability
baseline rerun (noise floor)	0.181	0.481	0.274	0.489	0.128	0.574	0.186	0.577
temperature 0.7, held-out	0.233	0.484	0.322	0.475	0.152	0.594	0.201	0.565
temperature 1.0	0.165	0.570	0.237	0.655	0.168	0.417	0.222	0.424
top-3 ban	0.273	0.403	0.410	0.410	0.300	0.361	0.330	0.329
new-algorithm prompt	0.138	0.463	0.190	0.583	0.059	0.414	0.069	0.456
2000-word novelty prompt	0.173	0.327	0.350	0.323	0.334	0.323	0.340	0.287
probe-weight direction $\alpha = 0.75$	0.000	0.500	0.000	0.500	0.000	0.250	0.000	0.250
probe-weight direction $\alpha = 0.892$	0.000	0.333	0.000	0.333	0.000	0.250	0.000	0.250
probe-weight direction $\alpha = 1.06$	0.000	0.250	0.000	0.250	0.000	0.250	0.000	0.250
probe-weight direction $\alpha = 1.26$	0.000	0.500	0.000	0.500	0.000	0.250	0.000	0.250
probe-weight direction $\alpha = 1.5$	0.000	0.500	0.000	0.500	0.000	0.250	0.000	0.250
difference-in-means direction $\alpha = 0.05$	0.109	0.526	0.153	0.564	0.162	0.553	0.226	0.553
difference-in-means direction $\alpha = 0.106$	0.000	0.333	0.009	0.333	0.209	0.401	0.183	0.394
difference-in-means direction $\alpha = 0.224$	0.000	0.333	0.000	0.333	0.000	0.250	0.000	0.250

difference-in-means direction $\alpha = 0.473$	0.000	0.333	0.000	0.333	0.000	0.250	0.000	0.250
difference-in-means direction $\alpha = 1$	0.000	0.500	0.000	0.500	0.000	0.333	0.000	0.333
rank-1 projection $a = 0.25$	0.169	0.645	0.292	0.668	0.127	0.509	0.177	0.522
rank-1 projection $a = 0.5$	0.108	0.462	0.224	0.516	0.147	0.360	0.221	0.398
rank-1 projection $a = 0.75$	0.162	0.431	0.242	0.476	0.116	0.336	0.169	0.384
rank-1 projection $a = 1$	0.175	0.436	0.227	0.466	0.093	0.273	0.144	0.281
LEACE erasure $a = 0.5$	0.290	0.282	0.408	0.427	0.117	0.294	0.116	0.288
LEACE erasure $a = 1$	0.085	0.392	0.192	0.526	0.114	0.286	0.081	0.292
novelty direction $\alpha = 0.2$	0.150	0.600	0.294	0.571	0.099	0.382	0.381	0.377
novelty direction $\alpha = 0.356$	0.146	0.394	0.290	0.544	0.034	0.337	0.311	0.422
novelty direction $\alpha = 0.632$	0.015	0.250	0.355	0.371	0.064	0.319	0.431	0.277
novelty direction $\alpha = 1.12$	0.003	0.250	0.172	0.245	0.202	0.250	0.001	0.250
novelty direction $\alpha = 2$	0.000	0.250	0.000	0.250	0.129	0.250	0.000	0.250

1.14 Novelty: the judge found no novel correct program in any condition

Every distinct passing program in the catch-all class of the classifier, across the sweep and the held-out draw, was sent to the judge: 70 programs. All 70 were judged to be known algorithms, most often Dijkstra written in an unusual way (41). The judge-confirmed novel-and-correct share is therefore 0 in every condition, including generate-then-select and the 2000-word novelty prompt. Over all 267 judge calls (audit plus novelty checks) the verdicts were: Dijkstra 81, BFS 45, other (known) 31, DFS 25, Bellman-Ford 23, A* 18, Floyd-Warshall 17, iterative deepening 8,

bidirectional 7, random walk 7, unparsed 4, greedy best-first search 1. Judge spend was $0.081 against a $1.5 cap.

Table 25. Judge verdicts on the 70 distinct passing programs in the catch-all class of the classifier.

judge verdict	open path search	weighted shortest path	total
Dijkstra	4	37	41
other (known)	11	1	12
Bellman-Ford	0	7	7
BFS	7	0	7
DFS	3	0	3
distinct novel algorithm	0	0	0
all	25	45	70

1.15 Noise floor and independent verification

Regenerating the baseline in differently composed batches flips 16.7% of open-task and 10.7% of weighted-task idea labels at the same seeds (the noise-floor rerun row in Table 10), because bf16 numerics depend on batch composition. Common random numbers therefore pair conditions only approximately, and the per-sample flip rates in Tables 16 and 17 have a floor of roughly 0.11 to 0.17. An independent re-derivation script, with its own oracle runner, a coarse independent classifier and a separate recomputation of every metric, reproduces the stored numbers. The placebo test also shows that the bootstrap CIs are somewhat anticonservative: random splits of the baseline produce a CI excluding 0 at a rate of 0.10 and 0.15 instead of 0.05, which matters for any claim resting on a CI that barely excludes 0. None of the iteration's verdicts rests on such a claim, because nothing came close to the survival margins.

Table 26. Independent verification and placebo checks.

check	result
independent oracle runner vs stored pass/fail	1.000 agreement on 147 programs
coarse independent classifier vs AST classifier	0.973 agreement
entropy difference, open path search, baseline vs noise-floor rerun (placebo)	95% CI [-0.182, 0.281], excludes 0: False
entropy difference, weighted shortest path, baseline vs noise-floor rerun (placebo)	95% CI [-0.189, 0.228], excludes 0: False
entropy difference, open path search, top-3 ban vs baseline (real effect)	95% CI [-0.035, 0.445], excludes 0: False
entropy difference, weighted shortest path, top-3 ban vs baseline (real effect)	95% CI [0.560, 1.103], excludes 0: True
rate at which a random split of the baseline yields a CI excluding 0	path 0.10, weighted 0.15 (nominal 0.05)

1.16 Bookkeeping

- Artifact: `art_aSN7z8WuA7Ir` (gen_art_experiment_3). 19,250 generations in total, in 111 setting files per task (baseline, calibration, sweep, held-out, post-hoc, generate-then-select). Pipeline stages and wall-clock times as recorded: baseline about 5 min, activation cache 2 min, probes 2 min, vectors 3 min, calibration 4 min, sweep about 65 min, post-sweep chain (judge, select, held-out, judge, analyse) about 25 min, post-hoc settings 5 min, verification 4 min; roughly two GPU-hours on one RTX PRO 4500 (32 GB).

> **Correction (iteration 2).** The file counts are not 111 per task. The open task has 111 generation files and 110 scored files; the weighted task has 81 and 80. The difference is the 30 calibration files (`calib_A4_*`, `calib_Hdim_*`, `calib_Hprobe_*`, ten strengths each), which exist only for the open task, and the generate-then-select stage-1 list file `gts_list`, which has no scored counterpart. The 19,250 total is correct.

- The first sweep launch was stopped after about 2,400 generations because of CUDA allocator thrashing and resumed from the per-setting files with `PYTORCH_CUDA_ALLOC_CONF=expandable_segments:True` (allocator only, no change to numerics). Two held-out files whose scoring thread crashed on a malformed return were scored afterwards by `score_backlog.py`.
- Deviations from the plan, all recorded in the artifact: the late-window probe feature and hook window; menus exclude the catch-all class, so $k = 2$ and erasure is rank 1; oracle runs only the requested code; code-block early stop for all conditions; $n = 150$ and 75 instead of the planned 100 and 50, with a 100-seed held-out draw; post-hoc low-strength flattening settings reported but excluded from survival; a single model.
- Kept on the run volume only (not in the published repository): `acts/` (0.9 GB baseline residuals, split into parts under 100 MB) and `caps/` (0.9 GB residuals captured during steering). Published: generations, scored rows, judge verdicts, vectors (`vectors/vectors.pt` and one LEACE eraser file per task, `vectors/leace_*.pt`), figures and results files.
- The external judge model was the only paid call: $0.081.
- Files: pre-registered rules `results/rules.json` (frozen 2026-10-15 15:39:37); all metrics and CIs `results/results.json`; independent re-derivation `verify_headline.py` → `results/verify_headline.json`; PCA view `figs/fig_pca_manifold.png`; artifact figures `figs/` (probe curve, three Pareto fronts, fraction-correct violins, novelty-direction dose-response, idea shares, PCA view); schema output `method_out.json` with `full_`, `mini_` and `preview_` variants.

1.17 What iteration 1 established, and what it did not

Established, on one model and two tasks.

1. The internal knobs tested here are selectors, not diversifiers. A single residual direction at layers 29 to 31, built from baseline activations, moves the open task from two-thirds BFS to 97% DFS without costing correctness. This is controllable choice among ideas the model already writes. No knob produced an idea outside the model's usual repertoire.

2. Under the pre-registered rules no internal knob beats the practitioner knobs: the difference-in-means direction reached 0.93x and 1.20x the entropy of the best eligible temperature on held-out seeds (2x required), menu erasure produced no novel correct program, and the cross-task novelty direction was not monotone in strength. The headline the rules prescribe is the prompting ceiling, with the top-3 ban atop the eligible entropy front and generate-then-select atop the full front.
3. Matched-norm random directions raise idea entropy about as much as targeted ones, and five of the ten highest-entropy eligible settings are random controls. Most of the diversity that steering buys here is non-specific perturbation.
4. Temperature is not a creativity knob for this model on these tasks. Its eligible range ends at $T = 0.7$; above that, correctness collapses while idea entropy gains a few tenths of a bit and no new idea class appears above 5% of samples.
5. Bans are mostly ignored by this 3B model (87% to 95% of programs still use a banned idea), and on the weighted task the ban moves mass from the correct menu idea to the incorrect one.
6. Novelty, defined strictly as a correct program the judge does not recognise as a known algorithm, was zero in every condition.
7. The idea is not linearly decodable from the prompt-final token or the first 8 generated tokens; it becomes decodable from the mean over the first 32 generated tokens, which cover the explanatory preamble, at about the same level at every layer.

> **Correction (iteration 2).** Three of these conclusions are revised by later evidence. Conclusion 3 (random parity) does not hold as stated once targeted settings are compared with the draw-averaged random entropy (Section 2.9). Conclusion 5 generalised from one prompt protocol to the model: non-compliance is 87% to 95% under the contract prompt with list-style wording, and 0% to 46% under the bare prompt with short-list wording (61% for the weighted-task top-5 list) (Section 2.4). Conclusion 6 is near-floor by construction on textbook tasks and only covered passing programs that the classifier put in its catch-all class; it is restated in the closing section as "no condition produced a correct program that a judge did not recognise". All seven conclusions are conditional on the contract prompt.

Dead ends recorded as such.

- *Flattening the idea utilities as a novelty knob (the main hypothesis's claim about direct utility edits).* Killed by Table 12, the idea-composition figure in Section 1.10 and Table 24: the flattening directions select the minority idea, the internal readout collapses to one idea, and the held-out entropy ratio is 0.93 and 1.20 against a required 2.
- *Menu erasure as a route to novelty (the menu-erasure alternative).* Killed by Tables 13 and 25: erasure removes idea decodability offline (weighted task 0.888 to 0.334) and online (0.27 to 0.44), but the outputs remain BFS, DFS and Dijkstra and the novel-and-correct share is 0.
- *A single portable unusualness dial (the unusualness-dial alternative).* Killed as a dial by Table 23 (Spearman 0.486 and -0.257, required 0.8), although the direction transfers perfectly as a readout (AUROC 1.0, Table 8). Reading whether novelty was requested is easy; turning that reading into atypical outputs did not work.

- *Probing the first 8 tokens for the choice.* The planned probe feature carried only task identity (Table 4).
- *Temperature arm and ban arm.* Not delivered; no evidence either way from them.

Open after iteration 1.

- Whether the temperature law holds (straight Arrhenius lines, no reordering) or the gradual-commitment alternative holds; the influence arm's temperature rows show no reordering among the two menu ideas but cover only $T = 0.7$ to 1.3 with usable output.
- Where the commitment point is per sample (forking-path resampling), and whether a utility vector read there predicts temperature, seed and ban effects.
- Seeds: untested.
- Where banned probability goes (nearest neighbour, far side, or proportional): the influence arm's single ban shows mostly non-compliance, which by itself argues that a 3B model may be the wrong place to test substitution structure, since there is little banned mass to redistribute.

> **Correction (iteration 2).** The ban arm's bare-prompt data, unreported at the time, show large compliance and structured substitution in this same 3B model (Section 2.4), so the argument that a 3B model has too little banned mass to redistribute is withdrawn.

- Whether the selector finding generalises to the reserved model and task.

Iteration 2: the bare prompt, the two unreported arms, and bans that act through the prohibition sentence

2.1 Why this iteration ran what it ran

The review of iteration 1 was blocking, and most of its must-fix items concerned the record, not the analysis. Two of the three arms had been written off as "not delivered", yet both had left thousands of generations on disk. The ban arm had 4,360 bare-prompt generations, labelled but never executed. The temperature arm had its full grid and its seed rows, generated but unlabelled. The influence arm, the only one reported, had appended an output contract (a fixed function signature) to every prompt, and that contract appeared to create the two-idea menus on which every steering conclusion rested. On the bare prompt, the ban arm's open-task baseline had four ideas above 3%, not two. The review also found three problems with the steering claims. The claim that random directions raise idea entropy as much as targeted ones compared single targeted cells with individual random draws at half the sample size. The closest published steering papers had not been compared. And conclusion 5 ("bans are mostly ignored by this 3B model") was contradicted by the ban arm's own data.

The hypothesis was revised in response (title: "Temperature flattens ideas, context reorders them"). It drops the claim that flattening the idea utilities is the creativity knob, and it moves every core test to the user's bare prompt. It also recasts a prompt ban as a displacement of an internal idea state rather than a restriction of the choice set. It states four groups of testable claims, which we

refer to by name throughout this section:

- the *temperature law*: the log-odds of each idea against the modal idea are linear in 1/T, the top-3 ideas never swap rank for $T \leq 1.3$, and within-idea phrasing diversity rises at least 3 times faster with T than between-idea diversity;
- three ban claims: the *regularity test* (a ban can lower the share of an idea it does not name, which a model where the ban only removes options cannot produce); the *substitution-prediction test* (a predictor built only from internal geometry predicts where banned mass goes, at least 20% better in Jensen-Shannon divergence (JSD) than independence of irrelevant alternatives (IIA), with the correct top substitute in at least 70% of single bans); and the *transplant test* (adding the ban's internal displacement to a placebo run reproduces at least 50% of the ban's effect, against at most 15% for random displacements);
- the *seed-shock test*: random seed strings act as small displacements, and the greedy flip rate they cause can be predicted within ± 0.10, with under 5% flips on the weighted task;
- the *off-menu-correct test*: a ban or an internal displacement reaches a share of correct off-menu programs at least 3 times that of the best temperature, at no less than 0.8 times the baseline pass rate.

We define *off-menu correct share* as the fraction of programs that pass the execution oracle and implement an idea outside the task's baseline menu (ideas at 3% or more in the baseline). On textbook tasks a correct algorithm that a judge does not recognise is near-floor by construction, so this share replaces judge-confirmed invention as the novelty observable.

The hypothesis also imposed screening discipline. Existing data (the ban arm and the influence arm) are the screen. Predictors, layers and thresholds are fixed and hashed on the screen before any confirmation sample exists. Confirmation uses conditions never generated before: rank-5 to rank-8 single bans, random ban pairs, fresh seed strings, a reserved fourth task (k-th smallest element) and a second model family (gemma-2-2b-it [20]), all on fresh sample seeds.

Three artifacts were planned and ran, one per block of claims:

1. **A shared label space plus the temperature and seed questions**. This artifact relabelled every bare-prompt program with one audited classifier, executed all of them, and wrote up the ban arm. It also fitted and froze the temperature law on the iteration-1 grid and tested it on a fresh 3,615-generation re-draw, and froze a seed-shock predictor before generating 2,160 programs under 120 new seed strings. It was chosen first because it answers questions (a), (b) and (c) of the original request, the seeding question and the ban question from data that already existed, at almost no GPU cost.
2. **Bans as displacements**. This artifact read the ban's displacement from the prompt-final residual stream, fitted seven one-parameter predictors of where banned mass goes, froze them, and scored them on 5,930 never-generated confirmation generations including the k-th-smallest task. It then ran the causal transplant and drew the idea manifold the user asked for.
3. **Selection versus flattening on the bare prompt, and a second model**. This artifact re-analysed iteration 1's random-control claim against draw-averaged random entropy, and ran

a pre-registered select-or-flatten test of a difference-in-means "away from the default idea" direction on Qwen with 5 random draws per strength. It then repeated the ban-prediction, temperature and transplant tests on gemma-2-2b-it, with Gemma Scope sparse autoencoders [19] as a descriptive last step.

The model remains Qwen2.5-3B-Instruct [1] unless gemma is named. The tasks are open path search, sorting (`Implement a sorting algorithm`), weighted shortest path (the weighted task, for short) and the k-th-smallest task; tables label them by name. All bare-prompt work uses the ban arm's wording (`Implement a path search algorithm in Python`, `Implement a sorting algorithm`, `Implement a function that finds a shortest path in a weighted graph`, with no system prompt beyond Qwen's default and no output contract). We call this the *ban-arm wording*. The iteration-1 temperature grid used a slightly different wording, which we call the *grid wording*: a final full stop, "in Python." added to sorting, and a longer weighted-task sentence that names non-negative edge weights. Section 2.5 shows that this small difference matters. A *placebo* is the bare prompt followed by a prohibition of an irrelevant algorithm (`Do not use the Rabin-Karp algorithm.` on path search and the weighted task, `Do not use Kruskal's algorithm.` on sorting). Every ban is compared with the placebo, never with the plain baseline, because the placebo sentence alone shifts the idea distribution. A ban sentence has the form `Do not use X.`, `Do not use X or Y.` or `Do not use X, Y or Z.`.

2.2 Corrections made in place to iteration 1

The review's points about the record were fixed where they point, each marked as a correction in Sections 1.2, 1.4, 1.9, 1.10, 1.12, 1.16 and 1.17. In summary: the two "not delivered" arms are restated with their actual output; every influence-arm conclusion is marked as conditional on the contract prompt; the "bans are ignored" claim is restricted to the contract prompt with list wording; the random-parity claim is replaced by the draw-averaged comparison of Section 2.9; the novelty-direction Spearman interval is replaced by an exact permutation p-value; and the file counts are corrected.

2.3 One label space for all bare-prompt programs

Iteration 1 used two classifiers that were never compared: the influence arm's AST classifier and the ban arm's rule classifier. The first artifact adopted the ban arm's rule classifier (hash `d0d321215bc2`, which passes its fixture tests) as the single classifier for all bare-prompt programs, because it covers both graph and sorting ideas. It reproduced all 4,210 stored ban-arm labels exactly. Every program then ran through a sandboxed execution oracle that discovers the entry point and calling convention, since bare-prompt programs have no fixed signature. Graph tasks are checked against a reference BFS and Dijkstra on random graphs; sorting tasks on random arrays. Pass rates on the bare prompt are therefore not comparable with the contract-prompt pass rates of iteration 1. A share of failures are programs the oracle could not call (on the path-search baseline, 41 not callable and 38 load errors among 300).

Table 27. The unified label ledger (13,968 programs, all labelled by the rule classifier and executed).

source	programs	origin

temperature grid, grid wording	3,623	iteration-1 temperature arm (3 tasks x 6 T x 2 top-k x 100, plus greedy and repeat rows)
seed rows, grid wording	360	iteration-1 temperature arm (60 seed strings x greedy and T = 0.7, per task)
ban arm, ban-arm wording	4,210	iteration-1 ban arm (co-mention lists are not programs and are excluded)
temperature re-draw, ban-arm wording	3,615	new in iteration 2, seeds 1000 and up, 1,536 tokens
fresh seed strings, ban-arm wording	2,160	new in iteration 2, 120 strings x 3 tasks x (1 greedy + 5 samples at T = 0.7)
total	**13,968**	

The classifier was audited blind on a stratified sample of 200 programs by two LLM judges (gpt-4.1-mini and gemini-2.5-flash-lite) at a cost of $0.042. Agreement between the rule classifier and the first judge is high on sorting and the weighted task and moderate on the open task (Table 28). The judges agree with each other at kappa 0.92, which bounds what any classifier can reach. The influence arm's AST classifier agrees with the judge far less (kappa 0.53), and with the rule classifier at kappa 0.44 to 0.76 depending on the source, which confirms that labels from the two iteration-1 arms are not directly comparable. The rule classifier's main confusions with the judge were: other labelled DFS by the judge (10); greedy best-first labelled other by the judge (3); SPFA labelled BFS by the judge (2); IDDFS labelled other by the judge (2); A* labelled greedy best-first by the judge (2); Timsort/hybrid labelled other by the judge (1); shell sort labelled other by the judge (1); SPFA labelled other by the judge (1); other labelled Dijkstra by the judge (1); other labelled BFS by the judge (1); greedy best-first labelled A* by the judge (1); Dijkstra labelled A* by the judge (1); DFS labelled other by the judge (1); bucket sort labelled counting sort by the judge (1); BFS labelled other by the judge (1).

Table 28. Classifier audit: Cohen's kappa [10] of the rule classifier against blind judges, and of the influence-arm AST classifier against the rule classifier on every bare-prompt source.

comparison	**n**	**agreement**	**Cohen's kappa [95% CI]**
rule classifier vs judge 1 (gpt-4.1-mini), all tasks	200	0.855	0.842 [0.78, 0.89]
rule classifier vs judge 1, open path search	82	0.732	0.671 [0.55, 0.78]
rule classifier vs judge 1, sorting	80	0.963	0.958 [0.90, 1.00]
rule classifier vs judge 1, weighted shortest path	38	0.895	0.849 [0.71, 0.96]
rule classifier vs judge 2 (gemini-2.5-flash-lite)	200	0.840	0.827 [0.77, 0.88]
judge 1 vs judge 2 (ceiling)	200	0.925	0.919 [0.87, 0.96]
influence-arm AST classifier vs judge 1 (graph tasks)	120	0.575	0.527 [0.43, 0.62]
rule classifier vs judge 1, shuffled labels (placebo)	200		-0.018
influence-arm AST classifier vs rule classifier, open path search, ban arm (bare prompt)	1400	0.731	0.658 [0.63, 0.69]
influence-arm AST classifier vs rule classifier, open path search, temperature grid	1208	0.745	0.603 [0.57, 0.64]

influence-arm AST classifier vs rule classifier, open path search, temperature re-draw	1405	0.688	0.525 [0.49, 0.56]
influence-arm AST classifier vs rule classifier, open path search, iteration-1 seed rows	120	0.833	0.578 [0.44, 0.70]
influence-arm AST classifier vs rule classifier, open path search, fresh seed strings	720	0.828	0.622 [0.57, 0.68]
influence-arm AST classifier vs rule classifier, weighted shortest path, ban arm (bare prompt)	1400	0.869	0.757 [0.73, 0.79]
influence-arm AST classifier vs rule classifier, weighted shortest path, temperature grid	1207	0.824	0.437 [0.39, 0.48]
influence-arm AST classifier vs rule classifier, weighted shortest path, temperature re-draw	805	0.841	0.530 [0.48, 0.58]
influence-arm AST classifier vs rule classifier, weighted shortest path, iteration-1 seed rows	120	1.000	1.000 [1.00, 1.00]
influence-arm AST classifier vs rule classifier, weighted shortest path, fresh seed strings	720	0.965	0.104 [-0.00, 0.25]

2.4 The iteration-1 ban arm, written up

The ban arm generated, for each task, a baseline (n = 300), a placebo and ten ban conditions (n = 100 each) at T = 0.7 under vLLM [22], plus forced-idea cells (`... using X.`, n = 10 per idea) and 50 co-mention lists (`List 10 algorithms that could solve this task`). Ranks refer to the baseline: on path search ranks 1 to 4 are BFS, DFS, A* and Dijkstra, and the rare idea is SPFA. All counts below are re-derived from the stored generations with the rule classifier and the iteration-2 oracle.

The contract made the menu. Table 29 sets the two iteration-1 baselines side by side. On the open task the bare prompt gives four ideas above 3% and 1.76 bits of idea entropy, against two ideas and 1.15 bits under the contract. On the weighted task the bare prompt gives Dijkstra in 300 of 300 programs, while the contract gives a queue-based relaxation that the classifier labels BFS in 71 of 300. Part of this gap could be labelling, since the arms used different classifiers. The comparison nonetheless shows that the two-idea menus of Section 1.4, the rank-1 erasure and the explanation that flattening "overshoots the uniform point" all depend on the contract.

Table 29. Baseline idea distributions at T = 0.7 under the two iteration-1 prompts, with the iteration-2 HF re-run of the bare prompt as an engine check.

task	prompt (engine, classifier)	n	idea counts	idea entropy (bits)	pass rate
open path search	contract (HF, AST classifier)	300	BFS 205, DFS 80, other 12, Dijkstra 2, A* 1	1.145	0.700 (contract oracle)
open path search	bare, ban-arm wording (vLLM, rule classifier)	300	BFS 179, DFS 44, A* 39, Dijkstra 22, other 14, SPFA 2	1.764	0.27

open path search	bare, ban-arm wording (HF, rule classifier)	300	BFS 184, DFS 42, Dijkstra 36, A* 24, other 11, SPFA 3	1.730	0.31
weighted shortest path	contract (HF, AST classifier)	300	Dijkstra 225, BFS 71, SPFA 2, other 1, bidirectional 1	0.906	0.373 (contract oracle)
weighted shortest path	bare, ban-arm wording (vLLM, rule classifier)	300	Dijkstra 300	0.000	0.84
sorting	bare, ban-arm wording (vLLM, rule classifier)	300	bubble sort 179, quicksort 70, merge sort 31, selection sort 15, insertion sort 5	1.587	0.99
sorting	bare, ban-arm wording (HF, rule classifier)	299	bubble sort 186, quicksort 70, merge sort 27, selection sort 11, insertion sort 5	1.504	0.98

The HF and vLLM baselines agree within sampling noise (JSD 0.009 on path search and 0.001 on sorting, chi-square p 0.18 and 0.91). The answer to the request's question "does the model always design BFS?" is therefore prompt-dependent. Under the bare prompt it writes BFS in 60% of programs, DFS in 15%, A* in 13% and Dijkstra in 7%. Under the contract it writes BFS in 68% and DFS in 27%.

Compliance is high on the bare prompt, and substitution is structured. Tables 30 to 32 hold every ban-arm condition. Banning BFS on path search leaves 1 of 100 programs using BFS and moves the mass to DFS (66 against 17 under the placebo). Banning DFS gives BFS 98 of 100. Banning Dijkstra raises A* from 1 to 34. Banning the top three (BFS, DFS, A*) or the top five ideas produces iterative deepening DFS (IDDFS) in 42 of 100 programs, an idea absent from the baseline that combines depth-first search with level-by-level deepening. Non-compliance rises with list length: on path search, 1, 29 and 38 of 100 programs still use a banned idea under the top-1, top-3 and top-5 bans; on the weighted task, 12, 46 and 61. This rise is what the priming account of negative instructions predicts [5]. On sorting compliance is near total (0, 0 and 7 of 100), and the top-5 ban yields counting sort 45 and heap sort 26, ideas outside the comparison-sort family that was banned. On the weighted task, banning Dijkstra yields Bellman-Ford in 55 programs, 44 of them correct; banning A*, Bellman-Ford or Floyd-Warshall, none of which the model writes, leaves Dijkstra at 95 to 100.

Two features of these tables recur through the rest of the section. First, the placebo sentence alone moves the distribution: on path search, A* falls from 39 of 300 at baseline to 1 of 100 under the placebo, and on sorting bubble sort falls from 60% to 43%. Second, bans on path search and the weighted task cost correctness. The path-search top-5 ban raises entropy to 2.54 bits, but the pass rate falls to 0.05 and 61% of outputs hit the token cap.

Table 30. Ban arm, open path search (bare prompt, T = 0.7, vLLM; counts from the rule classifier; the iteration-2 oracle).

condition	banned ideas	n	idea counts	idea entropy (bits) [95% CI]	still banned	pass rate	truncated	off-menu	correct (by idea)

baseline (n = 300)	-	300	BFS 179, DFS 44, A* 39, Dijkstra 22, other 14, SPFA 2	1.764 [1.59, 1.89]	0	0.27	0.45	1 (SPFA 1)
placebo	-	100	BFS 67, DFS 17, Dijkstra 11, other 4, A* 1	1.424 [1.11, 1.64]	0	0.45	0.16	0 (-)
ban rank-1	BFS	100	DFS 66, other 15, A* 11, Dijkstra 7, BFS 1	1.491 [1.20, 1.71]	1	0.27	0.19	0 (-)
ban rank-2	DFS	100	BFS 98, A* 2	0.141 [-0.00, 0.29]	0	0.54	0.07	0 (-)
ban rank-3	A*	100	BFS 69, DFS 17, Dijkstra 8, SPFA 4, other 2	1.394 [1.07, 1.62]	0	0.40	0.23	0 (-)
ban rank-4	Dijkstra	100	BFS 44, A* 34, DFS 16, other 3, SPFA 2, Dijkstra 1	1.804 [1.56, 1.97]	1	0.24	0.44	0 (-)
ban top-3	BFS, DFS, A*	100	IDDFS 42, DFS 18, Dijkstra 16, A* 8, other 6, bidirectional 4, BFS 3, greedy best-first 2, Bellman-Ford 1	2.446 [2.11, 2.63]	29	0.09	0.57	1 (IDDFS 1)
ban top-5	BFS, DFS, A*, Dijkstra, SPFA	100	IDDFS 42, A* 16, DFS 12, greedy best-first 8, other 8, Dijkstra 7, BFS 3, bidirectional 3, Bellman-Ford 1	2.537 [2.21, 2.72]	38	0.05	0.61	1 (IDDFS 1)
ban rare idea	SPFA	100	Dijkstra 50, BFS 42, A* 6, Bellman-Ford 1, DFS 1	1.402 [1.16, 1.57]	0	0.64	0.45	1 (Bellman-Ford 1)
paraphrase 1 of rank-1 ban	BFS	100	DFS 68, other 17, Dijkstra 9, A* 6	1.369 [1.11, 1.58]	0	0.28	0.21	0 (-)
paraphrase 2 of rank-1 ban	BFS	100	DFS 59, A* 16, other 15, Dijkstra 6, IDDFS 3, greedy best-first 1	1.744 [1.44, 1.96]	0	0.27	0.28	0 (-)

Table 31. Ban arm, sorting.

condition	banned ideas	n	idea counts	idea entropy (bits) [95% CI]	still banned	pass rate	truncated	off-menu correct (by idea)

baseline (n = 300)	-	300	bubble sort 179, quicksort 70, merge sort 31, selection sort 15, insertion sort 5	1.587 [1.43, 1.71]	0	0.99	0.07	5 (insertion sort 5)
placebo	-	100	bubble sort 43, merge sort 26, quicksort 18, selection sort 8, insertion sort 5	1.982 [1.77, 2.11]	0	0.96	0.12	5 (insertion sort 5)
ban rank-1	bubble sort	100	quicksort 71, merge sort 27, heap sort 1, insertion sort 1	0.994 [0.76, 1.18]	0	0.96	0.02	2 (heap sort 1, insertion sort 1)
ban rank-2	quicksort	100	insertion sort 41, merge sort 30, bubble sort 15, selection sort 13, heap sort 1	1.908 [1.71, 2.02]	0	1.00	0.00	42 (insertion sort 41, heap sort 1)
ban rank-3	merge sort	100	bubble sort 45, quicksort 23, insertion sort 14, selection sort 14, heap sort 4	1.986 [1.75, 2.11]	0	1.00	0.02	18 (insertion sort 14, heap sort 4)
ban rank-4	selection sort	100	bubble sort 41, merge sort 39, quicksort 10, insertion sort 9, heap sort 1	1.768 [1.54, 1.91]	0	1.00	0.03	10 (insertion sort 9, heap sort 1)
ban top-3	bubble sort, quicksort, merge sort	100	selection sort 55, insertion sort 42, counting sort 2, heap sort 1	1.179 [0.99, 1.33]	0	0.99	0.01	44 (insertion sort 42, counting sort 1, heap sort 1)
ban top-5	bubble sort, quicksort, merge sort, selection sort, insertion sort	100	counting sort 45, heap sort 26, shell sort 7, other 6, bubble sort 6, bucket sort 5, radix sort 4, merge sort 1	2.248 [1.91, 2.45]	7	0.69	0.03	59 (heap sort 26, counting sort 24, shell sort 7, bucket sort 2)
ban rare idea	insertion sort	100	quicksort 66, merge sort 27, heap sort 4, selection sort 2, bubble sort 1	1.271 [0.98, 1.48]	0	0.98	0.01	4 (heap sort 4)

paraphrase 1 of rank-1 ban	bubble sort	100	quicksort 49, merge sort 46, insertion sort 2, selection sort 2, Timsort/hybrid 1	1.312 [1.07, 1.49]	0	1.00	0.02	3 (insertion sort 2, Timsort/ hybrid 1)
paraphrase 2 of rank-1 ban	bubble sort	100	merge sort 50, quicksort 47, heap sort 1, selection sort 1, insertion sort 1	1.211 [1.00, 1.38]	0	1.00	0.00	2 (heap sort 1, insertion sort 1)

Table 32. Ban arm, weighted shortest path.

condition	banned ideas	n	idea counts	idea entropy (bits) [95% CI]	still banned	pass rate	truncated	off-menu correct (by idea)
baseline (n = 300)	-	300	Dijkstra 300	-0.000 [-0.00, 0.00]	0	0.84	0.86	0 (-)
placebo	-	100	Dijkstra 100	-0.000 [-0.00, 0.00]	0	0.88	0.74	0 (-)
ban rank-1	Dijkstra	100	Bellman-Ford 55, BFS 19, Dijkstra 12, A* 8, bidirectional 2, greedy best-first 1, Floyd-Warshall 1, other 1, SPFA 1	1.967 [1.61, 2.19]	12	0.50	0.57	44 (Bellman-Ford 44)
ban rank-2	A*	100	Dijkstra 95, BFS 4, SPFA 1	0.322 [0.08, 0.53]	0	0.82	0.70	0 (-)
ban rank-3	Bellman-Ford	100	Dijkstra 100	-0.000 [-0.00, 0.00]	0	0.85	0.73	0 (-)
ban rank-4	Floyd-Warshall	100	Dijkstra 100	-0.000 [-0.00, 0.00]	0	0.85	0.69	0 (-)
ban top-3	Dijkstra, A*, Bellman-Ford	100	BFS 32, Dijkstra 25, Bellman-Ford 21, SPFA 7, Floyd-Warshall 5, other 3, DFS 3, greedy best-first 1, IDDFS 1, unparseable 1, bidirectional 1	2.553 [2.21, 2.73]	46	0.19	0.67	6 (Bellman-Ford 3, BFS 2, SPFA 1)

ban top-5	Dijkstra, A*, Bellman-Ford, Floyd-Warshall, bidirectional	100	Dijkstra 41, BFS 26, Bellman-Ford 16, SPFA 10, bidirectional 2, A* 1, greedy best-first 1, other 1, Floyd-Warshall 1, DFS 1	2.233 [1.89, 2.41]	61	0.21	0.64	10 (SPFA 2, BFS 3, Bellman-Ford 5)
ban rare idea	greedy best-first	100	Dijkstra 98, DFS 1, BFS 1	0.161 [-0.00, 0.33]	0	0.79	0.68	0 (-)
paraphrase 1 of rank-1 ban	Dijkstra	100	Bellman-Ford 54, A* 16, BFS 15, Dijkstra 9, other 2, Floyd-Warshall 1, bidirectional 1, DFS 1, SPFA 1	2.005 [1.65, 2.22]	9	0.50	0.53	44 (Bellman-Ford 43, SPFA 1)
paraphrase 2 of rank-1 ban	Dijkstra	100	Bellman-Ford 52, BFS 13, A* 11, Dijkstra 10, SPFA 8, Floyd-Warshall 4, DFS 1, bidirectional 1	2.166 [1.84, 2.37]	10	0.42	0.66	38 (Bellman-Ford 36, BFS 1, SPFA 1)

Regularity violations already appear on the screen. A ban that only removes options can raise but never lower the share of an unbanned idea. Table 33 lists the cells where an unbanned idea lost share against the placebo at Holm-corrected $p < 0.05$ (Fisher exact tests over 28, 34 and 26 cells). The largest are sorting: banning insertion sort, which the model writes in 5% of baseline programs, takes bubble sort from 43 to 1, and banning quicksort takes it from 43 to 15. On path search, banning the rare SPFA takes DFS from 17 to 1. The replicate files the plan relied on for a noise floor do not exist, so the floor comes from splitting the baseline into thirds (open path search largest per-class difference between baseline thirds 0.09; sorting largest per-class difference between baseline thirds 0.03; weighted shortest path largest per-class difference between baseline thirds 0.00). Every listed drop exceeds that floor.

Table 33. Screen regularity violations: unbanned ideas that lost share under a ban, against the placebo ($n = 100$ each).

task	ban	idea that lost share	placebo count (n = 100)	ban count (n = 100)	Fisher p	Holm p
open path search	ban rank-2 (DFS)	Dijkstra	11	0	0.00073	0.015
open path search	ban rank-4 (Dijkstra)	BFS	67	44	0.0017	0.032
open path search	ban rare idea (SPFA)	BFS	67	42	0.00062	0.013
open path search	ban rare idea (SPFA)	DFS	17	1	7.5e-05	0.0016

sorting	ban rank-2 (quicksort)	bubble sort	43	15	2e-05	0.00052
sorting	ban rare idea (insertion sort)	bubble sort	43	1	1.9e-14	6.4e-13

Wording within the bare prompt matters little for one ban; the grid wording matters a lot for sorting. Two paraphrases of the rank-1 ban (`Avoid using X.` and `Your solution must not use X or close variants of it.`) give distributions within JSD 0.012 to 0.054 of the plain ban (Table 34). They differ significantly only on sorting, where the split between quicksort and merge sort moves. The grid wording, by contrast, moves the sorting baseline from bubble sort to quicksort as the mode (JSD 0.107, permutation $p < 0.001$). The temperature grid was therefore run on a different sorting distribution from everything else in this section.

Table 34. Wording checks: paraphrased bans against the rank-1 ban, and the grid wording against the ban-arm wording at $T = 0.7$.

task	comparison	Jensen-Shannon divergence [95% CI]	permutation p	chi-square p	classes compared
open path search	paraphrase 1 of rank-1 ban vs rank-1 ban	0.012 [0.004, 0.058]	0.618	0.579	A*, DFS, Dijkstra, other
open path search	paraphrase 2 of rank-1 ban vs rank-1 ban	0.030 [0.013, 0.077]	0.287	0.526	A*, DFS, Dijkstra, other
sorting	paraphrase 1 of rank-1 ban vs rank-1 ban	0.054 [0.020, 0.117]	0.004	0.006	merge sort, quicksort
sorting	paraphrase 2 of rank-1 ban vs rank-1 ban	0.048 [0.021, 0.121]	0.005	0.003	merge sort, quicksort
weighted shortest path	paraphrase 1 of rank-1 ban vs rank-1 ban	0.026 [0.018, 0.103]	0.865	0.467	A*, BFS, Bellman-Ford, Dijkstra
weighted shortest path	paraphrase 2 of rank-1 ban vs rank-1 ban	0.052 [0.034, 0.133]	0.143	0.281	A*, BFS, Bellman-Ford, Dijkstra
open path search	grid wording (n = 100, T = 0.7, no top-k: BFS 72, DFS 10, A* 7, Dijkstra 7, other 4) vs ban-arm wording (n = 300)	0.018 [0.007, 0.059]	0.2205	0.23	A*, BFS, DFS, Dijkstra
sorting	grid wording (n = 100, T = 0.7, no top-k: quicksort 55, bubble sort 34, merge sort 11) vs ban-arm wording (n = 300)	0.107 [0.064, 0.168]	0.0001	1e-08	bubble sort, merge sort, quicksort
weighted shortest path	grid wording (n = 100, T = 0.7, no top-k: Dijkstra 100) vs ban-arm wording (n = 300)	0.000 [0.000, 0.000]	1.0000	n/a	Dijkstra only

The forced-idea cells check that the model can write each idea when asked, and that the classifier

recognises it (Table 35). Purity is 0.8 or higher for every sorting idea and for most graph ideas. The exceptions are greedy best-first search on path search (0.5, the rest A*) and the weighted task (0.2), and BFS (0.5) and DFS (0.6) on the weighted task, where a request for BFS or DFS on a weighted graph often yields SPFA or Dijkstra. With n = 10 per cell these purities are rough.

Table 35. Forced-idea cells (`<task> using X.`, n = 10 each, T = 0.7).

task	forced idea	n	purity (share implementing the requested idea)	pass rate	idea counts
open path search	A*	10	0.9	0.0	A* 9, other 1
open path search	BFS	10	1.0	0.4	BFS 10
open path search	Bellman-Ford	10	1.0	0.4	Bellman-Ford 10
open path search	bidirectional	10	0.7	0.0	bidirectional 7, A* 2, BFS 1
open path search	DFS	10	0.9	0.1	DFS 9, other 1
open path search	Dijkstra	10	1.0	0.8	Dijkstra 10
open path search	Floyd-Warshall	10	1.0	0.0	Floyd-Warshall 10
open path search	greedy best-first	10	0.5	0.0	greedy best-first 5, A* 4, BFS 1
open path search	IDDFS	10	0.9	0.1	IDDFS 9, BFS 1
open path search	SPFA	10	0.8	0.4	SPFA 8, Dijkstra 2
sorting	bubble sort	10	1.0	1.0	bubble sort 10
sorting	bucket sort	10	1.0	0.6	bucket sort 10
sorting	counting sort	10	1.0	0.8	counting sort 10
sorting	heap sort	10	1.0	1.0	heap sort 10
sorting	insertion sort	10	1.0	1.0	insertion sort 10
sorting	merge sort	10	1.0	1.0	merge sort 10
sorting	quicksort	10	1.0	0.9	quicksort 10
sorting	radix sort	10	1.0	0.0	radix sort 10
sorting	selection sort	10	1.0	1.0	selection sort 10
sorting	shell sort	10	0.9	0.9	shell sort 9, other 1
sorting	Timsort/ hybrid	10	0.8	0.3	Timsort/hybrid 8, quicksort 1, merge sort 1

weighted shortest path	A*	10	1.0	0.0	A* 10
weighted shortest path	BFS	10	0.5	0.0	BFS 5, SPFA 3, Dijkstra 2
weighted shortest path	Bellman-Ford	10	1.0	0.9	Bellman-Ford 10
weighted shortest path	bidirectional	10	0.8	0.0	bidirectional 8, BFS 2
weighted shortest path	DFS	10	0.6	0.1	DFS 6, other 1, BFS 1, Bellman-Ford 1, Dijkstra 1
weighted shortest path	Dijkstra	10	1.0	0.9	Dijkstra 10
weighted shortest path	Floyd-Warshall	10	1.0	0.0	Floyd-Warshall 10
weighted shortest path	greedy best-first	10	0.2	0.0	Dijkstra 6, A* 2, greedy best-first 2
weighted shortest path	IDDFS	10	0.8	0.0	IDDFS 8, BFS 2
weighted shortest path	SPFA	10	0.9	0.1	SPFA 9, BFS 1

The co-mention lists show what the model knows about each task, as opposed to what it writes. On path search all 50 lists name BFS, DFS, Dijkstra and A*, 48 name bidirectional and greedy best-first search, and 29 name IDDFS (Table 36). On sorting every list names seven sorting algorithms. The menu the model writes from is far narrower than the menu it can list.

Table 36. Co-mention lists: how many of 50 lists (`List 10 algorithms that could solve this task`) name each idea.

task	lists	number of lists naming each idea
open path search	50	BFS 50, DFS 50, Dijkstra 50, A* 50, bidirectional 48, greedy best-first 48, other 44, IDDFS 29, SPFA 1
sorting	50	quicksort 50, insertion sort 50, bubble sort 50, heap sort 50, merge sort 50, counting sort 50, selection sort 50, radix sort 49, bucket sort 41, Timsort/hybrid 32, shell sort 14, other 1
weighted shortest path	50	other 50, Bellman-Ford 50, Floyd-Warshall 50, Dijkstra 50, A* 50, bidirectional 38, greedy best-first 22, BFS 9, DFS 3, SPFA 2

What the ban arm settles is behavioural. In this 3B model, at $T = 0.7$ and on the bare prompt, a short prohibition is obeyed in most programs. Banned mass goes to particular neighbours: DFS for BFS, A* for Dijkstra, IDDFS for the BFS/DFS/A* triple, Bellman-Ford for Dijkstra on weighted graphs, and non-comparison sorts when the comparison sorts are banned. Bans can also lower unbanned ideas. The geometry behind these moves is the subject of Section 2.8.

2.5 The temperature arm, labelled, and a frozen re-draw

The iteration-1 grid has 100 programs per cell over T in {0.3, 0.5, 0.7, 1.0, 1.3, 1.6} and top-k in {10, none}, with top-p = 1, on the grid wording, at a 768-token cap and one fixed base seed per sample index. The first artifact labelled and executed it, fitted the temperature law on it, and froze the fits (sha256 `d1dabe95…`). It then generated a confirmation re-draw on the ban-arm wording: the same grid on path search and sorting, no top-k only on the weighted task, seeds 1000 to 1099, and a 1,536-token cap to cut truncation. Tables 37 and 38 give every cell.

Idea diversity is measured with Hill numbers [11]: richness (the order-0 Hill number, the count of distinct ideas), the effective number of species (the order-1 Hill number, the exponential of the Shannon entropy) and the inverse Simpson index (the order-2 Hill number). We call the rarefied number of distinct ideas in a random draw of 10 programs the *expected ideas among 10*; it reads off the request's question about "10 reasoning trajectories" directly. We define *within-idea diversity* (alpha_1) as the order-1 similarity-sensitive diversity of programs inside one idea, computed on normalised code with edit-distance similarity in rarefied draws of 8 programs per idea and averaged over ideas. We define *between-idea diversity* (beta_1) as the ratio of total to within-idea diversity.

Table 37. Temperature grid, grid wording (n = 100 per cell, 768-token cap). Unparseable outputs count as a class in the entropy here, which is why the T = 1.6, no top-k cells show nonzero entropy with almost no code.

task	T	top-k	idea counts	idea entropy (bits)	Hill numbers: richness / order 1 / order 2	expected ideas among 10	pass rate	truncated	within-idea diversity (alpha_1)
open path search	0.3	none	BFS 82, A* 9, DFS 5, Dijkstra 2, other 1, SPFA 1	1.009	6 / 2.01 / 1.46	2.44	0.39	0.31	1.68
open path search	0.3	10	BFS 82, A* 9, DFS 5, Dijkstra 2, other 1, SPFA 1	1.009	6 / 2.01 / 1.46	2.44	0.38	0.31	1.67
open path search	0.5	none	BFS 73, DFS 12, Dijkstra 7, A* 5, SPFA 2, other 1	1.362	6 / 2.57 / 1.80	2.98	0.46	0.31	2.01
open path search	0.5	10	BFS 73, DFS 12, Dijkstra 7, A* 5, SPFA 2, other 1	1.362	6 / 2.57 / 1.80	2.98	0.46	0.33	1.97
open path search	0.7	none	BFS 72, DFS 10, A* 7, Dijkstra 7, other 4	1.396	5 / 2.63 / 1.85	3.08	0.27	0.40	2.23
open path search	0.7	10	BFS 67, DFS 13, A* 8, Dijkstra 8, other 4	1.539	5 / 2.90 / 2.08	3.28	0.33	0.33	2.37

open path search	1.0	none	BFS 54, DFS 15, Dijkstra 12, A* 10, other 8, SPFA 1	1.948	6 / 3.86 / 2.90	3.91	0.26	0.47	2.81
open path search	1.0	10	BFS 59, A* 13, DFS 12, Dijkstra 11, other 4, SPFA 1	1.801	6 / 3.49 / 2.54	3.66	0.29	0.48	2.70
open path search	1.3	none	BFS 44, other 35, DFS 12, Dijkstra 7, unparseable 1, A* 1	1.820	5 / 3.38 / 2.92	3.46	0.06	0.86	3.21
open path search	1.3	10	BFS 59, DFS 19, Dijkstra 11, A* 5, other 5, SPFA 1	1.753	6 / 3.37 / 2.49	3.53	0.14	0.63	3.14
open path search	1.6	none	unparseable 65, other 31, DFS 2, BFS 1, Dijkstra 1	1.174	4 / 1.61 / 1.27	2.37	0.00	0.91	n/a
open path search	1.6	10	BFS 46, DFS 19, other 16, Dijkstra 11, greedy best-first 4, A* 2, SPFA 2	2.155	7 / 4.45 / 3.47	4.17	0.03	0.75	3.49
sorting	0.3	none	quicksort 80, bubble sort 18, merge sort 1, selection sort 1	0.836	4 / 1.78 / 1.49	2.08	0.99	0.03	1.03
sorting	0.3	10	quicksort 79, bubble sort 17, merge sort 2, selection sort 2	0.929	4 / 1.90 / 1.53	2.24	0.99	0.06	1.02
sorting	0.5	none	quicksort 68, bubble sort 23, merge sort 9	1.179	3 / 2.26 / 1.91	2.57	1.00	0.00	1.20
sorting	0.5	10	quicksort 66, bubble sort 24, merge sort 9, selection sort 1	1.269	4 / 2.41 / 1.99	2.67	0.99	0.04	1.13
sorting	0.7	none	quicksort 55, bubble sort 34, merge sort 11	1.354	3 / 2.56 / 2.32	2.69	0.96	0.05	1.31
sorting	0.7	10	quicksort 55, bubble sort 30, merge sort 14, other 1	1.459	4 / 2.75 / 2.43	2.87	0.96	0.05	1.31
sorting	1.0	none	quicksort 45, bubble sort 32, merge sort 15, selection sort 5, insertion sort 3	1.823	5 / 3.54 / 3.02	3.49	0.97	0.07	1.38

sorting	1.0	10	quicksort 51, bubble sort 33, merge sort 11, insertion sort 3, selection sort 2	1.638	5 / 3.11 / 2.62	3.16	0.99	0.06	1.37
sorting	1.3	none	quicksort 37, bubble sort 35, merge sort 18, selection sort 6, insertion sort 2, built-in sort 1, other 1	1.995	7 / 3.99 / 3.38	3.73	0.87	0.28	1.61
sorting	1.3	10	bubble sort 40, quicksort 37, merge sort 16, selection sort 4, insertion sort 3	1.820	5 / 3.53 / 3.08	3.45	0.96	0.10	1.54
sorting	1.6	none	unparseable 35, quicksort 20, bubble sort 18, selection sort 8, merge sort 6, insertion sort 6, other 5, built-in sort 1, shell sort 1	2.567	8 / 5.71 / 4.76	4.93	0.24	0.91	2.11
sorting	1.6	10	bubble sort 37, quicksort 34, merge sort 18, selection sort 9, insertion sort 2	1.931	5 / 3.81 / 3.41	3.68	0.87	0.12	1.67
weighted shortest path	0.3	none	Dijkstra 100	-0.000	1 / 1.00 / 1.00	1.00	0.96	0.93	1.44
weighted shortest path	0.3	10	Dijkstra 100	-0.000	1 / 1.00 / 1.00	1.00	0.97	0.92	1.45
weighted shortest path	0.5	none	Dijkstra 100	-0.000	1 / 1.00 / 1.00	1.00	0.92	0.85	1.62
weighted shortest path	0.5	10	Dijkstra 100	-0.000	1 / 1.00 / 1.00	1.00	0.92	0.83	1.60
weighted shortest path	0.7	none	Dijkstra 100	-0.000	1 / 1.00 / 1.00	1.00	0.87	0.87	1.72
weighted shortest path	0.7	10	Dijkstra 100	-0.000	1 / 1.00 / 1.00	1.00	0.86	0.87	1.70

weighted shortest path	1.0	none	Dijkstra 99, other 1	0.081	2 / 1.06 / 1.02	1.10	0.68	0.78	2.21
weighted shortest path	1.0	10	Dijkstra 100	-0.000	1 / 1.00 / 1.00	1.00	0.76	0.84	2.11
weighted shortest path	1.3	none	Dijkstra 67, other 30, unparseable 2, DFS 1	1.088	3 / 1.95 / 1.78	2.27	0.17	0.92	3.20
weighted shortest path	1.3	10	Dijkstra 97, other 2, SPFA 1	0.222	3 / 1.17 / 1.06	1.29	0.41	0.86	2.91
weighted shortest path	1.6	none	unparseable 55, other 41, Dijkstra 4	1.188	2 / 1.35 / 1.19	2.34	0.00	0.95	n/a
weighted shortest path	1.6	10	Dijkstra 83, other 9, BFS 4, SPFA 1, Bellman-Ford 1, DFS 1, Floyd-Warshall 1	0.987	7 / 1.98 / 1.43	2.38	0.11	0.92	3.32

Table 38. Temperature re-draw, ban-arm wording (n = 100 per cell, 1,536-token cap, seeds 1000..1099).

task	**T**	**top-k**	**idea counts**	**idea entropy (bits)**	**Hill numbers: richness / order 1 / order 2**	**expected ideas among 10**	**pass rate**	**truncated**	**within-idea diversity (alpha_1)**
open path search	0.3	none	BFS 64, Dijkstra 14, A* 9, DFS 8, SPFA 3, IDDFS 1, other 1	1.698	7 / 3.24 / 2.25	3.48	0.49	0.00	1.78
open path search	0.3	10	BFS 64, Dijkstra 14, A* 9, DFS 8, SPFA 3, IDDFS 1, other 1	1.698	7 / 3.24 / 2.25	3.48	0.50	0.00	1.80
open path search	0.5	none	BFS 66, A* 12, DFS 12, Dijkstra 6, other 4	1.559	5 / 2.95 / 2.13	3.30	0.32	0.02	2.09
open path search	0.5	10	BFS 65, A* 14, Dijkstra 10, DFS 9, SPFA 1, other 1	1.579	6 / 2.99 / 2.17	3.29	0.36	0.02	2.05
open path search	0.7	none	BFS 61, DFS 21, Dijkstra 9, A* 7, other 2	1.602	5 / 3.04 / 2.33	3.27	0.35	0.00	2.32

open path search	0.7	10	BFS 61, DFS 16, Dijkstra 11, A* 8, other 3, SPFA 1	1.718	6 / 3.29 / 2.40	3.50	0.35	0.00	2.29
open path search	1.0	none	BFS 55, Dijkstra 14, DFS 14, A* 8, other 7, bidirectional 1, SPFA 1	1.962	7 / 3.89 / 2.83	3.91	0.19	0.01	2.94
open path search	1.0	10	BFS 59, DFS 18, Dijkstra 11, A* 7, other 5	1.729	5 / 3.32 / 2.50	3.53	0.32	0.00	2.69
open path search	1.3	none	BFS 39, other 35, DFS 12, Dijkstra 11, greedy best-first 3	1.929	5 / 3.81 / 3.31	3.70	0.07	0.65	3.44
open path search	1.3	10	BFS 58, DFS 21, Dijkstra 11, A* 7, other 2, SPFA 1	1.727	6 / 3.31 / 2.51	3.45	0.28	0.01	2.97
open path search	1.6	none	unparseable 59, other 38, BFS 2, greedy best-first 1	1.159	3 / 1.36 / 1.16	2.28	0.00	0.95	n/a
open path search	1.6	10	BFS 47, DFS 26, Dijkstra 11, other 4, A* 4, SPFA 3, greedy best-first 3, IDDFS 1, bidirectional 1	2.175	9 / 4.52 / 3.27	4.11	0.06	0.04	3.50
sorting	0.3	none	bubble sort 88, quicksort 5, merge sort 4, selection sort 2, insertion sort 1	0.743	5 / 1.67 / 1.28	2.06	0.99	0.00	1.09
sorting	0.3	10	bubble sort 89, quicksort 4, merge sort 4, selection sort 2, insertion sort 1	0.700	5 / 1.63 / 1.26	1.99	0.99	0.00	1.08
sorting	0.5	none	bubble sort 68, quicksort 18, merge sort 10, selection sort 3, insertion sort 1	1.374	5 / 2.59 / 1.98	2.92	0.98	0.00	1.10
sorting	0.5	10	bubble sort 68, quicksort 18, merge sort 10, selection sort 3, insertion sort 1	1.374	5 / 2.59 / 1.98	2.92	0.98	0.00	1.10

sorting	0.7	none	bubble sort 62, quicksort 20, merge sort 13, selection sort 3, insertion sort 2	1.539	5 / 2.91 / 2.26	3.14	1.00	0.01	1.23
sorting	0.7	10	bubble sort 63, quicksort 20, merge sort 13, insertion sort 2, selection sort 2	1.493	5 / 2.81 / 2.20	3.06	0.99	0.01	1.24
sorting	1.0	none	bubble sort 47, quicksort 23, merge sort 18, selection sort 7, insertion sort 3, other 1, heap sort 1	1.998	7 / 3.99 / 3.20	3.82	0.99	0.00	1.32
sorting	1.0	10	bubble sort 46, quicksort 31, merge sort 12, selection sort 5, insertion sort 5, Timsort/hybrid 1	1.905	6 / 3.74 / 3.06	3.65	0.97	0.00	1.31
sorting	1.3	none	bubble sort 41, quicksort 28, merge sort 14, insertion sort 10, selection sort 4, other 3	2.108	6 / 4.31 / 3.59	4.05	0.86	0.30	1.66
sorting	1.3	10	bubble sort 42, quicksort 32, merge sort 18, selection sort 5, insertion sort 3	1.865	5 / 3.64 / 3.18	3.55	0.96	0.00	1.53
sorting	1.6	none	unparseable 51, bubble sort 16, quicksort 15, other 10, merge sort 4, shell sort 2, insertion sort 1, selection sort 1	2.093	7 / 4.69 / 3.98	4.07	0.11	0.93	2.41
sorting	1.6	10	bubble sort 35, quicksort 32, merge sort 17, selection sort 11, insertion sort 3, heap sort 2	2.106	6 / 4.30 / 3.74	4.00	0.82	0.01	1.96

weighted shortest path	0.3	none	Dijkstra 100	-0.000	1 / 1.00 / 1.00	1.00	0.91	0.00	1.43
weighted shortest path	0.5	none	Dijkstra 99, A* 1	0.081	2 / 1.06 / 1.02	1.10	0.93	0.00	1.54
weighted shortest path	0.7	none	Dijkstra 99, A* 1	0.081	2 / 1.06 / 1.02	1.10	0.88	0.01	1.63
weighted shortest path	1.0	none	Dijkstra 98, Bellman-Ford 1, DFS 1	0.161	3 / 1.12 / 1.04	1.20	0.64	0.00	2.37
weighted shortest path	1.3	none	Dijkstra 63, other 29, greedy best-first 2, BFS 2, DFS 2, unparseable 1, Floyd-Warshall 1	1.409	6 / 2.53 / 2.03	2.75	0.09	0.77	3.27
weighted shortest path	1.6	none	unparseable 58, other 37, Dijkstra 5	1.203	2 / 1.44 / 1.27	2.41	0.00	0.93	n/a

[Figure fig_temp_bare: generated in the report phase, not in this draft]

Temperature adds ideas slowly and breaks code fast. On path search (re-draw, no top-k), idea entropy rises from 1.70 bits at $T = 0.3$ to 1.96 at $T = 1.0$, while the pass rate falls from 0.49 to 0.19. At $T = 1.3$ a third of outputs are "other" and 65% hit the token cap, and at $T = 1.6$ almost nothing parses. Top-k $= 10$ keeps $T = 1.6$ parseable and adds a few rare ideas (9 classes on path search), but at a pass rate of 0.06 on path search (0.82 on sorting). Sorting is the one task where heat buys ideas at little cost: entropy rises from 0.74 bits at $T = 0.3$ to 2.00 at $T = 1.0$ at pass 0.99. Even there, the new mass goes to the same five comparison sorts. The weighted task stays at Dijkstra up to $T = 1.0$, and above that the extra entropy comes from broken output, not from new algorithms.

The temperature law fails as stated. Table 39 gives every fitted line of idea log-odds against $1/T$, and Table 40 the multinomial-logit fits with a curvature test. Table 41 applies the pre-registered criteria. No task-and-top-k combination meets all of them, on either draw. Three findings matter more than the binary verdict.

First, $R^2 \geq 0.85$ is a weak criterion at $n = 100$ per cell. A simulation from a truly linear law with the observed shares gives R^2 below 0.85 in 25% to 87% of draws (Table 42). The curvature likelihood-ratio test is the better check. It does not reject linearity on the grid for path search (p 0.97 and 0.97) and rejects it weakly for sorting without top-k (p 0.034). On the re-draw it rejects linearity only for path search without top-k (p 0.016).

Second, the path-search intercepts are significantly below zero, for example A* at -2.38 [-3.19, -1.56] on the grid. A pure temperature-scaling model, in which each idea has a fixed utility and T

only divides the utilities, predicts zero intercepts. Some ideas therefore lose ground at every temperature for reasons that 1/T does not capture. Lines for A* and, on the re-draw, Dijkstra are flat in 1/T: their share relative to BFS does not change with temperature.

Third, rank reversals among the top three occur at $T \leq 1.3$, but only among the minority ideas (A* against DFS on the path-search grid; DFS against Dijkstra on the re-draw; quicksort against bubble sort at $T = 1.3$ on the sorting grid, not significant). The mode changes only once below $T = 1.6$: on the sorting grid with top-k 10 at $T = 1.3$, bubble sort (40) edges past quicksort (37), within noise.

Table 39. Fitted lines of idea log-odds (against the modal idea) versus 1/T, weighted least squares, on the grid and the re-draw. "Rest (pooled)" pools all other classes including unparseable output and is not part of the criteria.

draw	**task, top-k**	**reference (mode)**	**idea**	**slope vs 1/T (SE)**	**intercept (SE)**	**Rˆ2**	**counts at T = 0.3, 0.5, 0.7, 1.0, 1.3, 1.6 (reference counts)**
grid	open path search, none	BFS	DFS	-0.60 (0.18)	-0.73 (0.31)	0.725	5, 12, 10, 15, 12, 2 (82, 73, 72, 54, 44, 1)
grid	open path search, none	BFS	Dijkstra	-0.77 (0.22)	-0.89 (0.34)	0.760	2, 7, 7, 12, 7, 1 (82, 73, 72, 54, 44, 1)
grid	open path search, none	BFS	A*	-0.09 (0.22)	-1.99 (0.47)	0.035	9, 5, 7, 10, 1, 0 (82, 73, 72, 54, 44, 1)
grid	open path search, none	BFS	rest (pooled) (not eligible)	-1.71 (0.81)	0.88 (1.08)	0.523	2, 3, 4, 9, 36, 96 (82, 73, 72, 54, 44, 1)
grid	open path search, 10	BFS	A*	-0.08 (0.22)	-1.95 (0.43)	0.033	9, 5, 8, 13, 5, 2 (82, 73, 67, 59, 59, 46)
grid	open path search, 10	BFS	DFS	-0.62 (0.10)	-0.65 (0.15)	0.908	5, 12, 13, 12, 19, 19 (82, 73, 67, 59, 59, 46)
grid	open path search, 10	BFS	Dijkstra	-0.71 (0.06)	-0.99 (0.09)	0.968	2, 7, 8, 11, 11, 11 (82, 73, 67, 59, 59, 46)
grid	open path search, 10	BFS	rest (pooled) (not eligible)	-1.08 (0.42)	-0.65 (0.55)	0.626	2, 3, 4, 5, 6, 22 (82, 73, 67, 59, 59, 46)
grid	sorting, none	quicksort	bubble sort	-0.55 (0.07)	0.26 (0.12)	0.944	18, 23, 34, 32, 35, 18 (80, 68, 55, 45, 37, 20)
grid	sorting, none	quicksort	merge sort	-1.04 (0.16)	-0.05 (0.22)	0.911	1, 9, 11, 15, 18, 6 (80, 68, 55, 45, 37, 20)
grid	sorting, none	quicksort	selection sort	-1.21 (0.43)	-0.66 (0.54)	0.661	1, 0, 0, 5, 6, 8 (80, 68, 55, 45, 37, 20)
grid	sorting, none	quicksort	insertion sort	-1.70 (0.58)	-0.68 (0.65)	0.683	0, 0, 0, 3, 2, 6 (80, 68, 55, 45, 37, 20)
grid	sorting, none	quicksort	rest (pooled) (not eligible)	-2.84 (1.33)	1.91 (1.24)	0.533	0, 0, 0, 0, 2, 42 (80, 68, 55, 45, 37, 20)

grid	sorting, 10	quicksort	bubble sort	-0.60 (0.09)	0.35 (0.14)	0.926	17, 24, 30, 33, 40, 37 (79, 66, 55, 51, 37, 34)
grid	sorting, 10	quicksort	merge sort	-0.95 (0.15)	-0.14 (0.21)	0.908	2, 9, 14, 11, 16, 18 (79, 66, 55, 51, 37, 34)
grid	sorting, 10	quicksort	insertion sort	-1.19 (0.34)	-1.67 (0.40)	0.756	0, 0, 0, 3, 3, 2 (79, 66, 55, 51, 37, 34)
grid	sorting, 10	quicksort	rest (pooled) (not eligible)	-0.81 (0.36)	-1.34 (0.54)	0.563	2, 1, 1, 2, 4, 9 (79, 66, 55, 51, 37, 34)
grid	weighted shortest path, none	Dijkstra	rest (pooled) (not eligible)	-3.27 (1.94)	2.17 (1.78)	0.416	0, 0, 0, 1, 33, 96 (100, 100, 100, 99, 67, 4)
grid	weighted shortest path, 10	Dijkstra	rest (pooled) (not eligible)	-1.87 (0.85)	-0.77 (0.78)	0.547	0, 0, 0, 0, 3, 17 (100, 100, 100, 100, 97, 83)
re-draw	open path search, none	BFS	Dijkstra	-0.08 (0.17)	-1.44 (0.34)	0.053	14, 6, 9, 14, 11, 0 (64, 66, 61, 55, 39, 2)
re-draw	open path search, none	BFS	DFS	-0.35 (0.12)	-0.83 (0.21)	0.696	8, 12, 21, 14, 12, 0 (64, 66, 61, 55, 39, 2)
re-draw	open path search, none	BFS	A*	0.07 (0.19)	-2.03 (0.41)	0.031	9, 12, 7, 8, 0, 0 (64, 66, 61, 55, 39, 2)
re-draw	open path search, none	BFS	rest (pooled) (not eligible)	-1.20 (0.74)	0.64 (1.14)	0.398	5, 4, 2, 9, 38, 98 (64, 66, 61, 55, 39, 2)
re-draw	open path search, 10	BFS	DFS	-0.53 (0.12)	-0.51 (0.19)	0.819	8, 9, 16, 18, 21, 26 (64, 65, 61, 59, 58, 47)
re-draw	open path search, 10	BFS	Dijkstra	0.01 (0.07)	-1.62 (0.12)	0.002	14, 10, 11, 11, 11, 11 (64, 65, 61, 59, 58, 47)
re-draw	open path search, 10	BFS	A*	0.14 (0.13)	-2.13 (0.24)	0.227	9, 14, 8, 7, 7, 4 (64, 65, 61, 59, 58, 47)
re-draw	open path search, 10	BFS	rest (pooled) (not eligible)	-0.33 (0.29)	-1.75 (0.50)	0.237	5, 2, 4, 5, 3, 12 (64, 65, 61, 59, 58, 47)
re-draw	sorting, none	bubble sort	quicksort	-0.90 (0.09)	0.30 (0.14)	0.959	5, 18, 20, 23, 28, 15 (88, 68, 62, 47, 41, 16)
re-draw	sorting, none	bubble sort	merge sort	-0.75 (0.11)	-0.41 (0.17)	0.920	4, 10, 13, 18, 14, 4 (88, 68, 62, 47, 41, 16)
re-draw	sorting, none	bubble sort	selection sort	-0.62 (0.18)	-1.61 (0.30)	0.743	2, 3, 3, 7, 4, 1 (88, 68, 62, 47, 41, 16)
re-draw	sorting, none	bubble sort	insertion sort	-1.10 (0.37)	-1.01 (0.52)	0.688	1, 1, 2, 3, 10, 1 (88, 68, 62, 47, 41, 16)
re-draw	sorting, none	bubble sort	rest (pooled) (not eligible)	-3.29 (1.65)	2.44 (1.55)	0.499	0, 0, 0, 2, 3, 63 (88, 68, 62, 47, 41, 16)

re-draw	sorting, 10	bubble sort	quicksort	-1.02 (0.10)	0.54 (0.14)	0.965	4, 18, 20, 31, 32, 32 (89, 68, 63, 46, 42, 35)
re-draw	sorting, 10	bubble sort	merge sort	-0.83 (0.08)	-0.28 (0.11)	0.968	4, 10, 13, 12, 18, 17 (89, 68, 63, 46, 42, 35)
re-draw	sorting, 10	bubble sort	selection sort	-0.90 (0.27)	-1.04 (0.40)	0.729	2, 3, 2, 5, 5, 11 (89, 68, 63, 46, 42, 35)
re-draw	sorting, 10	bubble sort	insertion sort	-0.78 (0.21)	-1.75 (0.30)	0.778	1, 1, 2, 5, 3, 3 (89, 68, 63, 46, 42, 35)
re-draw	sorting, 10	bubble sort	rest (pooled) (not eligible)	-0.99 (0.37)	-2.46 (0.51)	0.645	0, 0, 0, 1, 0, 2 (89, 68, 63, 46, 42, 35)
re-draw	weighted shortest path, none	Dijkstra	rest (pooled) (not eligible)	-3.40 (1.66)	2.38 (1.61)	0.514	0, 1, 1, 2, 37, 95 (100, 99, 99, 98, 63, 5)

Table 40. Multinomial-logit fits with a curvature likelihood-ratio test (linear vs quadratic in 1/T).

draw	task, top-k	curvature likelihood-ratio test (df), p	idea	intercept [95% CI], p	slope [95% CI]
grid	open path search, none	0.22 (3), 0.975	DFS	-0.76 [-1.39, -0.12], 0.019	-0.61 [-1.00, -0.23]
grid	open path search, none	0.22 (3), 0.975	Dijkstra	-0.96 [-1.74, -0.17], 0.017	-0.79 [-1.30, -0.28]
grid	open path search, none	0.22 (3), 0.975	A*	-2.38 [-3.19, -1.56], 1e-08	0.03 [-0.35, 0.41]
grid	open path search, 10	0.27 (3), 0.966	A*	-2.18 [-2.81, -1.54], 1.8e-11	-0.03 [-0.36, 0.31]
grid	open path search, 10	0.27 (3), 0.966	DFS	-0.66 [-1.14, -0.17], 0.0086	-0.64 [-0.98, -0.31]
grid	open path search, 10	0.27 (3), 0.966	Dijkstra	-0.99 [-1.60, -0.38], 0.0014	-0.76 [-1.20, -0.32]
grid	sorting, none	10.41 (4), 0.034	bubble sort	0.26 [-0.14, 0.66], 0.2	-0.56 [-0.79, -0.33]
grid	sorting, none	10.41 (4), 0.034	merge sort	-0.07 [-0.69, 0.54], 0.82	-1.07 [-1.52, -0.62]
grid	sorting, none	10.41 (4), 0.034	selection sort	0.18 [-1.17, 1.53], 0.8	-2.41 [-3.80, -1.03]
grid	sorting, none	10.41 (4), 0.034	insertion sort	1.79 [-1.07, 4.65], 0.22	-5.15 [-8.82, -1.47]
grid	sorting, 10	4.77 (3), 0.189	bubble sort	0.36 [-0.01, 0.73], 0.056	-0.62 [-0.84, -0.39]
grid	sorting, 10	4.77 (3), 0.189	merge sort	-0.17 [-0.72, 0.38], 0.54	-0.96 [-1.37, -0.56]

grid	sorting, 10	4.77 (3), 0.189	insertion sort	-0.66 [-2.89, 1.56], 0.56	-2.76 [-5.32, -0.20]
re-draw	open path search, none	10.27 (3), 0.016	Dijkstra	-1.53 [-2.16, -0.89], 2.7e-06	-0.08 [-0.41, 0.24]
re-draw	open path search, none	10.27 (3), 0.016	DFS	-0.88 [-1.46, -0.29], 0.0032	-0.35 [-0.67, -0.02]
re-draw	open path search, none	10.27 (3), 0.016	A*	-2.49 [-3.28, -1.69], 7.8e-10	0.22 [-0.15, 0.58]
re-draw	open path search, 10	5.41 (3), 0.144	DFS	-0.50 [-0.95, -0.05], 0.029	-0.57 [-0.88, -0.26]
re-draw	open path search, 10	5.41 (3), 0.144	Dijkstra	-1.66 [-2.17, -1.15], 1.5e-10	0.01 [-0.27, 0.28]
re-draw	open path search, 10	5.41 (3), 0.144	A*	-2.20 [-2.80, -1.61], 4.4e-13	0.14 [-0.17, 0.44]
re-draw	sorting, none	2.22 (4), 0.695	quicksort	0.33 [-0.16, 0.81], 0.19	-0.93 [-1.25, -0.61]
re-draw	sorting, none	2.22 (4), 0.695	merge sort	-0.44 [-1.02, 0.13], 0.13	-0.76 [-1.12, -0.39]
re-draw	sorting, none	2.22 (4), 0.695	selection sort	-1.75 [-2.69, -0.80], 0.00029	-0.64 [-1.23, -0.05]
re-draw	sorting, none	2.22 (4), 0.695	insertion sort	-0.93 [-2.07, 0.21], 0.11	-1.40 [-2.34, -0.47]
re-draw	sorting, 10	3.67 (4), 0.453	quicksort	0.53 [0.10, 0.96], 0.015	-1.03 [-1.33, -0.72]
re-draw	sorting, 10	3.67 (4), 0.453	merge sort	-0.29 [-0.81, 0.23], 0.27	-0.85 [-1.21, -0.49]
re-draw	sorting, 10	3.67 (4), 0.453	selection sort	-1.04 [-1.86, -0.23], 0.012	-1.04 [-1.67, -0.41]
re-draw	sorting, 10	3.67 (4), 0.453	insertion sort	-1.79 [-2.86, -0.72], 0.001	-0.93 [-1.73, -0.14]

Table 41. Pre-registered criteria of the temperature law, per draw and task. The within/between slope ratio compares the slope of ln alpha_1 with the slope of ln beta_1 against T; the law requires at least 3.

draw	task, top-k	R^2 per eligible line	all R^2 >= 0.85	top-3 rank reversals at T <= 1.3	within/between slope ratio [95% CI]	curvature p	criterion met
grid	open path search, none	DFS 0.72, Dijkstra 0.76, A* 0.03	no	none	1.15 [0.74, 1.76]	0.975	no

grid	open path search, 10	A* 0.03, DFS 0.91, Dijkstra 0.97	no	T = 0.5: DFS above A* (0.12 vs 0.05, one-sided $p < 0.05$); T = 0.7: DFS above A* (0.13 vs 0.08); T = 1.3: DFS above A* (0.19 vs 0.05, one-sided $p < 0.05$)	1.05 [0.68, 1.36]	0.966	no
grid	sorting, none	bubble sort 0.94, merge sort 0.91, selection sort 0.66, insertion sort 0.68	no	none	0.58 [0.46, 0.72]	0.034	no
grid	sorting, 10	bubble sort 0.93, merge sort 0.91, insertion sort 0.76	no	T = 1.3: bubble sort above quicksort (0.40 vs 0.37)	0.73 [0.55, 1.01]	0.189	no
grid	weighted shortest path, none	not eligible: fewer than 2 ideas at 3% or more at T = 1					
grid	weighted shortest path, 10	not eligible: fewer than 2 ideas at 3% or more at T = 1					
re-draw	open path search, none	Dijkstra 0.05, DFS 0.70, A* 0.03	no	T = 0.3: Dijkstra above DFS (0.14 vs 0.08)	2.64 [1.27, 9.49]	0.016	no
re-draw	open path search, 10	DFS 0.82, Dijkstra 0.00, A* 0.23	no	T = 0.3: Dijkstra above DFS (0.14 vs 0.08); T = 0.5: Dijkstra above DFS (0.10 vs 0.09)	2.23 [1.12, 8.43]	0.144	no
re-draw	sorting, none	quicksort 0.96, merge sort 0.92, selection sort 0.74, insertion sort 0.69	no	none	0.79 [0.54, 1.03]	0.695	no
re-draw	sorting, 10	quicksort 0.97, merge sort 0.97, selection sort 0.73, insertion sort 0.78	no	none	0.68 [0.50, 0.86]	0.453	no

re-draw	weighted shortest path, none	not eligible: fewer than 2 ideas at 3% or more at T = 1

Table 42. Simulation under a true linear law with the observed shares (200 simulated grids per line): how often R^2 falls below 0.85, and how often the curvature test falsely rejects.

task, top-k	share of simulated draws with R^2 < 0.85 under a true linear law, per idea	curvature test false-rejection rate
open path search, none	DFS 0.275, Dijkstra 0.350, A* 0.365	0.085
open path search, 10	A* 0.295, DFS 0.250, Dijkstra 0.265	0.045
sorting, none	bubble sort 0.710, merge sort 0.370, selection sort 0.615, insertion sort 0.865	0.070
sorting, 10	bubble sort 0.600, merge sort 0.410, insertion sort 0.810	0.095

Ideas against phrasings: the answer to questions (a), (b) and (c) of the original request. Table 43 gives the diversity partition and Table 44 the "10 trajectories" view. At T = 0.3 on the open task, 10 programs contain about 2.4 (grid) to 3.5 (re-draw) distinct ideas and 8.3 to 9.1 distinct phrasings (normalised code at edit-distance similarity below 0.85). The model gives a few ideas, each in many phrasings, even at the tightest temperature; that is the user's case (a). Case (b), ten ideas with one phrasing each, never occurs: the largest expected number of ideas among 10 in any cell is 4.9 (sorting, T = 1.6, no top-k, where a third of outputs are unparseable). As T rises the number of ideas grows slowly (path-search re-draw: 3.5 to 3.9 from T = 0.3 to 1.0) and phrasings saturate at 10 of 10 by T = 1.0. Sorting differs. At T = 0.3 the sorting programs are written almost identically every time (2.4 to 2.8 distinct phrasings among 10), and heat adds both ideas and phrasings, with between-idea diversity rising faster than within-idea diversity (slope ratio 0.58 to 0.79). Phrasing grows at least 3 times faster than ideas in no cell; the largest ratio is 2.64, on the path-search re-draw.

Table 43. Diversity partition: log within-idea (alpha_1) and log between-idea (beta_1) diversity per temperature, with slopes against T.

draw	task, top-k	ln alpha_1 at T = 0.3, 0.5, 0.7, 1.0, 1.3, 1.6	ln beta_1 at the same T	slope of ln alpha_1 [95% CI]	slope of ln beta_1 [95% CI]	ratio
grid	open path search, none	0.52, 0.70, 0.80, 1.03, 1.17, n/a	0.70, 0.94, 0.97, 1.35, 1.22, 0.47	0.65 [0.50, 0.69]	0.56 [0.34, 0.76]	1.15
grid	open path search, 10	0.52, 0.68, 0.86, 0.99, 1.14, 1.25	0.70, 0.94, 1.07, 1.25, 1.22, 1.49	0.56 [0.43, 0.57]	0.53 [0.37, 0.70]	1.05
grid	sorting, none	0.03, 0.19, 0.27, 0.32, 0.47, 0.75	0.58, 0.82, 0.94, 1.26, 1.38, 1.74	0.49 [0.38, 0.54]	0.85 [0.67, 0.96]	0.58

grid	sorting, 10	0.02, 0.12, 0.27, 0.32, 0.43, 0.52	0.64, 0.88, 1.01, 1.14, 1.26, 1.34	0.37 [0.31, 0.43]	0.50 [0.36, 0.64]	0.73
grid	weighted shortest path, none	0.36, 0.48, 0.54, 0.79, 1.16, n/a	0.00, 0.00, 0.00, 0.06, 0.67, 0.30	0.78 [0.66, 0.85]	0.59 [0.48, 0.68]	1.31
grid	weighted shortest path, 10	0.37, 0.47, 0.53, 0.75, 1.07, 1.20	0.00, 0.00, 0.00, 0.00, 0.15, 0.68	0.67 [0.59, 0.70]	0.44 [0.28, 0.55]	1.52
re-draw	open path search, none	0.58, 0.74, 0.84, 1.08, 1.24, n/a	1.18, 1.08, 1.11, 1.36, 1.34, 0.31	0.66 [0.48, 0.70]	0.25 [0.05, 0.46]	2.64
re-draw	open path search, 10	0.59, 0.72, 0.83, 0.99, 1.09, 1.25	1.18, 1.09, 1.19, 1.20, 1.20, 1.51	0.50 [0.39, 0.53]	0.22 [0.05, 0.39]	2.23
re-draw	sorting, none	0.08, 0.10, 0.20, 0.28, 0.51, 0.88	0.52, 0.95, 1.07, 1.38, 1.46, 1.55	0.59 [0.42, 0.61]	0.74 [0.53, 0.86]	0.79
re-draw	sorting, 10	0.08, 0.10, 0.21, 0.27, 0.42, 0.67	0.49, 0.95, 1.03, 1.32, 1.29, 1.46	0.44 [0.36, 0.47]	0.65 [0.49, 0.81]	0.68
re-draw	weighted shortest path, none	0.36, 0.43, 0.49, 0.86, 1.18, n/a	0.00, 0.06, 0.06, 0.11, 0.93, 0.37	0.85 [0.71, 0.88]	0.81 [0.62, 0.94]	1.06

Table 44. The user's "10 trajectories" view: expected distinct ideas and distinct phrasings in a random draw of 10 programs. A cell is labelled (a) when it has at most 3 ideas and at least 7 phrasings, (b) when it has at least 7 ideas, and mixed otherwise. The last row of each block is the change from T = 0.3 to 1.3 with the verdict for the user's case (c).

draw	task, top-k	T	expected distinct ideas among 10	expected distinct phrasings among 10	phrasings per idea	cell verdict
grid	open path search, none	0.3	2.44	8.34	3.42	(a)
grid	open path search, none	0.5	2.98	9.51	3.19	(a)
grid	open path search, none	0.7	3.08	9.89	3.21	mixed
grid	open path search, none	1.0	3.91	10.00	2.56	mixed
grid	open path search, none	1.3	3.46	10.00	2.89	mixed
grid	open path search, none	1.6	2.37	4.27	1.80	mixed
grid	open path search, none	change 0.3 to 1.3	+1.02		-0.53	neither
grid	open path search, 10	0.3	2.44	8.30	3.41	(a)

grid	open path search, 10	0.5	2.98	9.46	3.17	(a)
grid	open path search, 10	0.7	3.28	9.90	3.02	mixed
grid	open path search, 10	1.0	3.66	10.00	2.73	mixed
grid	open path search, 10	1.3	3.53	10.00	2.83	mixed
grid	open path search, 10	1.6	4.17	10.00	2.40	mixed
grid	open path search, 10	change 0.3 to 1.3	+1.09		-0.58	neither
grid	sorting, none	0.3	2.08	2.35	1.13	mixed
grid	sorting, none	0.5	2.57	3.73	1.45	mixed
grid	sorting, none	0.7	2.69	4.82	1.79	mixed
grid	sorting, none	1.0	3.49	7.17	2.06	mixed
grid	sorting, none	1.3	3.73	8.16	2.19	mixed
grid	sorting, none	1.6	4.93	7.23	1.47	mixed
grid	sorting, none	change 0.3 to 1.3	+1.65		+1.06	c-b (more ideas)
grid	sorting, 10	0.3	2.24	2.58	1.15	mixed
grid	sorting, 10	0.5	2.67	3.56	1.33	mixed
grid	sorting, 10	0.7	2.87	5.16	1.80	mixed
grid	sorting, 10	1.0	3.16	6.53	2.07	mixed
grid	sorting, 10	1.3	3.45	7.96	2.31	mixed
grid	sorting, 10	1.6	3.68	9.02	2.45	mixed
grid	sorting, 10	change 0.3 to 1.3	+1.21		+1.16	c-a (few ideas, more phrasings each)
grid	weighted shortest path, none	0.3	1.00	7.04	7.04	(a)
grid	weighted shortest path, none	0.5	1.00	8.78	8.78	(a)
grid	weighted shortest path, none	0.7	1.00	9.06	9.06	(a)

grid	weighted shortest path, none	1.0	1.10	9.81	8.92	(a)
grid	weighted shortest path, none	1.3	2.27	9.98	4.40	(a)
grid	weighted shortest path, none	1.6	2.34	5.49	2.34	mixed
grid	weighted shortest path, none	change 0.3 to 1.3	+1.27		-2.63	neither
grid	weighted shortest path, 10	0.3	1.00	6.94	6.94	mixed
grid	weighted shortest path, 10	0.5	1.00	8.78	8.78	(a)
grid	weighted shortest path, 10	0.7	1.00	8.88	8.88	(a)
grid	weighted shortest path, 10	1.0	1.00	9.86	9.86	(a)
grid	weighted shortest path, 10	1.3	1.29	9.99	7.74	(a)
grid	weighted shortest path, 10	1.6	2.38	10.00	4.21	(a)
grid	weighted shortest path, 10	change 0.3 to 1.3	+0.29		+0.80	c-a (few ideas, more phrasings each)
re-draw	open path search, none	0.3	3.48	9.10	2.61	mixed
re-draw	open path search, none	0.5	3.30	9.76	2.95	mixed
re-draw	open path search, none	0.7	3.27	9.85	3.01	mixed
re-draw	open path search, none	1.0	3.91	10.00	2.56	mixed

re-draw	open path search, none	1.3	3.70	10.00	2.70	mixed
re-draw	open path search, none	1.6	2.28	4.90	2.14	mixed
re-draw	open path search, none	change 0.3 to 1.3	+0.22		+0.09	c-a (few ideas, more phrasings each)
re-draw	open path search, 10	0.3	3.48	9.18	2.64	mixed
re-draw	open path search, 10	0.5	3.29	9.32	2.83	mixed
re-draw	open path search, 10	0.7	3.50	9.89	2.82	mixed
re-draw	open path search, 10	1.0	3.53	10.00	2.83	mixed
re-draw	open path search, 10	1.3	3.45	10.00	2.90	mixed
re-draw	open path search, 10	1.6	4.11	10.00	2.43	mixed
re-draw	open path search, 10	change 0.3 to 1.3	-0.03		+0.27	c-a (few ideas, more phrasings each)
re-draw	sorting, none	0.3	2.06	2.82	1.37	mixed
re-draw	sorting, none	0.5	2.92	3.99	1.37	mixed
re-draw	sorting, none	0.7	3.14	4.83	1.54	mixed
re-draw	sorting, none	1.0	3.82	5.91	1.55	mixed
re-draw	sorting, none	1.3	4.05	8.68	2.14	mixed
re-draw	sorting, none	1.6	4.07	5.89	1.45	mixed
re-draw	sorting, none	change 0.3 to 1.3	+2.00		+0.77	c-b (more ideas)
re-draw	sorting, 10	0.3	1.99	2.69	1.36	mixed
re-draw	sorting, 10	0.5	2.92	4.03	1.38	mixed

re-draw	sorting, 10	0.7	3.06	4.85	1.59	mixed
re-draw	sorting, 10	1.0	3.65	6.22	1.70	mixed
re-draw	sorting, 10	1.3	3.55	7.85	2.21	mixed
re-draw	sorting, 10	1.6	4.00	9.36	2.34	mixed
re-draw	sorting, 10	change 0.3 to 1.3	+1.56		+0.86	c-b (more ideas)
re-draw	weighted shortest path, none	0.3	1.00	6.55	6.55	mixed
re-draw	weighted shortest path, none	0.5	1.10	7.36	6.69	(a)
re-draw	weighted shortest path, none	0.7	1.10	8.21	7.46	(a)
re-draw	weighted shortest path, none	1.0	1.20	9.96	8.30	(a)
re-draw	weighted shortest path, none	1.3	2.75	10.00	3.64	(a)
re-draw	weighted shortest path, none	1.6	2.41	4.82	2.00	mixed
re-draw	weighted shortest path, none	change 0.3 to 1.3	+1.75		-2.91	c-b (more ideas)

Out of sample, the frozen law helps on path search and hurts on sorting. Table 45 scores the frozen grid fits on the re-draw against two baselines. One is fixed utilities, in which the $T = 0.7$ shares are raised to the power $0.7/T$. The other is no temperature effect at all. On path search with top-k 10 the frozen law has a mean absolute error of 0.040 against 0.073 for fixed utilities, and without top-k 0.109 against 0.129. On sorting it loses (0.100 and 0.103 against 0.028 and 0.056). The sorting loss is largely the wording change: the grid was fitted with quicksort as the mode and the re-draw has bubble sort as the mode, so the frozen ordering is reversed at every temperature.

Table 45. Out-of-sample scoring of the frozen temperature law on the re-draw (mean absolute error over ideas and temperatures, calibration slope in brackets, and mean JSD per cell).

task, top-k	frozen law, anchored: MAE (calibration slope)	frozen law, unanchored	fixed-utility baseline (shares at 0.7 raised to 0.7/T)	no-temperature-effect baseline	mean JSD: anchored / unanchored / fixed-utility / no effect	reversals against the frozen order at T <= 1.3
open path search, none	0.109 (0.76)	0.091 (0.89)	0.129 (0.66)	0.125 (0.57)	0.133 / 0.095 / 0.160 / 0.170	T = 0.3 DFS/ Dijkstra
open path search, 10	0.040 (0.96)	0.040 (0.88)	0.073 (0.82)	0.047 (0.97)	0.019 / 0.018 / 0.042 / 0.019	T = 0.7 A*/DFS; T = 1.3 A*/DFS
sorting, none	0.103 (0.68)	0.136 (0.38)	0.056 (0.89)	0.088 (0.78)	0.120 / 0.173 / 0.071 / 0.100	T = 0.3 quicksort/ bubble sort; T = 0.5 quicksort/bubble sort; T = 0.7 quicksort/bubble sort; T = 1.3 quicksort/bubble sort
sorting, 10	0.100 (0.73)	0.121 (0.35)	0.028 (1.00)	0.075 (0.92)	0.064 / 0.135 / 0.011 / 0.033	T = 0.3 quicksort/ bubble sort; T = 0.5 quicksort/bubble sort; T = 0.7 quicksort/bubble sort; T = 1.3 quicksort/bubble sort

Correctness depends on the idea as well as on temperature (Table 46). On path search, Dijkstra programs pass most often (0.53 grid, 0.65 re-draw) and A* least often (0.04, 0.09). Each unit of T lowers the log-odds of passing by 1.7 to 4.6. Top-k 10 raises the pass rate on sorting (coefficient 1.25 to 1.40, $p \leq 2e\text{-}5$) and slightly on the path-search re-draw.

Table 46. Pass rate by idea, and the effect of T and top-k on passing (logistic regression over all cells of a draw).

draw	task	pass rate by idea (all T and top-k pooled)	temperature coefficient (log-odds per unit T), p	top-k 10 coefficient, p
grid	open path search	A* 0.04, BFS 0.31, DFS 0.23, Dijkstra 0.53, other 0.00, rare_pooled 0.47	-1.95, 8.2e-20	0.07, 0.65
grid	sorting	bubble sort 0.95, insertion sort 0.74, merge sort 0.87, quicksort 0.95, selection sort 0.79, rare_pooled 0.00	-3.80, 4e-17	1.40, 9e-07

grid	weighted shortest path	Dijkstra 0.73, other 0.00, rare_pooled 0.00	-4.08, 2.5e-54	0.30, 0.097
re-draw	open path search	A* 0.09, BFS 0.32, DFS 0.20, Dijkstra 0.65, other 0.00, rare_pooled 0.20	-1.70, 1.3e-17	0.30, 0.041
re-draw	sorting	bubble sort 0.97, insertion sort 0.91, merge sort 0.88, quicksort 0.88, selection sort 0.96, rare_pooled 0.15	-4.34, 1.5e-16	1.25, 1.8e-05
re-draw	weighted shortest path	Dijkstra 0.74, other 0.00, rare_pooled 0.00	-4.57, 3.7e-24	n/a (no top-k cells)

Truncation and reproducibility. Half of the grid rows (51.1%) stopped at the 768-token cap. Truncation is more likely at high T, for A* (odds ratio 182 against BFS) and for Dijkstra (7.7). The audit continued 150 truncated rows from their own tokens (Table 47). Labels flipped in 16 of them (10.7%). Most flips were bubble sort becoming merge sort: the model writes a short bubble sort as a first example and then a merge sort, and the classifier labels the final program. Regenerating 20 rows under vLLM reproduced none of the 768-token prefixes; the median common prefix was 2 tokens. The user's "always the same seed" protocol is therefore not bit-reproducible under vLLM batching, although 17 of 20 repeated rows kept their label.

Table 47. Truncation audit and reproducibility.

quantity	**value**
grid rows stopped at the 768-token cap	0.511 of 3600
odds ratio of truncation per unit T	4.36 (p 4.4e-32)
odds ratio of truncation, A* vs BFS	181.5 [25.0, 1320.4]
odds ratio of truncation, Dijkstra vs BFS	7.7 [4.7, 12.7]
odds ratio of truncation, top-k 10 vs none	0.67
truncated rows continued from their own tokens	150
label flips after continuation	16 (0.107, 95% CI [0.067, 0.166])
flip rate by T (n = 25 each)	T = 0.3: 0.16, T = 0.5: 0.12, T = 0.7: 0.16, T = 1.0: 0.04, T = 1.3: 0.04, T = 1.6: 0.12
flip rate by label before continuation	bubble sort 0.45 (n = 20), Dijkstra 0.00 (n = 52), A* 0.00 (n = 8), BFS 0.11 (n = 18), DFS 0.12 (n = 8), selection sort 1.00 (n = 2), SPFA 0.00 (n = 2), merge sort 0.00 (n = 7), quicksort 0.00 (n = 9), other 0.00 (n = 13), insertion sort 0.33 (n = 3), unparseable 0.14 (n = 7), greedy best-first 0.00 (n = 1)

transitions	bubble sort to merge sort 8, selection sort to merge sort 2, DFS to BFS 1, BFS to Dijkstra 1, BFS to other 1, bubble sort to quicksort 1, unparseable to other 1, insertion sort to merge sort 1
pass rate, truncated vs continued	0.533 vs 0.520
still truncated after continuation	23
768-token prefixes reproduced on regeneration	0 of 20 (median common prefix 2 tokens)
iteration-1 grid repeat rows: identical tokens / identical label	2/20 / 17/20

2.6 Seed strings narrow the ideas

The user asked whether prepending a random seed string helps. The iteration-1 seed rows used 60 strings per task of the two forms the user gave, a random-character string (`seed=` followed by 11 random characters, as in `seed=d32qxk4nt43`) and four random words (as in `seed=(horse chair roof love)`). The user's two literal strings were not among them. The first artifact froze a predictor of the greedy flip rate (sha256 `1cffcec3`…). The planned internal readout, the log-probability the model assigns to each idea's name token, failed its pre-registered coverage gate on path search and sorting (prefix coverage 0.10 to 0.37). The behavioural fallback was therefore primary: a Gaussian-shock model fitted to the screen's greedy flips. The artifact then generated 120 fresh strings per task (60 of each kind) on the ban-arm wording, with one greedy sample and five samples at $T = 0.7$ each.

The seed-shock test fails. The frozen prediction matches the observed greedy flip rate on path search (0.28 predicted, 0.25 observed) and on the weighted task (0.00 and 0.00, under the 5% bound), but not on sorting (0.39 predicted, 0.017 observed) (Table 48). The sorting miss traces to the wording change. On the grid wording the unseeded greedy program is quicksort, and seed strings flipped it to bubble sort in 88% of cases. On the ban-arm wording the unseeded greedy program is already bubble sort, and seeds keep it there. Random words flip the path-search idea far more often than random characters (0.42 against 0.08).

Table 48. Greedy flips under 120 fresh seed strings per task, against the frozen prediction.

task	unseeded greedy idea (5 repeats)	seed strings	greedy flips (rate, 95% CI)	flips, random-character strings	flips, random-word strings	frozen behavioural prediction	readout prediction (failed its gate)	greedy idea counts under seeds	screen flip rate (grid wording, n = 60)

open path search	BFS	120	30 (0.250, [0.181, 0.334])	0.083	0.417	0.281	0.950	BFS 90, DFS 23, other 6, SPFA 1	0.183 (unseeded greedy BFS; seeded BFS 49, DFS 6, other 4, SPFA 1)
sorting	bubble sort	120	2 (0.017, [0.005, 0.059])	0.033	0.000	0.394	0.108	bubble sort 118, quicksort 1, other 1	0.883 (unseeded greedy quicksort; seeded bubble sort 53, quicksort 7)
weighted shortest path	Dijkstra	120	0 (0.000, [0.000, 0.031])	0.000	0.000	0.000	0.000	Dijkstra 120	0.000 (unseeded greedy Dijkstra; seeded Dijkstra 60)

Seeds lower idea diversity. At $T = 0.7$, seeded samples have lower idea entropy than unseeded ones on path search (-0.34 bits [-0.53, -0.13]) and sorting (-0.68 [-0.92, -0.47]) (Table 49). Random words on sorting cut entropy by more than a bit and push 282 of 300 samples to bubble sort. Seed strings make the model more predictable: they push it toward the modal idea rather than away from it. The weighted task does not move at all (597 of 600 Dijkstra).

Table 49. Idea diversity of seeded and unseeded samples at $T = 0.7$ (5 samples per string; unseeded $n = 300$).

task	**seed kind**	**n seeded / unseeded**	**idea entropy seeded vs unseeded (bits)**	**difference [95% CI]**	**expected ideas among 10, seeded vs unseeded**	**seeded idea counts**	**unseeded idea counts**
open path search	both	600 / 300	1.283 vs 1.620	-0.337 [-0.529, -0.132]	2.76 vs 3.31	BFS 443, DFS 88, other 37, A* 14, Dijkstra 12, SPFA 6	BFS 193, DFS 45, Dijkstra 27, A* 26, other 5, SPFA 4

open path search	char	300 / 300	1.222 vs 1.620	-0.398 [-0.634, -0.158]	2.71 vs 3.31	BFS 226, DFS 39, other 20, Dijkstra 11, SPFA 2, A* 2	BFS 193, DFS 45, Dijkstra 27, A* 26, other 5, SPFA 4
open path search	word	300 / 300	1.296 vs 1.620	-0.324 [-0.538, -0.117]	2.78 vs 3.31	BFS 217, DFS 49, other 17, A* 12, SPFA 4, Dijkstra 1	BFS 193, DFS 45, Dijkstra 27, A* 26, other 5, SPFA 4
sorting	both	600 / 300	0.857 vs 1.534	-0.677 [-0.917, -0.467]	2.12 vs 3.14	bubble sort 519, quicksort 44, selection sort 11, built-in sort 9, other 6, insertion sort 5, merge sort 4, Timsort/hybrid 1, bucket sort 1	bubble sort 192, quicksort 56, merge sort 32, selection sort 12, insertion sort 8
sorting	char	300 / 300	1.118 vs 1.534	-0.416 [-0.698, -0.157]	2.47 vs 3.14	bubble sort 237, quicksort 41, selection sort 7, other 5, merge sort 4, insertion sort 3, Timsort/hybrid 1, bucket sort 1, built-in sort 1	bubble sort 192, quicksort 56, merge sort 32, selection sort 12, insertion sort 8
sorting	word	300 / 300	0.448 vs 1.534	-1.085 [-1.313, -0.873]	1.56 vs 3.14	bubble sort 282, built-in sort 8, selection sort 4, quicksort 3, insertion sort 2, other 1	bubble sort 192, quicksort 56, merge sort 32, selection sort 12, insertion sort 8
weighted shortest path	both	600 / 300	0.050 vs 0.032	+0.018 [-0.058, +0.090]	1.05 vs 1.03	Dijkstra 597, A* 2, BFS 1	Dijkstra 299, A* 1
weighted shortest path	char	300 / 300	0.032 vs 0.032	+0.000 [-0.081, +0.081]	1.03 vs 1.03	Dijkstra 299, BFS 1	Dijkstra 299, A* 1
weighted shortest path	word	300 / 300	0.058 vs 0.032	+0.026 [-0.058, +0.102]	1.07 vs 1.03	Dijkstra 298, A* 2	Dijkstra 299, A* 1

The narrowing is not driven by a few strings. Heterogeneity between seed strings is not significant on path search (permutation p 0.29) and is modest on sorting (p 0.004, intraclass correlation 0.16), and the non-modal share at T = 0.7 drops from 0.36 to 0.26 on path search and from 0.36 to 0.14 on sorting (Table 50).

Table 50. Non-mode share, heterogeneity between strings, and the intraclass correlation of the mode.

task	non-mode share at $T = 0.7$, unseeded vs seeded	chi-square p (seeded vs unseeded)	between-string heterogeneity permutation p	intraclass correlation of the mode
open path search	0.357 vs 0.262	5.1e-10	0.286	-0.021
sorting	0.360 vs 0.135	1.2e-18	0.004	0.160
weighted shortest path	0.003 vs 0.005	0.78	1.000	-0.003

The narrowing is not because seeds leave the residual stream untouched. At the last five chat-template tokens, a seed string displaces the residual stream 1.1 to 3.2 times more than a ban sentence on path search and sorting at layers 4 and above (Table 51). The cosine between the seed and ban displacements swings across layers (from -0.81 to 0.73), so the two are not aligned in any consistent way. On the weighted task the ban displaces more than the seed at layers 24 and above. A large displacement in an irrelevant direction does not change the idea; the ban's smaller displacement does. Section 2.8 shows that even the ban's displacement at these positions is not where the ban acts.

Table 51. Norm of the residual displacement at the last five chat-template tokens (mean over tokens), by layer: seed string against unseeded prompt, ban against placebo, and placebo against plain baseline.

task	layer	seed minus unseeded	ban minus placebo	placebo minus baseline	seed / ban ratio
open path search	8	15.6	9.3	26.5	1.69
open path search	16	19.8	9.2	35.0	2.17
open path search	24	29.7	24.2	40.7	1.23
open path search	28	39.8	26.6	43.3	1.50
open path search	31	215.9	68.7	244.5	3.14
open path search	32	227.0	71.7	245.3	3.16
open path search	34	224.6	74.6	240.5	3.01
open path search	36	66.1	41.7	60.4	1.59
sorting	8	16.3	8.6	29.0	1.89
sorting	16	21.3	11.0	39.6	1.93
sorting	24	33.0	22.8	46.2	1.45
sorting	28	45.6	26.7	49.1	1.70
sorting	31	240.0	78.8	518.2	3.04
sorting	32	254.1	81.2	520.6	3.13

sorting	34	252.0	89.3	509.4	2.82
sorting	36	71.5	52.5	93.8	1.36
weighted shortest path	8	13.0	5.5	27.9	2.36
weighted shortest path	16	16.9	7.1	36.1	2.37
weighted shortest path	24	28.3	30.7	44.3	0.92
weighted shortest path	28	33.7	44.0	45.4	0.77
weighted shortest path	31	96.7	137.0	253.9	0.71
weighted shortest path	32	103.8	144.0	267.6	0.72
weighted shortest path	34	106.6	150.5	255.6	0.71
weighted shortest path	36	52.6	80.0	62.6	0.66

2.7 The temperature reference for off-menu correct programs

The off-menu-correct test needs a temperature comparator. The first artifact froze it before the sibling artifacts used it (sha256 `783778b0…`). The rule picks, per task and source, the temperature with the highest off-menu correct share among cells whose pass rate is at least 0.8 times that at T = 0.7. No temperature produces more than 8% off-menu correct programs on any task (Table 52). On path search the best is SPFA in 2 of 100 programs; on sorting it is insertion sort, which is just below the 3% menu threshold at baseline; on the weighted task it is zero.

Table 52. Frozen temperature reference for the off-menu-correct test (menus: path search BFS, DFS, A*, Dijkstra; sorting bubble, quick, merge and selection sort; weighted task Dijkstra).

task	source	top-k	best T	pass rate	off-menu correct share	off-menu correct ideas	pass at T = 0.7	off-menu correct at T = 0.7
open path search	grid (768 tokens)	none	0.5	0.46	0.02	SPFA 2	0.27	0.00
open path search	grid (768 tokens)	10	0.5	0.46	0.02	SPFA 2	0.33	0.00
open path search	re-draw (1,536 tokens)	none	0.3	0.49	0.02	SPFA 2	0.35	0.00

open path search	re-draw (1,536 tokens)	10	0.3	0.50	0.02	SPFA 2	0.35	0.01
open path search	re-draw truncated to 768	none	0.3	0.45	0.02	SPFA 2	0.35	0.00
open path search	re-draw truncated to 768	10	0.3	0.46	0.02	SPFA 2	0.38	0.00
sorting	grid (768 tokens)	none	1.0	0.97	0.03	insertion sort 3	0.96	0.00
sorting	grid (768 tokens)	10	1.0	0.99	0.03	insertion sort 3	0.96	0.00
sorting	re-draw (1,536 tokens)	none	1.3	0.86	0.08	insertion sort 8	1.00	0.02
sorting	re-draw (1,536 tokens)	10	1.0	0.97	0.05	insertion sort 5	0.99	0.02
sorting	re-draw truncated to 768	none	1.3	0.85	0.08	insertion sort 8	0.97	0.02
sorting	re-draw truncated to 768	10	1.0	0.97	0.05	insertion sort 5	0.96	0.02
weighted shortest path	grid (768 tokens)	none	0.3	0.96	0.00	-	0.87	0.00
weighted shortest path	grid (768 tokens)	10	0.3	0.97	0.00	-	0.86	0.00
weighted shortest path	re-draw (1,536 tokens)	none	0.3	0.91	0.00	-	0.88	0.00
weighted shortest path	re-draw truncated to 768	none	0.3	0.90	0.00	-	0.87	0.00

2.8 Bans as displacements: what the internal state predicts, and what it does not carry

Design. The second artifact asked whether the internal change a ban sentence causes predicts which ideas gain share, and whether that change causes the behaviour. We define the *ban displacement* Δh as the residual-stream difference between the ban prompt and the placebo prompt. It is read at the five chat-template tokens that end every prompt (`<|im_end|>\n<|im_start|>assistant\n`), which are aligned across conditions, and averaged over layers 32 to 34. That band was chosen by a per-layer idea probe against shuffled labels. Probe accuracy was nearly flat across

all 35 layers (balanced accuracy 0.72 to 0.74), as in iteration 1, so the band choice carries little information. Idea centroids come from a *commitment window*, the mean residual over the 17 generated tokens starting at the first token that names or implements the idea (or the first 32 generated tokens when no such token is found), computed on baseline and forced-idea programs. The forced cells were extended by 810 new generations to 40 per idea.

Seven predictors of the post-ban idea distribution share one floor-and-compliance model and differ in one scalar each. The shared part keeps a fraction κ of the banned ideas' placebo mass on those ideas (non-compliance, fitted per ban-list length) and smooths zero shares with a small floor proportional to co-mention frequency. Each predictor then reweights the unbanned ideas' placebo shares. They were fitted by leave-one-ban-out on the 27 screen bans and frozen by sha256 and git commit before any confirmation generation:

- *IIA*: proportional renormalisation (the Luce rule) of the placebo distribution over the unbanned ideas;
- *displacement*: each unbanned idea's share is multiplied by $\exp(\beta\, s_j)$, where s_j is the projection of Δh onto that idea's unit centroid, scaled by the median screen displacement norm ($\beta = 19.75$, near the edge of the pre-registered grid of -20 to 20);
- *nested geometry*: substitution weighted by centroid cosine similarity, as in the iteration-1 plan;
- *name embedding* and *prompt embedding*: the same substitution weights read from bge text embeddings [21] of the algorithm names and descriptions, or of the prompt difference;
- *Qwen name embedding*: weights from Qwen's own input embeddings of the names;
- *co-mention*: weights from the pointwise mutual information of two ideas in the co-mention lists.

Confirmation used 5,930 never-generated vLLM generations at $n = 100$ per condition. On path search and sorting these were rank-5 to rank-8 single bans and six random ban pairs per task; on the weighted task, bans of SPFA and BFS. There were also three placebo blocks per task, an alternative placebo, a cross-domain ban, *mention controls* (we use this name for prompts that end in `You may use X.` instead of a ban), and the whole k-th-smallest task as pure transfer: baseline, forced cells, co-mention lists, rank-1 to rank-4, top-3, top-5 and rare bans, and four pairs. The k-th-smallest task needed its own classifier and oracle, audited by gpt-4.1-mini on 60 programs (kappa 0.79, $0.015). The placebo P0 used for confirmation is the mean of the three placebo blocks: open path search: BFS 0.63, DFS 0.19, other 0.07, Dijkstra 0.05, A* 0.03, SPFA 0.03; sorting: bubble sort 0.54, quicksort 0.18, merge sort 0.15, selection sort 0.09, insertion sort 0.03, heap sort 0.00, unparseable 0.00; weighted shortest path: Dijkstra 0.99, Bellman-Ford 0.01; k-th smallest: quickselect 0.77, sort then index 0.19, heap select 0.04, other 0.01.

Table 53 gives every confirmation condition with its observed counts and the JSD of four of the frozen predictors.

Table 53. Confirmation conditions (n = 100 each, never generated before the predictors were frozen), with the JSD between observed and predicted distributions.

task	condition	banned ideas	n	idea counts	still banned	JSD, IIA	JSD, displacement	JSD, nested geometry	JSD, name embedding	displacement top substitute correct
open path search	alternative placebo	-	100	BFS 62, DFS 19, Dijkstra 13, A* 4, other 2	n/a	0.043	0.041	0.043	0.043	no
open path search	single ban	Bellman-Ford	100	Dijkstra 56, BFS 40, DFS 2, A* 2	0	0.313	0.270	0.312	0.230	yes
open path search	single ban	bidirectional	100	BFS 67, DFS 16, A* 8, other 6, Dijkstra 2, greedy best-first 1	0	0.032	0.053	0.033	0.028	no
open path search	single ban	greedy best-first	100	A* 58, BFS 23, Dijkstra 8, DFS 7, other 3, greedy best-first 1	1	0.341	0.286	0.329	0.243	no
open path search	single ban	IDDFS	100	A* 49, BFS 41, Dijkstra 4, DFS 2, other 2, greedy best-first 1, SPFA 1	0	0.273	0.328	0.278	0.379	no
open path search	cross-domain ban	-	100	BFS 75, DFS 18, other 4, Dijkstra 2, A* 1	n/a	0.032	0.026	0.032	0.032	no
open path search	mention: "You may use A*"	-	100	A* 95, Dijkstra 5	n/a	0.858	0.832	0.858	0.858	no
open path search	mention: "You may use BFS"	-	100	BFS 97, SPFA 3	n/a	0.201	0.037	0.201	0.201	yes
open path search	random pair ban	bidirectional, Floyd-Warshall	100	BFS 55, Dijkstra 26, A* 8, DFS 6, other 4, SPFA 1	0	0.104	0.089	0.102	0.110	yes
open path search	random pair ban	DFS, Bellman-Ford	100	BFS 73, A* 19, Dijkstra 8	0	0.135	0.119	0.143	0.142	no
open path search	random pair ban	Dijkstra, SPFA	100	BFS 63, A* 30, Bellman-Ford 3, DFS 2, bidirectional 1, SPFA 1	1	0.221	0.221	0.215	0.202	no
open path search	random pair ban	IDDFS, Dijkstra	100	BFS 75, A* 21, Dijkstra 3, DFS 1	3	0.190	0.238	0.190	0.230	no

open path search	random pair ban	IDDFS, Floyd-Warshall	100	BFS 83, A* 12, DFS 2, greedy best-first 1, SPFA 1, other 1	0	0.141	0.160	0.143	0.202	no
open path search	random pair ban	SPFA, Floyd-Warshall	100	BFS 64, Dijkstra 31, DFS 2, A* 2, SPFA 1	1	0.178	0.100	0.167	0.131	yes
sorting	alternative placebo	-	100	bubble sort 62, quicksort 14, merge sort 14, selection sort 8, insertion sort 2	n/a	0.013	0.013	0.013	0.013	yes
sorting	single ban	counting sort	100	quicksort 35, merge sort 32, bubble sort 26, insertion sort 5, selection sort 2	0	0.105	0.083	0.101	0.109	no
sorting	single ban	heap sort	100	bubble sort 51, quicksort 17, selection sort 14, insertion sort 12, merge sort 6	0	0.043	0.066	0.044	0.054	no
sorting	single ban	radix sort	100	merge sort 39, quicksort 36, bubble sort 15, selection sort 8, insertion sort 2	0	0.153	0.130	0.144	0.191	no
sorting	single ban	shell sort	100	merge sort 49, bubble sort 28, quicksort 15, selection sort 6, insertion sort 2	0	0.109	0.096	0.108	0.183	no
sorting	cross-domain ban	-	100	bubble sort 71, quicksort 14, merge sort 6, selection sort 5, insertion sort 4	n/a	0.038	0.047	0.038	0.038	no
sorting	mention: "You may use bubble sort"	-	100	bubble sort 100	n/a	0.286	0.044	0.286	0.286	yes
sorting	mention: "You may use merge sort"	-	100	merge sort 100	n/a	0.680	0.099	0.680	0.680	yes
sorting	random pair ban	insertion sort, bubble sort	100	quicksort 58, merge sort 41, heap sort 1	0	0.209	0.130	0.220	0.233	yes
sorting	random pair ban	insertion sort, shell sort	100	merge sort 63, quicksort 35, heap sort 1, selection sort 1	0	0.443	0.238	0.458	0.516	no

sorting	random pair ban	merge sort, heap sort	100	bubble sort 65, insertion sort 24, selection sort 10, quicksort 1	0	0.167	0.137	0.183	0.213	no
sorting	random pair ban	merge sort, selection sort	100	insertion sort 51, quicksort 17, heap sort 17, bubble sort 13, radix sort 1, other 1	0	0.430	0.405	0.425	0.444	no
sorting	random pair ban	quicksort, bubble sort	100	insertion sort 54, merge sort 38, heap sort 4, selection sort 4	0	0.316	0.362	0.319	0.359	no
sorting	random pair ban	quicksort, shell sort	100	insertion sort 71, selection sort 14, merge sort 10, other 3, heap sort 2	0	0.634	0.599	0.631	0.643	no
weighted shortest path	single ban	BFS	100	Dijkstra 100	0	0.012	0.017	0.012	0.008	no
weighted shortest path	single ban	SPFA	100	Dijkstra 98, A* 1, Bellman-Ford 1	0	0.006	0.005	0.006	0.006	no
k-th smallest	ban rank-1	quickselect	100	sort then index 49, quickselect 19, heap select 15, median of medians 14, counting/bucket 3	19	0.134	0.156	0.138	0.157	no
k-th smallest	ban rank-2	sort then index	100	quickselect 89, sort then index 10, heap select 1	10	0.051	0.036	0.054	0.049	yes
k-th smallest	ban rank-3	heap select	100	quickselect 74, sort then index 25, median of medians 1	0	0.012	0.130	0.010	0.019	no
k-th smallest	ban rank-4	median of medians	100	quickselect 81, sort then index 19	0	0.026	0.095	0.026	0.038	no
k-th smallest	ban rare idea	partial selection	100	quickselect 89, sort then index 11	0	0.038	0.040	0.038	0.057	yes
k-th smallest	ban top-3	quickselect, sort then index, heap select	100	quickselect 51, median of medians 32, sort then index 11, counting/bucket 3, other 2, heap select 1	63	0.299	0.291	0.303	0.365	no

k-th smallest	ban top-5	quickselect, sort then index, heap select, median of medians, counting/ bucket	100	quickselect 75, sort then index 16, median of medians 4, partial selection 2, other 2, heap select 1	96	0.369	0.372	0.369	0.373	no
k-th smallest	random pair ban	heap select, binary search on value	100	quickselect 75, sort then index 25	0	0.016	0.113	0.014	0.022	no
k-th smallest	random pair ban	median of medians, partial selection	100	quickselect 90, sort then index 10	0	0.040	0.043	0.038	0.025	yes
k-th smallest	random pair ban	sort then index, binary search on value	100	quickselect 83, sort then index 14, median of medians 1, counting/bucket 1, other 1	14	0.041	0.029	0.043	0.049	yes
k-th smallest	random pair ban	sort then index, median of medians	100	quickselect 92, sort then index 8	8	0.032	0.019	0.036	0.022	yes

The regularity test passes. In 8 of 46 tested cells an unbanned idea lost share beyond the noise floor at Holm-corrected $p < 0.05$, spread over 6 of the confirmation bans (Table 54). Testing placebo blocks against each other in the same way gives 0 violations in 45 cells, and shuffling ban and placebo labels gives 0 in each of 20 repetitions. The most striking cases are bans of ideas the model almost never writes. Banning greedy best-first search, written in 0 of 300 placebo programs, takes A* from 3% to 58% and BFS from 63% to 23%. Banning IDDFS, also absent, gives A* 49%. Banning Bellman-Ford takes Dijkstra from 5% to 56%. On sorting, banning radix, counting or shell sort, none of which the model writes, takes bubble sort from 54% to between 15% and 28%. A ban that only restricted the choice set could not move mass between ideas it does not name. Violations of IIA-type properties have also been reported for language-model choices in an economic setting [23]. These bans act more like an instruction to "do something other than the obvious", and the model's answer is a particular other idea.

Table 54. Confirmation regularity violations (placebo = three pooled placebo blocks, n = 300).

task	ban (confirmation)	idea that lost share	placebo count	ban count	drop	noise floor	Fisher p	Holm p
open path search	ban Bellman-Ford	BFS	190/300	40/100	0.233	0.10	3.8e-05	0.0015
open path search	ban Bellman-Ford	DFS	58/300	2/100	0.173	0.07	2e-06	8.4e-05
open path search	ban greedy best-first	BFS	190/300	23/100	0.403	0.10	1.3e-12	6.2e-11

open path search	ban IDDFS	BFS	190/300	41/100	0.223	0.10	7.7e-05	0.003
open path search	ban IDDFS	DFS	58/300	2/100	0.173	0.07	2e-06	8.4e-05
sorting	ban counting sort	bubble sort	162/300	26/100	0.280	0.16	7e-07	3.1e-05
sorting	ban radix sort	bubble sort	162/300	15/100	0.390	0.16	1.5e-12	6.6e-11
sorting	ban shell sort	bubble sort	162/300	28/100	0.260	0.16	4.2e-06	0.00017

The substitution-prediction test fails, narrowly on the primary criterion and clearly on the top-substitute criterion. On the 22 confirmation bans of the three original tasks, the displacement predictor has a mean JSD of 0.188 against 0.207 for IIA, a 9.2% reduction (95% CI [−0.6%, 19.6%]) against a required 20% (Tables 55 and 56). Its top predicted substitute is right in 20% of bans against a required 70%. It does predict the sign of share changes well: on the 52 cells where an idea moved by at least 10 points, it is right in 83% (balanced 0.79), where IIA can only predict gains (0.65, balanced 0.50). It also beats both bge text-embedding predictors and the co-mention predictor on the three original tasks (all CIs exclude 0), so the internal state carries information about substitution that the names alone do not. On the k-th-smallest task, pure transfer, it is 25% worse than IIA (CI [−102%, −0.5%]), and across all 33 bans the reduction is 2.8% (CI [−9.1%, 12.8%]). Anchoring the displacement predictor on the screen placebo instead of the confirmation placebo lowers its JSD to 0.165, with a top-substitute hit rate of 0.35 and sign accuracy of 0.84. Part of the error is therefore the placebo itself drifting between draws.

Table 55. Frozen predictors scored on the confirmation bans, by subset.

ban subset	n bans	predictor	mean JSD	mean log-likelihood	top-substitute hit rate (n)	sign accuracy	balanced sign accuracy
T1-T3 bans (primary)	22	IIA (proportional renormalisation)	0.207	-1.596	0.050 (20)	0.591	0.500
T1-T3 bans (primary)	22	displacement (internal, frozen)	0.188	-1.535	0.200 (20)	0.795	0.776
T1-T3 bans (primary)	22	displacement anchored on the screen placebo	0.165	-1.472	0.350 (20)	0.841	0.848
T1-T3 bans (primary)	22	nested geometry (centroid similarity)	0.207	-1.592	0.300 (20)	0.682	0.654
T1-T3 bans (primary)	22	text embedding of names and descriptions	0.221	-1.628	0.150 (20)	0.386	0.395
T1-T3 bans (primary)	22	text embedding of the prompt difference	0.216	-1.625	0.000 (20)	0.409	0.389
T1-T3 bans (primary)	22	Qwen input-embedding of names	0.211	-1.600	0.050 (20)	0.591	0.585

T1-T3 bans (primary)	22	co-mention similarity	0.239	-1.677	0.050 (20)	0.614	0.588
all 33 bans	33	IIA (proportional renormalisation)	0.170	-1.360	0.172 (29)	0.654	0.500
all 33 bans	33	displacement (internal, frozen)	0.165	-1.363	0.310 (29)	0.827	0.789
all 33 bans	33	nested geometry (centroid similarity)	0.171	-1.358	0.414 (29)	0.731	0.676
all 33 bans	33	text embedding of names and descriptions	0.183	-1.391	0.345 (29)	0.462	0.458
all 33 bans	33	text embedding of the prompt difference	0.176	-1.379	0.276 (29)	0.500	0.448
all 33 bans	33	Qwen input-embedding of names	0.174	-1.367	0.276 (29)	0.654	0.631
all 33 bans	33	co-mention similarity	0.191	-1.414	0.172 (29)	0.673	0.619
T1/T2 single bans	8	IIA (proportional renormalisation)	0.171	-1.803	0.000 (8)	0.471	0.500
T1/T2 single bans	8	displacement (internal, frozen)	0.164	-1.754	0.125 (8)	0.765	0.771
T1/T2 single bans	8	displacement anchored on the screen placebo	0.143	-1.730	0.500 (8)	0.765	0.764
T1/T2 single bans	8	nested geometry (centroid similarity)	0.169	-1.787	0.375 (8)	0.882	0.882
T1/T2 single bans	8	text embedding of names and descriptions	0.177	-1.753	0.250 (8)	0.471	0.465
T1/T2 single bans	8	text embedding of the prompt difference	0.181	-1.830	0.000 (8)	0.471	0.465
T1/T2 single bans	8	Qwen input-embedding of names	0.170	-1.792	0.125 (8)	0.588	0.583
T1/T2 single bans	8	co-mention similarity	0.187	-1.854	0.000 (8)	0.588	0.590
random pairs	16	IIA (proportional renormalisation)	0.206	-1.411	0.188 (16)	0.690	0.500
random pairs	16	displacement (internal, frozen)	0.188	-1.383	0.375 (16)	0.828	0.783
random pairs	16	nested geometry (centroid similarity)	0.208	-1.413	0.375 (16)	0.586	0.456
random pairs	16	text embedding of names and descriptions	0.221	-1.478	0.312 (16)	0.379	0.367

random pairs	16	text embedding of the prompt difference	0.213	-1.434	0.250 (16)	0.414	0.300
random pairs	16	Qwen input-embedding of names	0.211	-1.420	0.188 (16)	0.621	0.572
random pairs	16	co-mention similarity	0.242	-1.496	0.188 (16)	0.655	0.567
T3 bans	2	IIA (proportional renormalisation)	0.009	-0.075	n/a (0)	n/a	n/a
T3 bans	2	displacement (internal, frozen)	0.011	-0.081	n/a (0)	n/a	n/a
T3 bans	2	displacement anchored on the screen placebo	0.021	-0.100	n/a (0)	n/a	n/a
T3 bans	2	nested geometry (centroid similarity)	0.009	-0.076	n/a (0)	n/a	n/a
T3 bans	2	text embedding of names and descriptions	0.007	-0.073	n/a (0)	n/a	n/a
T3 bans	2	text embedding of the prompt difference	0.011	-0.079	n/a (0)	n/a	n/a
T3 bans	2	Qwen input-embedding of names	0.010	-0.077	n/a (0)	n/a	n/a
T3 bans	2	co-mention similarity	0.010	-0.077	n/a (0)	n/a	n/a
T4 (pure transfer)	11	IIA (proportional renormalisation)	0.096	-0.889	0.444 (9)	1.000	1.000
T4 (pure transfer)	11	displacement (internal, frozen)	0.120	-1.021	0.556 (9)	1.000	1.000
T4 (pure transfer)	11	nested geometry (centroid similarity)	0.097	-0.890	0.667 (9)	1.000	1.000
T4 (pure transfer)	11	text embedding of names and descriptions	0.107	-0.918	0.778 (9)	0.875	0.875
T4 (pure transfer)	11	text embedding of the prompt difference	0.095	-0.887	0.889 (9)	1.000	1.000
T4 (pure transfer)	11	Qwen input-embedding of names	0.100	-0.901	0.778 (9)	1.000	1.000
T4 (pure transfer)	11	co-mention similarity	0.096	-0.889	0.444 (9)	1.000	1.000
all single bans	17	IIA (proportional renormalisation)	0.136	-1.313	0.154 (13)	0.609	0.500
all single bans	17	displacement (internal, frozen)	0.144	-1.345	0.231 (13)	0.826	0.798

all single bans	17	nested geometry (centroid similarity)	0.136	-1.306	0.462 (13)	0.913	0.909
all single bans	17	text embedding of names and descriptions	0.146	-1.310	0.385 (13)	0.565	0.563
all single bans	17	text embedding of the prompt difference	0.141	-1.328	0.308 (13)	0.609	0.599
all single bans	17	Qwen input-embedding of names	0.138	-1.317	0.385 (13)	0.696	0.690
all single bans	17	co-mention similarity	0.144	-1.337	0.154 (13)	0.696	0.671

Table 56. Paired differences: the displacement predictor against IIA (relative JSD reduction) and against each other predictor (difference in mean JSD; negative favours the displacement predictor).

ban subset	**displacement vs IIA: relative JSD reduction [95% CI]**	**displacement minus name embedding**	**minus prompt embedding**	**minus Qwen name embedding**	**minus co-mention**	**minus nested geometry**
T1-T3 bans (primary)	+0.092 [-0.006, +0.196]	-0.033 [-0.063, -0.008]	-0.029 [-0.062, -0.004]	-0.023 [-0.047, -0.004]	-0.051 [-0.086, -0.019]	-0.020 [-0.045, +0.001]
all 33 bans	+0.028 [-0.091, +0.128]	-0.017 [-0.041, +0.004]	-0.011 [-0.036, +0.010]	-0.008 [-0.028, +0.009]	-0.026 [-0.054, -0.001]	-0.005 [-0.026, +0.012]
T1/T2 single bans	+0.041 [-0.125, +0.145]	-0.013 [-0.044, +0.018]	-0.017 [-0.037, +0.006]	-0.006 [-0.024, +0.008]	-0.023 [-0.069, +0.015]	-0.005 [-0.025, +0.016]
random pairs	+0.089 [-0.061, +0.239]	-0.034 [-0.075, -0.002]	-0.025 [-0.072, +0.010]	-0.024 [-0.059, +0.005]	-0.054 [-0.099, -0.015]	-0.020 [-0.056, +0.009]
T3 bans	-0.210 [-0.418, +0.183]	+0.004 [-0.001, +0.009]	+0.000 [-0.002, +0.002]	+0.001 [-0.002, +0.004]	+0.001 [-0.002, +0.005]	+0.002 [-0.001, +0.004]
T4 (pure transfer)	-0.250 [-1.017, -0.005]	+0.013 [-0.014, +0.044]	+0.025 [-0.000, +0.056]	+0.020 [-0.005, +0.051]	+0.024 [+0.001, +0.053]	+0.023 [-0.001, +0.053]
all single bans	-0.060 [-0.275, +0.063]	-0.002 [-0.024, +0.022]	+0.003 [-0.017, +0.023]	+0.006 [-0.011, +0.025]	+0.001 [-0.029, +0.025]	+0.009 [-0.009, +0.027]

Every predictor did better on the screen than on confirmation, and the ranking changed between the two (Table 57). On the screen the displacement predictor had the lowest leave-one-ban-out JSD (0.183) and sign accuracy 0.95; on confirmation its advantage shrank to the margins of Table 56.

Table 57. Screen (leave-one-ban-out) against confirmation for every predictor.

predictor	screen leave-one-ban-out mean JSD	screen sign accuracy	confirmation mean JSD, T1-T3	confirmation mean JSD, T4
IIA (proportional renormalisation)	0.191	0.857	0.207	0.096
displacement (internal, frozen)	0.183	0.952	0.188	0.120
nested geometry (centroid similarity)	0.192	0.905	0.207	0.097
text embedding of names and descriptions	0.184	0.881	0.221	0.107
text embedding of the prompt difference	0.193	0.786	0.216	0.095
Qwen input-embedding of names	0.187	0.905	0.211	0.100
co-mention similarity	0.193	0.881	0.239	0.096

The mention controls test the opposite instruction (Table 58). `You may use A*` takes A* from 3% to 95% of path-search programs, and `You may use merge sort` takes merge sort to 100 of 100 on sorting. The displacement predictor gets the direction right in all four mention conditions. It badly underestimates the size of the shift for A* (predicted 0.02, observed 0.92). The alternative placebo and the cross-domain ban fall within the placebo noise floor on sorting but not on path search. Ban and mention displacements point the same way on the user's tokens (cosine 0.70 to 0.85) but not at the suffix tokens (0.06 to 0.38) (Table 59). Whatever the prompt-final state encodes, it is not mainly "the named algorithm".

Table 58. Confound and mention controls (JSD against the confirmation placebo; noise floor = largest JSD between placebo blocks).

task	control	observed JSD to placebo	placebo noise floor (JSD)	within floor	observed shift toward named idea	displacement-predicted shift	direction match
open path search	alternative placebo	0.041	0.015	no	-	-	-
open path search	cross-domain ban	0.027	0.015	no	-	-	-
open path search	mention: "You may use A*"	0.863	0.015	no	0.923	0.021	yes

open path search	mention: "You may use BFS"	0.196	0.015	no	0.337	0.314	yes
sorting	alternative placebo	0.008	0.066	yes	-	-	-
sorting	cross-domain ban	0.033	0.066	yes	-	-	-
sorting	mention: "You may use bubble sort"	0.280	0.066	no	0.460	0.375	yes
sorting	mention: "You may use merge sort"	0.679	0.066	no	0.850	0.666	yes

Table 59. Ban against mention displacement for the same idea.

task, idea	screen ban	cosine of ban and mention displacement, suffix tokens	cosine, user-token mean	norm of ban displacement (suffix)	norm of mention displacement (suffix)
open path search, BFS	ban rank-1	0.06	0.85	9.0	12.3
open path search, A*	ban rank-3	0.27	0.76	8.9	12.8
sorting, bubble sort	ban rank-1	0.33	0.70	10.8	20.9
sorting, merge sort	ban rank-3	0.38	0.70	9.7	22.1

The transplant test fails, and decoding-time attention to the ban sentence restores the placebo behaviour. The artifact added Δh to the placebo run at the five suffix tokens at layer 32 during prefill, with common random numbers and $n = 60$ per cell, for six screen bans. *Recovery* is $1 - \text{JSD(patched, ban)} / \text{JSD(placebo, ban)}$. Mean recovery is -0.001, the same as 5 matched-norm random displacements (-0.006) (Table 60). *Restoration* is the matching quantity for the reverse edit, $1 - \text{JSD(edited ban run, placebo)} / \text{JSD(ban, placebo)}$. Subtracting Δh from the ban run restores nothing (0.01). Overwriting the suffix residual at all 35 layers, which forces the whole prompt-final state to equal the ban's, recovers 0.01 and 0.04 on the two bans tried. Adding Δh at every prompt position, in the style of Stolfo et al. [17], recovers 0.08 and -0.03. Patching at layers 8 and 18 does no better. An exploratory cell then knocked out attention from every generated token to the ban sentence's tokens during decoding. That restored the placebo distribution with a mean restoration of 0.72 over four bans (0.57 to 0.80). The ban is read from the sentence tokens while the program is being written, not from a summary state at the end of the prompt. The plan's causal claim about the prompt-final idea state is therefore dead, while the claim that the ban acts

through attention to the sentence is supported by one exploratory manipulation.

Table 60. Transplant, ceiling, layer sweep, random and knockout cells (n = 60 per cell; recovery and restoration with 95% CIs).

ban	JSD placebo vs ban (reference, n=60)	transplant at layer 32: recovery [95% CI]	off-target real displacement	35-layer suffix ceiling	add at all positions	transplant at layer 8	transplant at layer 18	random draws, mean (5 draws)	subtract from ban: restoration	knock out ban sentence during decoding: restoration [95% CI]
open path search, ban BFS	0.525	0.03 [-0.29, 0.28]	-0.03 [-0.36, 0.25]	0.01 [-0.35, 0.28]	0.08 [-0.23, 0.32]	-0.08 [-0.44, 0.18]	0.04 [-0.32, 0.30]	-0.039 (-0.03, 0.06, -0.09, -0.04, -0.09)	-0.02 [-0.15, 0.08]	0.75 [0.52, 0.86]
open path search, ban Dijkstra	0.156	-0.15 [-0.44, 0.17]	-0.02 [-0.30, 0.28]	-	-	-	-	0.010 (-0.03, 0.26, 0.03, -0.04, -0.18)	-0.15 [-1.13, 0.38]	0.57 [-0.01, 0.84]
open path search, ban top-3	0.541	-0.02 [-0.30, 0.15]	-0.19 [-0.40, 0.01]	-	-	-	-	-0.031 (0.00, -0.07, -0.01, -0.01, -0.07)	0.03 [-0.32, 0.27]	not run
sorting, ban insertion sort	0.544	0.13 [-0.22, 0.38]	0.08 [-0.25, 0.33]	-	-	-	-	0.035 (0.01, -0.04, 0.14, 0.06, 0.00)	-0.01 [-0.07, 0.04]	0.80 [0.62, 0.90]
sorting, ban top-5	0.821	-0.00 [-0.04, 0.03]	-0.03 [-0.06, 0.01]	0.04 [-0.06, 0.12]	-0.03 [-0.07, 0.00]	-	-	-0.010 (-0.03, 0.00, -0.01, -0.02, -0.00)	0.02 [-0.19, 0.19]	0.76 [0.58, 0.85]
weighted shortest path, ban Dijkstra	0.833	0.00 [-0.05, 0.05]	0.00 [-0.05, 0.05]	-	-	-	-	-0.003 (0.00, -0.03, -0.03, -0.03, 0.06)	0.22 [-0.01, 0.38]	not run

Table 61. Idea counts in the main transplant cells (n = 60), with pass rates.

ban	reference placebo	reference ban	placebo + displacement	ban - displacement	ban with ban sentence knocked out	pass: placebo / ban / transplant / restore
open path search, ban BFS	BFS 40, DFS 10, other 8, Dijkstra 2	DFS 34, A* 11, other 9, Dijkstra 5, IDDFS 1	BFS 40, DFS 14, other 5, Dijkstra 1	DFS 36, A* 11, other 9, Dijkstra 2, greedy best-first 2	BFS 35, A* 11, DFS 7, Dijkstra 4, other 2, greedy best-first 1	0.40 / 0.27 / 0.35 / 0.15
open path search, ban Dijkstra	BFS 40, DFS 10, other 8, Dijkstra 2	BFS 32, A* 16, DFS 8, other 3, Dijkstra 1	BFS 39, DFS 11, other 8, SPFA 2	BFS 31, A* 16, DFS 11, other 1, Dijkstra 1	BFS 46, DFS 9, other 2, SPFA 2, A* 1	0.40 / 0.32 / 0.28 / 0.20
open path search, ban top-3	BFS 40, DFS 10, other 8, Dijkstra 2	IDDFS 19, DFS 10, Dijkstra 9, other 8, bidirectional 5, A* 4, greedy best-first 2, BFS 2, SPFA 1	BFS 41, DFS 12, other 3, SPFA 2, A* 1, Dijkstra 1	IDDFS 21, other 11, Dijkstra 7, bidirectional 7, BFS 4, greedy best-first 4, DFS 4, A* 2	not run	0.40 / 0.15 / 0.37 / 0.02
sorting, ban insertion sort	bubble sort 41, merge sort 8, quicksort 8, insertion sort 2, other 1	quicksort 36, merge sort 21, heap sort 2, selection sort 1	bubble sort 38, quicksort 11, merge sort 8, insertion sort 2, selection sort 1	quicksort 37, merge sort 19, heap sort 3, selection sort 1	bubble sort 24, quicksort 23, merge sort 9, selection sort 3, insertion sort 1	0.98 / 1.00 / 1.00 / 1.00
sorting, ban top-5	bubble sort 41, merge sort 8, quicksort 8, insertion sort 2, other 1	counting sort 28, heap sort 22, bubble sort 4, radix sort 4, other 1, shell sort 1	bubble sort 38, quicksort 12, merge sort 6, insertion sort 3, other 1	counting sort 32, heap sort 19, bubble sort 3, other 2, selection sort 1, bucket sort 1, merge sort 1, shell sort 1	bubble sort 33, insertion sort 11, selection sort 5, other 3, quicksort 2, heap sort 2, radix sort 1, Timsort/hybrid 1, merge sort 1, bucket sort 1	0.98 / 0.68 / 0.98 / 0.73

weighted shortest path, ban Dijkstra	Dijkstra 59, other 1	Bellman-Ford 31, BFS 14, A* 7, Dijkstra 3, other 2, Floyd-Warshall 1, SPFA 1, IDDFS 1	Dijkstra 59, other 1	Bellman-Ford 31, A* 10, Dijkstra 9, BFS 4, other 3, Floyd-Warshall 1, DFS 1, SPFA 1	not run	0.78 / 0.32 / 0.73 / 0.40

No priming in the suffix displacement. Decoding Δh through the unembedding (logit lens) gives top tokens such as `while`, `although`, `rather`, `another` and `nouvel`, plus rare multilingual tokens; the banned name ranks in the thousands to tens of thousands (median best rank about 17,600 over the 27 screen bans) (Table 62). The one exception is banning DFS on path search, where `DFS` ranks 72 at the band and first at layer 35. That is also the ban with the most extreme outcome (BFS 98 of 100). The suffix state says "do something else"; it does not prime the banned name, so the priming account of non-compliance [5] finds no support at these positions.

Table 62. Logit lens of the screen ban displacements: rank of the banned idea's name token among 151,936 vocabulary tokens, and the top tokens at the band.

task	ban	banned idea rank, band layers 32-34	banned idea rank, layer 35	top tokens at the band (first six)
open path search	ban rank-1	BFS 13619	BFS 1323	`while` 韙 瘗 `слиш` `realistically` `nouvel`
open path search	ban rank-2	DFS 72	DFS 1	`sino` 深层 纵深 `_altern` `onStop` 深层次
open path search	ban rank-3	A* 53520	A* 2577	蟮 觇 廨 `nouvel` 芟 `secondo`
open path search	ban rank-4	Dijkstra 70873	Dijkstra 13858	`while` `While` `\twhile` `although` 觇 `while`
open path search	ban top-3	BFS 31701, DFS 15334, A* 40271	BFS 2468, DFS 3116, A* 6094	`given` `due` `despite` `debido` `unless` `because`
open path search	ban top-5	BFS 9930, DFS 18072, A* 41781, Dijkstra 51983, SPFA 44136	BFS 6524, DFS 5433, A* 18628, Dijkstra 19172, SPFA 44278	`слиш` 韙 諮詢 蟮 觇 贲
open path search	ban rare idea	SPFA 20269	SPFA 17	`слиш` `ngOn` 嚅 觇 蟮 掊
open path search	paraphrase 1 of rank-1 ban	BFS 16354	BFS 1277	`слиш` 蟮 廨 `nouvel` 贲 觇
open path search	paraphrase 2 of rank-1 ban	BFS 29147	BFS 2699	`devis` `understanding` `due` 难度 `unless` `challenging`

sorting	ban rank-1	bubble sort 4925	bubble sort 16840	还有一些 趿 一口气 掩 咻 蛮
sorting	ban rank-2	quicksort 21787	quicksort 2829	咻 掩 嗵 嚯 噗 呸
sorting	ban rank-3	merge sort 83858	merge sort 18392	还有一些 嗵 咻 嚯 呸 噗
sorting	ban rank-4	selection sort 16699	selection sort 13448	掩 呸 嗵 哐 嚯 嗞
sorting	ban top-3	bubble sort 16522, quicksort 62179, merge sort 122031	bubble sort 39105, quicksort 22711, merge sort 63441	另一种 还有一个 还有一些 `alternate` `Alternate` 另
sorting	ban top-5	bubble sort 43258, quicksort 5551, merge sort 68728, selection sort 25874, insertion sort 60841	bubble sort 28014, quicksort 1677, merge sort 53561, selection sort 7227, insertion sort 27212	`another` 一个新的 困难 瘗 `nouvel` `algu`
sorting	ban rare idea	insertion sort 12020	insertion sort 1664	还有一些 性能 一封 `leans` 还有一个 `éal`
sorting	paraphrase 1 of rank-1 ban	bubble sort 2563	bubble sort 5242	掩 趿 咻 嚯 嗵 噗
sorting	paraphrase 2 of rank-1 ban	bubble sort 10400	bubble sort 13690	掩 邋 趿 呸 咻 ﷺ
weighted shortest path	ban rank-1	Dijkstra 26720	Dijkstra 790	`although` `while` `realistically` 虽然 `while` `though`
weighted shortest path	ban rank-2	A* 71482	A* 47816	觇 `realistically` `/AFP` 畀 `quanto` `thèque`
weighted shortest path	ban rank-3	Bellman-Ford 20532	Bellman-Ford 12959	`Möglichkeit` `unless` 觇 饩 `frei` `while`
weighted shortest path	ban rank-4	Floyd-Warshall 17598	Floyd-Warshall 31542	畀 `Simmons` 有序 觇 饩 `frei`
weighted shortest path	ban top-3	Dijkstra 60937, A* 57255, Bellman-Ford 14155	Dijkstra 17910, A* 33701, Bellman-Ford 10365	韪 觇 贲 `frei` 创新发展 諮詢
weighted shortest path	ban top-5	Dijkstra 65809, A* 53841, Bellman-Ford 17591, Floyd-Warshall 17591, bidirectional 66767	Dijkstra 22902, A* 26231, Bellman-Ford 9775, Floyd-Warshall 9775, bidirectional 51292	觇 韪 贲 `слиш` `nouvel` 諮詢

weighted shortest path	ban rare idea	greedy best-first 29086	greedy best-first 696	觇 蟮 畀 剴 nouvel 諮詢
weighted shortest path	paraphrase 1 of rank-1 ban	Dijkstra 41539	Dijkstra 994	韙 觇 frei avoiding realistically 畀
weighted shortest path	paraphrase 2 of rank-1 ban	Dijkstra 26652	Dijkstra 2084	realistically although despite challenging devis albeit

Edge or far side: no consistent direction. The user asked whether, after the top ideas are banned, mass moves to the edge of the banned region or to the opposite side. The *edge statistic* compares the internal similarity between the post-ban mass and the banned centroids with what the placebo mass would give; positive values mean the mass lands near the banned ideas. On the screen it is positive for most path-search bans (BFS, DFS, Dijkstra, the top-3 and the rare ban: 0.14 to 0.29) and near zero or negative for sorting and the weighted task (-0.16 to 0.07) (Table 63). No single statistic reaches permutation $p < 0.05$ except the first paraphrase on path search (p 0.029). The top-5 bans are near zero on all three tasks (-0.035, 0.006, 0.003). Over all 56 bans (screen and confirmation), 33 statistics are positive, with mean 0.047; among the 30 confirmation bans, 17 are positive. The same statistic from text embeddings is near zero everywhere. Bans on path search send mass to internal neighbours of the banned ideas; bans on sorting do not; top-5 bans do neither in a measurable way.

Table 63. Edge-versus-far-side statistic for the screen bans (positive = post-ban mass closer to the banned centroids than the placebo mass).

task	**screen ban**	**banned ideas**	**internal edge statistic [95% CI]**	**permutation p**	**text-embedding edge statistic**	**permutation p**
open path search	ban rank-1	BFS	+0.239 [+0.150, +0.314]	0.065	+0.0049	0.828
open path search	ban rank-2	DFS	+0.139 [+0.113, +0.156]	0.192	+0.0164	0.178
open path search	ban rank-3	A*	-0.027 [-0.060, +0.006]	0.311	+0.0009	0.765

open path search	ban rank-4	Dijkstra	+0.231 [+0.169, +0.293]	0.188	-0.0171	0.488
open path search	ban top-3	BFS, DFS, A*	+0.160 [+0.107, +0.210]	0.242	-0.0048	0.793
open path search	ban top-5	BFS, DFS, A*, Dijkstra, SPFA	-0.035 [-0.041, -0.029]	0.674	-0.0133	0.406
open path search	ban rare idea	SPFA	+0.289 [+0.225, +0.351]	0.150	+0.0363	0.226
open path search	paraphrase 1 of rank-1 ban	BFS	+0.264 [+0.186, +0.337]	0.029	+0.0080	0.664
open path search	paraphrase 2 of rank-1 ban	BFS	+0.182 [+0.097, +0.261]	0.141	-0.0030	0.899
sorting	ban rank-1	bubble sort	-0.061 [-0.070, -0.050]	0.491	-0.0060	0.753
sorting	ban rank-2	quicksort	-0.012 [-0.078, +0.052]	0.823	+0.0003	0.993
sorting	ban rank-3	merge sort	+0.016 [-0.046, +0.083]	0.582	-0.0010	0.882
sorting	ban rank-4	selection sort	+0.001 [-0.040, +0.042]	0.999	-0.0057	0.400
sorting	ban top-3	bubble sort, quicksort, merge sort	+0.014 [+0.006, +0.021]	0.340	+0.0049	0.096
sorting	ban top-5	bubble sort, quicksort, merge sort, selection sort, insertion sort	+0.006 [-0.007, +0.019]	0.933	-0.0088	0.398
sorting	ban rare idea	insertion sort	-0.160 [-0.175, -0.142]	0.299	-0.0006	0.994

sorting	paraphrase 1 of rank-1 ban	bubble sort	-0.066 [-0.081, -0.047]	0.225	-0.0039	0.659
sorting	paraphrase 2 of rank-1 ban	bubble sort	-0.076 [-0.088, -0.060]	0.175	-0.0051	0.610
weighted shortest path	ban rank-1	Dijkstra	+0.070 [+0.059, +0.081]	0.652	+0.0553	0.059
weighted shortest path	ban rank-2	A*	+0.013 [-0.004, +0.034]	0.409	+0.0004	0.920
weighted shortest path	ban rank-3	Bellman-Ford	-0.007 [-0.007, -0.007]	0.406	+0.0025	0.121
weighted shortest path	ban rank-4	Floyd-Warshall	-0.011 [-0.011, -0.011]	0.203	+0.0013	0.466
weighted shortest path	ban top-3	Dijkstra, A*, Bellman-Ford	-0.071 [-0.088, -0.052]	0.628	+0.0200	0.378
weighted shortest path	ban top-5	Dijkstra, A*, Bellman-Ford, Floyd-Warshall, bidirectional	+0.003 [-0.005, +0.012]	0.878	+0.0142	0.402
weighted shortest path	paraphrase 1 of rank-1 ban	Dijkstra	+0.068 [+0.058, +0.078]	0.628	+0.0499	0.086
weighted shortest path	paraphrase 2 of rank-1 ban	Dijkstra	+0.066 [+0.057, +0.076]	0.582	+0.0538	0.036

The manifold. The user asked for a map of the path-search ideas with each centroid sized by how often the model implements it. The artifact projected commitment-window activations at the band layers onto principal components per task and wrote the coordinates to its results files (Tables 64 to 66). The structure is driven by ideas, not by surface features: the idea label explains 77%, 82%, 57% and 28% of PC1 variance on the four tasks (eta^2), and no surface feature explains more than 17% (output length, truncation, number of headers, prose before code, a main guard). On path search, PC1 separates the shortest-path family (Dijkstra at 33.8, SPFA 23.1, Bellman-Ford 63.6, Floyd-Warshall 68.4) from the unweighted searches (BFS −27.9, DFS −24.7, IDDFS −26.1). PC2 separates the heuristic searches (A* 38.6, greedy best-first 37.9, bidirectional 33.3) from the rest.

BFS, the default, sits next to DFS and IDDFS, which is where its ban sends mass. Under the top-5 ban the mean point moves from $(-26.2, -7.7)$ at baseline toward the upper middle $(-12.4, 6.4)$, between the heuristic cluster and the IDDFS/DFS cluster; it does not cross to the far shortest-path side. The iteration-1 review noted that the influence arm's PCA view (Section 1.9) showed a spread of about 70 units along PC1 that was not aligned with idea labels. That view used the contract prompt and a different window. On the bare prompt PC1 is label-dominated.

[Figure fig_manifold_T1: generated in the report phase, not in this draft]

Table 64. Manifold summary per task (commitment-window activations, band layers).

task	explained variance PC1 / PC2 / PC3	etaˆ2 of idea label on PC1 / PC2 / PC3	largest Rˆ2 of PC1 on a surface feature
open path search	0.317 / 0.148 / 0.075	0.77 / 0.75 / 0.39	0.079 (n headers)
sorting	0.237 / 0.144 / 0.087	0.82 / 0.69 / 0.56	0.118 (n headers)
weighted shortest path	0.228 / 0.152 / 0.118	0.57 / 0.68 / 0.69	0.163 (prose before code)
k-th smallest	0.440 / 0.125 / 0.067	0.28 / 0.08 / 0.70	0.106 (prose before code)

Table 65. Idea centroids on the first three principal components, with baseline counts (n = 300) and the number of programs (baseline, ban and forced) behind each centroid.

task	idea	PC1	PC2	PC3	baseline count (n=300)	programs used for the centroid
open path search	BFS	-27.9	-10.2	6.1	179	226
open path search	A*	8.6	38.6	-11.8	39	98
open path search	DFS	-24.7	-2.5	12.6	44	83
open path search	Dijkstra	33.8	-2.1	7.6	22	71
open path search	Bellman-Ford	63.6	-35.1	-3.0	0	40
open path search	Floyd-Warshall	68.4	-34.3	-8.7	0	40
open path search	SPFA	23.1	-11.3	-22.0	2	36
open path search	IDDFS	-26.1	10.2	-24.4	0	34
open path search	other	-16.7	7.2	7.1	14	29
open path search	bidirectional	14.1	33.3	10.1	0	25
open path search	greedy best-first	0.6	37.9	-10.4	0	18
sorting	bubble sort	34.6	19.1	-3.7	179	219
sorting	quicksort	7.5	-34.7	10.7	70	111
sorting	merge sort	0.3	-29.7	12.4	31	73
sorting	selection sort	7.7	14.3	16.1	15	55

sorting	insertion sort	1.3	13.6	7.3	5	45
sorting	bucket sort	-48.9	11.4	-8.1	0	40
sorting	counting sort	-38.6	6.5	-35.5	0	40
sorting	heap sort	-25.1	-9.3	23.2	0	40
sorting	radix sort	-54.5	13.6	-27.9	0	40
sorting	shell sort	-29.9	4.4	-0.1	0	37
sorting	Timsort/hybrid	-28.0	-16.5	-14.3	0	37
sorting	other	-28.4	-1.0	-5.3	0	3
weighted shortest path	Dijkstra	1.0	-12.5	8.2	300	361
weighted shortest path	Bellman-Ford	40.4	7.8	-22.8	0	41
weighted shortest path	A*	-27.8	47.3	27.0	0	40
weighted shortest path	Floyd-Warshall	44.6	18.3	-34.7	0	37
weighted shortest path	bidirectional	-24.9	32.7	-7.8	0	34
weighted shortest path	BFS	-31.0	4.5	-27.1	0	33
weighted shortest path	SPFA	-9.7	5.8	-15.9	0	14
weighted shortest path	IDDFS	-44.3	8.3	-26.3	0	8
weighted shortest path	DFS	-34.2	14.2	-17.2	0	6
weighted shortest path	other	25.8	19.1	-28.5	0	4
weighted shortest path	greedy best-first	-44.2	33.3	0.6	0	2
k-th smallest	quickselect	-17.2	-5.7	0.2	226	312
k-th smallest	sort then index	5.3	8.0	-1.3	59	113
k-th smallest	heap select	38.4	7.0	-34.4	14	57
k-th smallest	median of medians	-2.3	0.8	30.3	1	42
k-th smallest	counting/bucket	50.2	4.4	14.8	0	39

k-th smallest	binary search on value	46.8	18.1	11.9	0	11
k-th smallest	partial selection	36.6	14.3	8.1	0	6

Table 66. Mean point of each screen condition on PC1 and PC2, the norm of its prompt displacement against the placebo, and its idea counts.

task	condition	mean point PC1, PC2	prompt displacement norm (suffix)	idea counts
open path search	baseline (n = 300)	-26.2, -7.7	16.04	BFS 179, DFS 44, A* 39, Dijkstra 22, other 14, SPFA 2
open path search	placebo	-18.2, -8.0	0.00	BFS 67, DFS 17, Dijkstra 11, other 4, A* 1
open path search	ban rank-1	-16.4, 3.3	8.98	DFS 66, other 15, A* 11, Dijkstra 7, BFS 1
open path search	ban rank-2	-28.7, -8.8	8.37	BFS 98, A* 2
open path search	ban rank-3	-23.7, -1.9	8.93	BFS 69, DFS 17, Dijkstra 8, SPFA 4, other 2
open path search	ban rank-4	-11.1, 10.4	10.29	BFS 44, A* 34, DFS 16, other 3, SPFA 2, Dijkstra 1
open path search	ban top-3	-13.0, 8.4	15.86	IDDFS 42, DFS 18, Dijkstra 16, A* 8, other 6, bidirectional 4, BFS 3, greedy best-first 2, Bellman-Ford 1
open path search	ban top-5	-12.4, 6.4	19.81	IDDFS 42, A* 16, DFS 12, greedy best-first 8, other 8, Dijkstra 7, BFS 3, bidirectional 3, Bellman-Ford 1
open path search	ban rare idea	0.4, -8.4	8.94	Dijkstra 50, BFS 42, A* 6, Bellman-Ford 1, DFS 1
sorting	baseline (n = 300)	28.9, -2.5	23.42	bubble sort 179, quicksort 70, merge sort 31, selection sort 15, insertion sort 5
sorting	placebo	29.4, -11.3	0.00	bubble sort 43, merge sort 26, quicksort 18, selection sort 8, insertion sort 5
sorting	ban rank-1	20.6, -34.1	10.84	quicksort 71, merge sort 27, heap sort 1, insertion sort 1
sorting	ban rank-2	20.4, -11.5	11.25	insertion sort 41, merge sort 30, bubble sort 15, selection sort 13, heap sort 1
sorting	ban rank-3	30.9, -11.1	9.70	bubble sort 45, quicksort 23, insertion sort 14, selection sort 14, heap sort 4

sorting	ban rank-4	26.2, -17.1	11.78	bubble sort 41, merge sort 39, quicksort 10, insertion sort 9, heap sort 1
sorting	ban top-3	15.9, -8.7	13.48	selection sort 55, insertion sort 42, counting sort 2, heap sort 1
sorting	ban top-5	3.0, -11.1	17.93	counting sort 45, heap sort 26, shell sort 7, bubble sort 6, other 6, bucket sort 5, radix sort 4, merge sort 1
sorting	ban rare idea	15.5, -33.7	11.02	quicksort 66, merge sort 27, heap sort 4, selection sort 2, bubble sort 1
weighted shortest path	baseline (n = 300)	-0.8, -17.6	15.58	Dijkstra 300
weighted shortest path	placebo	-3.4, -17.2	0.00	Dijkstra 100
weighted shortest path	ban rank-1	-28.2, -13.6	19.19	Bellman-Ford 55, BFS 19, Dijkstra 12, A* 8, bidirectional 2, Floyd-Warshall 1, greedy best-first 1, other 1, SPFA 1
weighted shortest path	ban rank-2	-25.9, -12.5	8.84	Dijkstra 95, BFS 4, SPFA 1
weighted shortest path	ban rank-3	-13.0, -19.5	9.28	Dijkstra 100
weighted shortest path	ban rank-4	-7.4, -18.0	6.74	Dijkstra 100
weighted shortest path	ban top-3	-30.5, -13.3	21.33	BFS 32, Dijkstra 25, Bellman-Ford 21, SPFA 7, Floyd-Warshall 5, DFS 3, other 3, bidirectional 1, greedy best-first 1, IDDFS 1, unparseable 1
weighted shortest path	ban top-5	-24.1, -10.3	20.01	Dijkstra 41, BFS 26, Bellman-Ford 16, SPFA 10, bidirectional 2, A* 1, DFS 1, Floyd-Warshall 1, greedy best-first 1, other 1
weighted shortest path	ban rare idea	-8.0, -8.2	8.30	Dijkstra 98, BFS 1, DFS 1

The centroid geometry is stable within a run (split-half median similarity 0.973, 0.981 and 0.954 on path search, sorting and the weighted task) but agrees only partly between natural and forced programs of the same idea (Spearman 0.36 over 5 ideas on path search and 0.83 over 4 on sorting; the weighted task has too few natural ideas to compare). Centroids from forced programs are a noisy stand-in for how the model represents an idea it chose on its own.

2.9 Steering on the bare prompt: selection confirmed, flattening not seen, and a second model

2.9.1 Iteration 1's random-control claim, re-analysed

The third artifact first re-analysed the influence arm's random-control comparison as the review required. Each of the 18 targeted settings (three strengths per knob family on each of the two tasks, the strengths at which random draws exist) was compared with the mean entropy of the three random draws at the same strength. The comparison used Miller-Madow-corrected entropy on n-matched seeds 0 to 74 and a hierarchical bootstrap that resamples draws, then samples within draws. Three settings beat the draw mean, five are significantly below it and ten are undetermined (Tables 67 and 68). The below-mean settings are four difference-in-means and probe-weight settings on the open task, which select DFS, and the probe-weight direction at $\alpha = 1.5$ on the weighted task; the settings above the mean are the novelty direction at $\alpha = 0.632$ on the open task and the difference-in-means direction at $\alpha = 0.224$ and 0.473 on the weighted task. With only three random draws at $n = 75$, draw-level variance is poorly estimated, so most comparisons cannot be resolved. The statement that random directions raise idea entropy as much as targeted ones is withdrawn. The defensible statement is that, where targeted directions select, they lower entropy below random, and elsewhere the data cannot separate them.

Table 67. Iteration-1 targeted settings against the draw-averaged random entropy (primary specification: Miller-Madow, n-matched seeds 0..74).

task	setting	targeted entropy (bits, Miller-Madow, n=75)	random draw mean (draws 0, 1, 2)	difference [95% CI]	verdict	iteration-1 comparison (pooled draws): difference, verdict
open path search	probe-weight direction, alpha = 1.06	0.760	1.177 (1.11, 1.10, 1.31)	-0.417 [-0.815, -0.063]	below	-0.498, below
open path search	probe-weight direction, alpha = 1.26	1.060	1.229 (0.79, 1.26, 1.64)	-0.169 [-0.715, +0.233]	undetermined	-0.463, below
open path search	probe-weight direction, alpha = 1.5	1.060	1.297 (0.73, 1.26, 1.90)	-0.237 [-0.877, +0.306]	undetermined	-0.634, below
open path search	difference-in-means direction, alpha = 0.224	0.223	1.113 (0.99, 1.21, 1.14)	-0.890 [-1.162, -0.623]	below	-0.892, below

open path search	difference-in-means direction, alpha = 0.473	0.659	1.135 (1.04, 1.36, 1.01)	-0.477 [-0.829, -0.141]	below	-0.713, below
open path search	difference-in-means direction, alpha = 1	0.000	1.222 (1.12, 1.23, 1.31)	-1.222 [-2.037, -1.025]	below	-0.287, below
open path search	novelty direction, alpha = 0.632	1.889	1.215 (1.12, 1.52, 1.00)	+0.674 [+0.262, +1.000]	beats	+0.496, beats
open path search	novelty direction, alpha = 1.12	1.427	1.299 (1.19, 1.17, 1.54)	+0.128 [-0.203, +0.450]	undetermined	+0.092, undetermined
open path search	novelty direction, alpha = 2	1.285	1.568 (1.56, 1.47, 1.67)	-0.283 [-0.529, +0.110]	undetermined	-0.219, undetermined
weighted shortest path	probe-weight direction, alpha = 1.06	1.495	1.146 (1.26, 0.90, 1.28)	+0.348 [-0.001, +0.705]	undetermined	+0.278, beats
weighted shortest path	probe-weight direction, alpha = 1.26	1.203	1.436 (1.69, 1.15, 1.47)	-0.234 [-0.649, +0.228]	undetermined	-0.201, undetermined
weighted shortest path	probe-weight direction, alpha = 1.5	0.919	1.428 (1.29, 1.38, 1.61)	-0.510 [-0.984, -0.067]	below	-0.763, below
weighted shortest path	difference-in-means direction, alpha = 0.224	1.446	1.033 (1.11, 0.85, 1.15)	+0.413 [+0.057, +0.709]	beats	+0.374, beats
weighted shortest path	difference-in-means direction, alpha = 0.473	1.530	1.111 (1.00, 1.04, 1.29)	+0.419 [+0.023, +0.715]	beats	+0.302, beats
weighted shortest path	difference-in-means direction, alpha = 1	1.453	1.105 (1.01, 0.78, 1.53)	+0.349 [-0.486, +0.841]	undetermined	-0.210, undetermined

weighted shortest path	novelty direction, alpha = 0.632	1.298	1.008 (0.96, 0.95, 1.12)	+0.290 [-0.106, +0.652]	undetermined	+0.118, undetermined
weighted shortest path	novelty direction, alpha = 1.12	1.123	1.220 (1.58, 0.72, 1.36)	-0.097 [-0.626, +0.461]	undetermined	-0.454, below
weighted shortest path	novelty direction, alpha = 2	0.672	1.042 (0.95, 1.02, 1.16)	-0.370 [-0.713, +0.078]	undetermined	-0.456, below

Table 68. How the verdict depends on the specification (counts out of 18 targeted settings).

specification	beats draw mean	below draw mean	undetermined	n
Miller-Madow, n-matched seeds 0..74 (primary)	3	5	10	18
Miller-Madow, targeted n = 150	5	5	8	18
plug-in, targeted n = 150	5	6	7	18
plug-in, n-matched	4	5	9	18
plug-in, pooled draws (iteration-1 comparison)	4	9	5	18
plug-in, single draw 0 (iteration-1 survival rule)	7	4	7	18
rule-classifier relabel, Miller-Madow n-matched	4	7	7	18
rule-classifier relabel, plug-in n = 150	6	5	7	18

The novelty-direction dose-response is not significant by exact permutation (Table 69).

Table 69. Exact permutation test of the novelty direction's Spearman correlation between strength and atypicality (6 strength points).

task	Spearman rho	exact two-sided p (720 permutations)	one-sided p (positive)	original CI	corrected CI
open path search	0.486	0.356	0.178	[-0.029, 0.543]	[-0.429, 0.600]
weighted shortest path	-0.257	0.658	0.718	[-0.486, 0.143]	[-1.000, 0.086]

How this compares with published steering results. The selector finding agrees with the reliability literature. Tan et al. [13] find that steerability varies widely across inputs and that steering vectors can fail on some concepts. Turner et al. [7] report that a matched-norm random vector barely changes outputs. Our iteration-1 random draws did change the idea mix, by up to 0.65 bits of pooled entropy in a single draw (Table 15), but the draw-averaged re-analysis no longer supports parity. Olson et al. [14] extract a direction from "boring" against "creative" prompts and report that adding it raises the creativity of text. That is close to our cross-task novelty direction, which transferred perfectly as a readout (AUROC 1.0) but did not raise atypicality

monotonically in code (Table 69). STARS [15] and Riemannian activation steering [16] report diversity gains over plain sampling. They do so by optimising several steering directions jointly across concurrent generations at test time, which is a different intervention from one fixed direction. What this run adds to these comparisons is execution-verified correctness, algorithm-level labels in place of lexical diversity, five matched-norm random draws per strength, and a fresh-seed re-draw. Candidate reasons for the disagreement with the creativity-steering results are the task (code with a small menu of textbook ideas, not open text), the model size (3B), the late layers (29 to 34 of 36), and the window of the first 64 generated tokens.

2.9.2 Select or flatten: the pre-registered test on the bare prompt

Design. On the bare prompt, with menus of four or more ideas, a direction that flattens and a direction that selects make different predictions. A flattener lowers the mode share without making any single other idea dominant, so the distance from uniform falls steadily with strength. A selector hands most of the mass to one other idea. The artifact built the *away-from-mode direction* as the difference in means of commitment-window activations at layers 32 to 34 between non-modal and modal programs, unit-normalised per layer. It added the direction at the prompt-final token and the first 64 generated tokens with strength α times the layer's median residual norm divided by three, as in iteration 1. Comparators were five matched-norm random directions per strength, an off-target real direction (the other task's away-from-mode direction), temperature 1.0 and 1.3, directions toward the centroids of two off-menu ideas (IDDFS and Bellman-Ford on path search, counting and heap sort on sorting), and simply asking for those ideas. The frozen rules (sha256 files) were:

- strength s1: the α whose mode share is closest to half the baseline mode share;
- strength s2: the α with the lowest mode share whose pass rate stays at least 0.8 times the baseline's (calibrated on $n = 30$);
- SELECT: at s2 one non-modal idea takes at least half the samples, that idea was a predicted gainer, and the distance from uniform rises again from s1 to s2;
- FLATTEN: the distance from uniform falls from baseline to s1 to s2 and no non-modal idea reaches half the samples.

Predicted gainers came from two frozen rankings. One is the projection of the away-from-mode direction onto each centroid's offset from the mode; the other is each centroid's nearest-neighbour similarity to the modal centroid. For path search, the projection ranking is A*, Dijkstra, other, DFS, Bellman-Ford, IDDFS (scores A* 0.843, Dijkstra 0.742, other 0.685, DFS 0.656, Bellman-Ford 0.401, IDDFS 0.363); the nearest-neighbour ranking is IDDFS, Bellman-Ford, DFS, other, Dijkstra, A*. For sorting, the projection ranking is quicksort, merge sort, heap sort, selection sort, insertion sort, counting sort (scores quicksort 0.973, merge sort 0.887, heap sort 0.521, selection sort 0.488, insertion sort 0.480, counting sort 0.429); the nearest-neighbour ranking is selection sort, insertion

sort, counting sort, heap sort, merge sort, quicksort. Main runs used seeds 0 to 149 and a re-draw used seeds 1000 to 1149, both on the HF engine.

Result: SELECT on both tasks, replicated on both draws. On path search at s2 ($\alpha = 0.224$), BFS falls from 61% to 13% and A* rises to 57% (main) and 63% (re-draw). A* was the projection ranking's top predicted gainer. None of the five random draws makes A* the top gainer, and they leave BFS at 61% to 72%. The non-mode share is 0.54 above the random-draw mean [0.47, 0.61]. The cost is correctness: the pass rate falls to 0.17 against 0.34 for the random-draw mean, below the 0.8 bar the $n = 30$ calibration had predicted, and 78% of outputs hit the token cap. On sorting at s2 ($\alpha = 0.473$), bubble sort falls from 62% to 1% and 0%, and quicksort rises to 65% and 68%, again the projection ranking's prediction, at a pass rate of 0.92. Sorting entropy at s2 is 0.71 bits below the random-draw mean $[-0.94, -0.44]$, the signature of selection. On path search entropy at s2 is above the random mean on the main draw ($+0.32$ [0.04, 0.56]) and undetermined on the re-draw ($+0.10$ $[-0.19, 0.36]$), because the path-search baseline mode is so dominant that moving mass to A* while BFS, DFS and Dijkstra keep 11% to 15% each raises entropy even though one idea takes over. The nearest-neighbour ranking (IDDFS on path search, selection sort on sorting) was wrong on both tasks. The direction lands on the idea along its own projection, not on the idea that sits closest to the mode.

Tables 70 to 73 give every cell and every comparison with the random-draw mean. Some random draws also move sorting a lot: draw 3 at s2 gives selection sort 72 of 150, and draw 0 gives quicksort 75. The random-draw mode share at s2 ranges from 0.19 to 0.69. A single random cell would have misled either way, which is why the comparison is against the mean of five draws.

[Figure fig_selector: generated in the report phase, not in this draft]

Table 70. Selector test, open path search (Qwen, HF engine, bare prompt). The last five rows are the $n = 30$ strength calibration.

setting	n	idea counts	idea entropy (bits)	mode share	pass rate	truncated	off-menu correct
baseline (seeds 0..299)	300	BFS 184, DFS 42, Dijkstra 36, A* 24, other 11, SPFA 3	1.730	0.61	0.31	0.42	0
away-from-mode direction, s1	150	BFS 78, A* 27, Dijkstra 24, DFS 11, other 8, SPFA 2	1.944	0.52	0.35	0.46	1
away-from-mode direction, s2	150	A* 85, DFS 22, BFS 19, Dijkstra 17, other 5, SPFA 1, greedy best-first 1	1.864	0.13	0.17	0.78	0

away-from-mode, s1, re-draw	150	BFS 85, A* 26, DFS 18, Dijkstra 14, other 5, bidirectional 1, SPFA 1	1.849	0.57	0.33	0.44	1
away-from-mode, s2, re-draw	150	A* 95, BFS 21, DFS 19, Dijkstra 11, other 2, SPFA 1, greedy best-first 1	1.648	0.14	0.17	0.77	0
off-target real direction (other task), s2	150	BFS 56, A* 40, DFS 32, Dijkstra 12, other 8, SPFA 1, bidirectional 1	2.128	0.37	0.21	0.52	0
temperature 1.0	150	BFS 91, DFS 28, Dijkstra 14, A* 8, other 8, SPFA 1	1.708	0.61	0.23	0.49	1
temperature 1.3	150	BFS 63, other 36, DFS 24, Dijkstra 16, A* 5, SPFA 3, greedy best-first 1	2.121	0.42	0.05	0.81	0
random direction draw 0, s1	100	BFS 59, DFS 20, Dijkstra 13, A* 8	1.588	0.59	0.29	0.42	0
random direction draw 0, s2	150	BFS 92, Dijkstra 21, A* 15, DFS 15, other 6, SPFA 1	1.728	0.61	0.28	0.41	0
random direction draw 1, s1	100	BFS 63, A* 13, DFS 13, Dijkstra 7, SPFA 2, other 2	1.680	0.63	0.26	0.45	1
random direction draw 1, s2	150	BFS 103, DFS 22, A* 12, Dijkstra 7, other 6	1.462	0.69	0.31	0.33	0
random direction draw 2, s1	100	BFS 70, DFS 13, Dijkstra 9, A* 8	1.347	0.70	0.32	0.38	0
random direction draw 2, s2	150	BFS 98, Dijkstra 21, DFS 20, other 5, A* 3, SPFA 3	1.575	0.65	0.41	0.37	2
random direction draw 3, s1	100	BFS 64, A* 13, DFS 13, Dijkstra 8, other 2	1.582	0.64	0.20	0.46	0
random direction draw 3, s2	150	BFS 98, DFS 21, Dijkstra 19, A* 6, other 5, greedy best-first 1	1.573	0.65	0.35	0.39	0
random direction draw 4, s1	100	BFS 71, Dijkstra 11, A* 10, DFS 8	1.325	0.71	0.30	0.45	0
random direction draw 4, s2	150	BFS 108, Dijkstra 17, A* 10, DFS 9, SPFA 3, other 3	1.427	0.72	0.34	0.48	1
toward IDDFS centroid, s2	150	DFS 53, BFS 49, A* 24, Dijkstra 13, other 11	2.063	0.33	0.30	0.43	0

toward Bellman-Ford centroid, s2	150	BFS 87, Dijkstra 52, A* 8, DFS 2, other 1	1.342	0.58	0.55	0.46	0
prompt: "... using IDDFS."	60	IDDFS 55, BFS 4, other 1	0.474	0.07	0.03	0.32	2
prompt: "... using Bellman-Ford."	60	Bellman-Ford 60	-0.000	0.00	0.75	0.60	45
baseline, seeds 0..149	150	BFS 89, DFS 27, Dijkstra 18, A* 11, other 4, SPFA 1	1.723	0.59	0.26	0.42	0
calibration, away-from-mode alpha = 0.05	30	BFS 13, A* 7, Dijkstra 6, DFS 4	1.865	0.43	0.27	0.60	0
calibration, away-from-mode alpha = 0.106	30	BFS 14, Dijkstra 6, A* 5, DFS 3, other 2	2.001	0.47	0.33	0.37	0
calibration, away-from-mode alpha = 0.224	30	A* 14, Dijkstra 8, BFS 4, DFS 4	1.797	0.13	0.33	0.77	0
calibration, away-from-mode alpha = 0.473	30	A* 27, Dijkstra 2, DFS 1	0.561	0.00	0.07	0.90	0
calibration, away-from-mode alpha = 1	30	A* 21, Dijkstra 4, BFS 2	0.968	0.07	0.10	0.93	0

Table 71. Selector test, sorting.

setting	n	idea counts	idea entropy (bits)	mode share	pass rate	truncated	off-menu correct
baseline (seeds 0..299)	300	bubble sort 186, quicksort 70, merge sort 27, selection sort 11, insertion sort 5	1.504	0.62	0.98	0.07	5
away-from-mode direction, s1	150	quicksort 76, bubble sort 43, merge sort 21, selection sort 9, insertion sort 1	1.703	0.29	0.99	0.05	1
away-from-mode direction, s2	150	quicksort 98, merge sort 50, selection sort 1, bubble sort 1	1.026	0.01	0.92	0.06	0
away-from-mode, s1, re-draw	150	quicksort 65, bubble sort 47, merge sort 28, insertion sort 6, selection sort 4	1.825	0.31	0.99	0.02	6

away-from-mode, s2, re-draw	150	quicksort 102, merge sort 45, selection sort 3	1.012	0.00	0.91	0.08	0
off-target real direction (other task), s2	150	quicksort 52, merge sort 50, bubble sort 35, selection sort 11, other 1	1.876	0.23	0.98	0.07	0
temperature 1.0	150	bubble sort 65, quicksort 47, selection sort 17, merge sort 16, insertion sort 4, Timsort/hybrid 1	1.935	0.43	1.00	0.07	5
temperature 1.3	150	bubble sort 71, quicksort 38, merge sort 15, selection sort 15, insertion sort 5, other 5	2.008	0.47	0.81	0.32	5
random direction draw 0, s1	100	bubble sort 54, quicksort 28, merge sort 8, insertion sort 5, selection sort 5	1.718	0.54	0.98	0.09	5
random direction draw 0, s2	150	quicksort 75, bubble sort 33, merge sort 21, selection sort 18, insertion sort 2, Timsort/hybrid 1	1.876	0.22	0.97	0.09	2
random direction draw 1, s1	100	bubble sort 57, quicksort 26, merge sort 8, selection sort 6, insertion sort 3	1.654	0.57	0.98	0.07	3
random direction draw 1, s2	150	bubble sort 85, quicksort 47, merge sort 8, insertion sort 6, selection sort 4	1.540	0.57	0.99	0.05	6
random direction draw 2, s1	100	bubble sort 61, quicksort 25, merge sort 10, selection sort 2, insertion sort 2	1.493	0.61	0.99	0.11	2
random direction draw 2, s2	150	bubble sort 103, merge sort 22, quicksort 18, selection sort 5, insertion sort 2	1.392	0.69	0.98	0.07	2
random direction draw 3, s1	100	bubble sort 55, quicksort 20, merge sort 12, selection sort 8, insertion sort 5	1.813	0.55	1.00	0.05	5
random direction draw 3, s2	150	selection sort 72, merge sort 30, bubble sort 28, quicksort 15, insertion sort 4, other 1	1.944	0.19	0.98	0.07	4
random direction draw 4, s1	100	bubble sort 63, quicksort 21, merge sort 7, insertion sort 5, selection sort 4	1.563	0.63	1.00	0.03	5

random direction draw 4, s2	150	bubble sort 60, merge sort 51, quicksort 25, selection sort 10, insertion sort 4	1.889	0.40	0.99	0.09	3
toward counting sort centroid, s2	150	quicksort 64, merge sort 54, insertion sort 19, bubble sort 6, selection sort 3, counting sort 2, bucket sort 1, other 1	1.911	0.04	0.98	0.09	22
toward heap sort centroid, s2	150	heap sort 137, merge sort 10, Timsort/hybrid 2, insertion sort 1	0.511	0.00	0.95	0.16	135
prompt: "... using counting sort."	60	counting sort 60	-0.000	0.00	0.33	0.12	20
prompt: "... using heap sort."	60	heap sort 60	-0.000	0.00	1.00	0.08	60
baseline, seeds 0..149	150	bubble sort 92, quicksort 39, merge sort 12, selection sort 4, insertion sort 3	1.482	0.61	0.98	0.10	3
calibration, away-from-mode alpha = 0.05	30	quicksort 16, bubble sort 7, merge sort 5, selection sort 1, insertion sort 1	1.732	0.23	0.97	0.10	1
calibration, away-from-mode alpha = 0.106	30	quicksort 20, selection sort 3, merge sort 3, insertion sort 2, bubble sort 2	1.575	0.07	0.97	0.07	1
calibration, away-from-mode alpha = 0.224	30	quicksort 15, merge sort 15	1.000	0.00	1.00	0.07	0
calibration, away-from-mode alpha = 0.473	30	quicksort 21, merge sort 9	0.881	0.00	1.00	0.03	0
calibration, away-from-mode alpha = 1	30	quicksort 5	-0.000	0.00	0.13	0.90	0

Table 72. Open path search: targeted settings against the mean of five matched-norm random draws at the same strength (hierarchical bootstrap, 10,000 resamples for entropy, 5,000 for shares).

comparison	idea entropy, targeted vs random draw mean (5 draws): difference [95% CI], verdict	non-mode share: difference [95% CI], verdict	pass rate: difference [95% CI], verdict

main at s1	1.968 vs 1.530: +0.438 [+0.175, +0.691], beats	0.480 vs 0.346: +0.134 [+0.035, +0.228], beats	0.353 vs 0.274: +0.079 [-0.013, +0.171], undetermined
redraw at s1	1.878 vs 1.530: +0.348 [+0.064, +0.613], beats	0.433 vs 0.346: +0.087 [-0.008, +0.182], undetermined	0.327 vs 0.274: +0.053 [-0.038, +0.145], undetermined
both draws at s1	1.923 vs 1.530: +0.393 [+0.156, +0.623], beats	0.457 vs 0.346: +0.111 [+0.026, +0.197], beats	0.340 vs 0.274: +0.066 [-0.009, +0.144], undetermined
main at s2	1.893 vs 1.576: +0.317 [+0.039, +0.561], beats	0.873 vs 0.335: +0.539 [+0.468, +0.605], beats	0.173 vs 0.337: -0.164 [-0.245, -0.084], below
redraw at s2	1.677 vs 1.576: +0.100 [-0.188, +0.358], undetermined	0.860 vs 0.335: +0.525 [+0.452, +0.596], beats	0.167 vs 0.337: -0.171 [-0.249, -0.092], below
both draws at s2	1.785 vs 1.576: +0.209 [-0.063, +0.454], undetermined	0.867 vs 0.335: +0.532 [+0.473, +0.590], beats	0.170 vs 0.337: -0.167 [-0.236, -0.103], below
off-target direction at s2	2.157 vs 1.576: +0.581 [+0.354, +0.783], beats	0.627 vs 0.335: +0.292 [+0.201, +0.381], beats	0.207 vs 0.337: -0.131 [-0.213, -0.045], below

Table 73. Sorting: targeted settings against the random-draw mean.

comparison	**idea entropy, targeted vs random draw mean (5 draws): difference [95% CI], verdict**	**non-mode share: difference [95% CI], verdict**	**pass rate: difference [95% CI], verdict**
main at s1	1.722 vs 1.677: +0.045 [-0.169, +0.269], undetermined	0.713 vs 0.420: +0.293 [+0.202, +0.382], beats	0.987 vs 0.990: -0.003 [-0.027, +0.016], undetermined
redraw at s1	1.844 vs 1.677: +0.167 [-0.031, +0.389], undetermined	0.687 vs 0.420: +0.267 [+0.173, +0.358], beats	0.993 vs 0.990: +0.003 [-0.015, +0.020], undetermined
both draws at s1	1.783 vs 1.677: +0.106 [-0.088, +0.316], undetermined	0.700 vs 0.420: +0.280 [+0.203, +0.359], beats	0.990 vs 0.990: +0.000 [-0.017, +0.017], undetermined
main at s2	1.040 vs 1.749: -0.709 [-0.944, -0.444], below	0.993 vs 0.588: +0.405 [+0.235, +0.577], beats	0.920 vs 0.984: -0.064 [-0.111, -0.021], below
redraw at s2	1.022 vs 1.749: -0.727 [-0.954, -0.451], below	1.000 vs 0.588: +0.412 [+0.243, +0.583], beats	0.913 vs 0.984: -0.071 [-0.121, -0.027], below
both draws at s2	1.031 vs 1.749: -0.718 [-0.928, -0.466], below	0.997 vs 0.588: +0.409 [+0.240, +0.582], beats	0.917 vs 0.984: -0.067 [-0.101, -0.035], below

off-target direction at s2	1.895 vs 1.749: +0.146 [-0.083, +0.396], undetermined	0.765 vs 0.588: +0.177 [-0.002, +0.360], undetermined	0.980 vs 0.984: -0.004 [-0.032, +0.019], undetermined

Asking beats steering for off-menu ideas. Directions toward the centroids of IDDFS, Bellman-Ford and counting sort produce 0% to 1% of the target idea. Asking for the idea in the prompt (`... using X.`) produces 92% to 100% (Table 74). The exception is heap sort: the direction toward its centroid at $\alpha = 0.473$ yields heap sort in 91% of programs, 89% correct, at a pass rate of 0.95. This mirrors the AxBench finding that prompting is a strong baseline for steering [18].

Table 74. Steering toward an off-menu idea's centroid against asking for it.

task	target idea	α	steered: target share	steered: correct target share	steered: pass rate	asked: target share	asked: correct target share
open path search	IDDFS	0.224	0.00	0.00	0.30	0.92	0.03
open path search	Bellman-Ford	0.224	0.00	0.00	0.55	1.00	0.75
sorting	counting sort	0.224	0.01	0.01	0.98	1.00	0.33
sorting	heap sort	0.473	0.91	0.89	0.95	1.00	1.00

The off-menu-correct test passes on sorting only. The rule requires a setting with pass rate at least 0.8 times baseline whose off-menu correct rate is at least 3 times that of the best HF temperature, with one-sided Fisher $p < 0.05$. On path search nothing passes: the best temperature gives 1 off-menu correct program in 150, and no steering setting or iteration-1 ban does better than 1 in 100 (Table 75). On sorting it passes through the heap-sort direction (135 of 150, ratio 24.6), the counting-sort direction (22 of 150, of which 19 are insertion sort, ratio 4.1), and three of the iteration-1 bans (rank-2, rank-3 and top-3; up to 44 of 100) (Table 76). The result is the same with built-in sorts excluded. On sorting the "off-menu" ideas these settings reach are insertion and heap sort, textbook algorithms just outside a narrow baseline menu.

Table 75. Off-menu-correct test, open path search (menu BFS, DFS, A*, Dijkstra, other).

setting	off-menu correct k/n	rate	pass rate	pass eligible	ratio to best temperature	Fisher p	passes	engine
temperatures (HF)	T=0.7: 0/300, T=1.0: 1/150, T=1.3: 0/150							best T = 1.0
away-from-mode direction, s1	1/150	0.010	0.35	yes	1.00	0.75	no	HF

away-from-mode direction, s2	0/150	0.003	0.17	no	0.33	1	no	HF
toward IDDFS centroid, s2	0/150	0.003	0.30	yes	0.33	1	no	HF
toward Bellman-Ford centroid, s2	0/150	0.003	0.55	yes	0.33	1	no	HF
iteration-1 bare-prompt ban rank-1	0/100	0.005	0.27	yes	0.50	1	no	vLLM (cross-engine)
iteration-1 bare-prompt ban rank-2	0/100	0.005	0.54	yes	0.50	1	no	vLLM (cross-engine)
iteration-1 bare-prompt ban rank-3	0/100	0.005	0.40	yes	0.50	1	no	vLLM (cross-engine)
iteration-1 bare-prompt ban rank-4	0/100	0.005	0.24	no	0.50	1	no	vLLM (cross-engine)
iteration-1 bare-prompt ban top-3	1/100	0.015	0.09	no	1.50	0.64	no	vLLM (cross-engine)
iteration-1 bare-prompt ban top-5	1/100	0.015	0.05	no	1.50	0.64	no	vLLM (cross-engine)
iteration-1 bare-prompt ban rare idea	1/100	0.015	0.64	yes	1.50	0.64	no	vLLM (cross-engine)

Table 76. Off-menu-correct test, sorting (menu bubble, quick, merge, selection sort).

setting	**off-menu correct k/n**	**rate**	**pass rate**	**pass eligible**	**ratio to best temperature**	**Fisher p**	**passes**	**engine**
temperatures (HF)	T = 0.7: 5/300, T = 1.0: 5/150, T = 1.3: 5/150							best T = 1.0
away-from-mode direction, s1	1/150	0.010	0.99	yes	0.27	0.99	no	HF
away-from-mode direction, s2	0/150	0.003	0.92	yes	0.09	1	no	HF

toward counting sort centroid, s2	22/150	0.149	0.98	yes	4.09	0.00045	yes	HF
toward heap sort centroid, s2	135/150	0.897	0.95	yes	24.64	2e-60	yes	HF
iteration-1 bare-prompt ban rank-1	2/100	0.025	0.96	yes	0.68	0.85	no	vLLM (cross-engine)
iteration-1 bare-prompt ban rank-2	42/100	0.421	1.00	yes	11.55	9.1e-15	yes	vLLM (cross-engine)
iteration-1 bare-prompt ban rank-3	18/100	0.183	1.00	yes	5.03	0.00011	yes	vLLM (cross-engine)
iteration-1 bare-prompt ban rank-4	10/100	0.104	1.00	yes	2.85	0.03	no	vLLM (cross-engine)
iteration-1 bare-prompt ban top-3	44/100	0.441	0.99	yes	12.10	9.1e-16	yes	vLLM (cross-engine)
iteration-1 bare-prompt ban top-5	63/100	0.629	0.69	no	17.26	1e-26	no	vLLM (cross-engine)
iteration-1 bare-prompt ban rare idea	4/100	0.045	0.98	yes	1.22	0.52	no	vLLM (cross-engine)

2.9.3 The second model: gemma-2-2b-it

The plan reserved gemma-2-2b-it [20] for confirmation. The third artifact ran it on the HF engine with eager attention, at sample sizes cut for a 16 GB GPU: baseline n = 150, three placebo blocks of 50, forced cells of 20, bans of 50, temperature cells of 40 and transplant cells of 30, with a 768-token cap. The gemma classifier audit (100 programs, gpt-4.1-mini, $0.020) gives kappa 0.94 on sorting (55 programs) but 0.39 on path search (45 programs, agreement 0.47); pooled kappa is 0.71. Gemma's path-search labels are flagged as label-limited, and 87% of its path-search answers hit the token cap.

The baselines differ from Qwen's. Gemma's path-search baseline is split between A* (49) and BFS (48), with Dijkstra 27 and DFS 15; its sorting baseline is bubble sort 121 of 150 (Table 77). The placebo sentence alone moves gemma's sorting mode from bubble sort to merge sort (about 55%

across blocks). The placebo noise floors are large (18 points on path search, 12 on sorting), so the path-search confirmation is underpowered. Gemma obeys bans on sorting almost perfectly and barely on path search: under the top-3 ban (A*, BFS, Dijkstra) BFS still appears in 36 of 50 programs.

Table 77. Gemma screen: baseline, placebo blocks and screen bans, with screen JSD of IIA, the displacement logit and the text-embedding predictor (open path search: baseline pass 0.13, truncated 0.87, menu at 3%: BFS, Dijkstra, A*, DFS, SPFA, placebo noise floor 18 pp; sorting: baseline pass 0.99, truncated 0.11, menu at 3%: bubble sort, selection sort, merge sort, placebo noise floor 12 pp).

task	condition	banned ideas	n	idea counts	still banned	pass rate	JSD, IIA	JSD, displacement logit	JSD, text embedding
open path search	baseline	-	150	A* 49, BFS 48, Dijkstra 27, DFS 15, SPFA 7, greedy best-first 3, other 1	-	0.13			
open path search	placebo block 1	-	50	BFS 27, DFS 6, A* 6, greedy best-first 5, Dijkstra 3, SPFA 2, other 1	-				
open path search	placebo block 2	-	50	BFS 25, A* 10, DFS 6, Dijkstra 4, SPFA 3, other 2	-				
open path search	placebo block 3	-	50	BFS 18, DFS 11, A* 7, Dijkstra 7, SPFA 6, greedy best-first 1	-				
open path search	ban rank-1	A*	50	BFS 25, Dijkstra 13, A* 7, SPFA 2, greedy best-first 1, other 1, DFS 1	7	0.10	0.164	0.239	0.131
open path search	ban rank-2	BFS	50	DFS 31, BFS 9, A* 5, other 3, greedy best-first 1, SPFA 1	9	0.10	0.315	0.126	0.232

open path search	ban rank-3	Dijkstra	50	BFS 41, DFS 5, Dijkstra 2, SPFA 1, greedy best-first 1	2	0.08	0.175	0.034	0.129
open path search	ban rank-4	DFS	50	BFS 45, greedy best-first 2, DFS 1, SPFA 1, A* 1	1	0.16	0.189	0.166	0.129
open path search	ban top-3	A*, BFS, Dijkstra	50	BFS 36, DFS 13, Dijkstra 1	37	0.04	0.648	0.091	0.328
open path search	ban top-5	A*, BFS, Dijkstra, DFS, SPFA	50	BFS 36, other 6, DFS 4, Dijkstra 2, A* 1, SPFA 1	44	0.18	0.821	0.146	0.206
sorting	baseline	-	150	bubble sort 121, merge sort 17, selection sort 8, insertion sort 3, counting sort 1	-	0.99			
sorting	placebo block 1	-	50	merge sort 25, bubble sort 16, selection sort 3, quicksort 3, counting sort 2, shell sort 1	-				
sorting	placebo block 2	-	50	merge sort 31, bubble sort 13, insertion sort 2, quicksort 2, selection sort 1, counting sort 1	-				
sorting	placebo block 3	-	50	merge sort 26, bubble sort 17, insertion sort 3, counting sort 2, quicksort 2	-				

sorting	ban rank-1	bubble sort	50	merge sort 33, insertion sort 12, quicksort 3, selection sort 1, counting sort 1	0	0.96	0.079	0.229	0.091
sorting	ban rank-2	merge sort	50	bubble sort 20, insertion sort 12, quicksort 9, counting sort 7, selection sort 1, shell sort 1	0	0.88	0.095	0.286	0.133
sorting	ban rank-3	selection sort	50	merge sort 37, insertion sort 6, bubble sort 5, counting sort 1, quicksort 1	0	0.98	0.089	0.131	0.132
sorting	ban rank-4	insertion sort	50	merge sort 40, bubble sort 6, quicksort 3, selection sort 1	0	1.00	0.082	0.119	0.094
sorting	ban top-3	bubble sort, merge sort, selection sort	50	quicksort 15, insertion sort 15, counting sort 10, heap sort 6, unparseable 2, Timsort/hybrid 1, radix sort 1	0	0.70	0.099	0.663	0.291
sorting	ban top-5	bubble sort, merge sort, selection sort, insertion sort, counting sort	50	heap sort 20, quicksort 13, radix sort 8, unparseable 4, shell sort 2, bubble sort 1, Timsort/hybrid 1, other 1	1	0.58	0.287	0.763	0.484

Gemma can write every sorting idea on request, but on path search it does not produce greedy best-first or bidirectional search when asked, and SPFA only in 4 of 20 (Table 78).

Table 78. Gemma forced-idea cells (n = 20): compliance with the requested idea and correctness.

task	forced idea	n	compliance	correct
open path search	A*	20	0.95	0.00
open path search	BFS	20	1.00	0.10
open path search	Dijkstra	20	0.95	0.90
open path search	DFS	20	1.00	0.00
open path search	SPFA	20	0.20	0.10
open path search	greedy best-first	20	0.00	0.00
open path search	bidirectional	20	0.00	0.00
open path search	IDDFS	20	0.65	0.05
sorting	bubble sort	20	1.00	1.00
sorting	merge sort	20	1.00	1.00
sorting	selection sort	20	1.00	1.00
sorting	insertion sort	20	1.00	1.00
sorting	counting sort	20	1.00	0.00
sorting	quicksort	20	1.00	1.00
sorting	heap sort	20	1.00	0.80
sorting	radix sort	20	1.00	0.00

The displacement logit loses to IIA on held-out bans. The frozen gemma predictor (variant G, band layers 13 to 15) has a mean JSD of 0.121 on the 14 confirmation bans, against 0.088 for IIA and 0.093 for a MiniLM text embedding (Tables 79 and 80). Its relative JSD gain over IIA is -0.71 (CI over bans $[-1.29, -0.23]$) and over the text embedding -0.50. Its top-substitute hit rate is 7%. It predicts the same substitute for nearly every ban, and the independent re-check finds that scoring it against a mismatched ban's prediction does no worse. On gemma, the prompt-final displacement is dominated by a generic "do not use" component. On the screen it ties IIA (0.249 against 0.254) and wins only where gemma ignores the ban. Regularity holds in 71 of 72 cells; the exception is the heap-sort ban on sorting, where merge sort falls from 0.55 to 0.16 (Holm p 7e-5).

Table 79. Gemma confirmation bans (n = 50 each, seeds 1000 and up).

task	condition	banned ideas	n	idea counts	JSD, IIA	JSD, displacement logit	JSD, text embedding
open path search	ban rank-5	SPFA	50	BFS 20, Dijkstra 12, DFS 7, A* 7, other 2, greedy best-first 1, SPFA 1	0.048	0.148	0.039

open path search	ban rank-6	greedy best-first	50	A* 21, BFS 16, Dijkstra 5, SPFA 3, other 3, DFS 2	0.096	0.189	0.071
open path search	ban rank-7	bidirectional	50	BFS 18, DFS 15, A* 9, Dijkstra 3, SPFA 3, greedy best-first 1, other 1	0.034	0.031	0.032
open path search	ban rank-8	IDDFS	50	BFS 19, A* 15, Dijkstra 7, SPFA 4, other 4, greedy best-first 1	0.120	0.257	0.180
open path search	random pair 1	greedy best-first, IDDFS	50	A* 18, BFS 16, greedy best-first 5, SPFA 4, DFS 3, Dijkstra 3, other 1	0.115	0.139	0.110
open path search	random pair 2	IDDFS, Floyd-Warshall	50	BFS 19, A* 17, Dijkstra 5, greedy best-first 3, SPFA 3, other 2, DFS 1	0.077	0.153	0.094
open path search	random pair 3	bidirectional, IDDFS	50	BFS 18, DFS 9, A* 8, SPFA 6, Dijkstra 5, greedy best-first 2, other 2	0.014	0.065	0.044
sorting	ban rank-5	counting sort	50	merge sort 15, insertion sort 14, bubble sort 13, quicksort 4, selection sort 2, shell sort 2	0.124	0.122	0.129
sorting	ban rank-6	quicksort	50	merge sort 28, bubble sort 16, insertion sort 4, selection sort 1, counting sort 1	0.024	0.048	0.025
sorting	ban rank-7	heap sort	50	bubble sort 24, selection sort 11, merge sort 8, insertion sort 6, counting sort 1	0.207	0.140	0.202
sorting	ban rank-8	radix sort	50	merge sort 21, bubble sort 12, insertion sort 6, quicksort 5, selection sort 4, counting sort 2	0.053	0.048	0.056
sorting	random pair 1	heap sort, bucket sort	50	merge sort 15, bubble sort 13, selection sort 12, insertion sort 8, quicksort 1, counting sort 1	0.141	0.135	0.149

sorting	random pair 2	quicksort, Timsort/hybrid	50	merge sort 21, bubble sort 20, insertion sort 7, selection sort 1, counting sort 1	0.050	0.059	0.057
sorting	random pair 3	quicksort, bucket sort	50	merge sort 31, insertion sort 10, bubble sort 5, selection sort 4	0.128	0.167	0.114

Table 80. Gemma confirmation summary over 14 bans.

predictor	**mean JSD (14 bans)**	**mean log-likelihood per sample**	**top-substitute hit rate**	**sign accuracy (25 cells)**
IIA	0.088	n/a	0.000	0.56
IIA with compliance term	0.158	-1.842	0.000	0.44
displacement logit (primary, frozen)	0.121	-1.782	0.071	0.60
displacement logit + banned-class penalty	0.110	-1.757	0.071	0.60
text embedding (MiniLM)	0.093	-1.712	0.071	0.56

Temperature on gemma. On sorting, the log-odds of merge sort and selection sort against bubble sort are close to linear in 1/T (R^2 0.96 and 0.93; slopes -1.08 and -1.03), with only four temperature points and $n = 40$. At $T = 0.3$ gemma writes bubble sort in 40 of 40 programs with 19 distinct 32-token openings: one idea, many phrasings, the user's case (a) in its purest form (Tables 81 and 82). Hill-1 diversity of ideas rises from 1.0 at $T = 0.3$ to 1.5, 3.0 and 4.5 at $T = 0.7$, 1.0 and 1.3. Top-k 10 at $T = 1.0$ gives the same number of distinct ideas as no top-k (6 against 6 on sorting, 6 against 7 on path search). On path search, the top three reorder by $T = 1.0$ (Dijkstra first), and $T = 1.3$ is mostly degenerate (19 of 40 "other", pass 0.05).

Table 81. Gemma temperature cells ($n = 40$ per T).

task	**T**	**n**	**idea counts**	**Hill q0 / q1 / q2**	**pass rate**	**truncated**	**distinct 32-token openings per idea**
open path search	0.3	40	BFS 16, A* 16, Dijkstra 5, DFS 3	4 / 3.28 / 2.93	0.100	0.725	A* 8, BFS 2, DFS 2, Dijkstra 2
open path search	0.7	40	A* 14, BFS 13, Dijkstra 7, DFS 2, SPFA 2, greedy best-first 2	6 / 4.42 / 3.76	0.125	0.850	BFS 10, A* 10, Dijkstra 6, SPFA 2, greedy best-first 2, DFS 1

open path search	1.0	40	Dijkstra 12, BFS 11, A* 8, greedy best-first 4, SPFA 3, DFS 1, other 1	7 / 5.19 / 4.49	0.100	0.975	BFS 11, Dijkstra 10, A* 8, greedy best-first 4, SPFA 3, DFS 1, other 1
open path search	1.3	40	other 19, greedy best-first 8, BFS 7, DFS 2, Dijkstra 2, SPFA 2	6 / 4.18 / 3.29	0.050	0.425	other 19, greedy best-first 8, BFS 7, DFS 2, Dijkstra 2, SPFA 2
open path search	1.0, top-k 10	40	Dijkstra 12, BFS 11, A* 10, other 3, DFS 2, greedy best-first 2	distinct ideas 6 (none: 7)			BFS 11, Dijkstra 11, A* 10, other 3, DFS 2, greedy best-first 2
sorting	0.3	40	bubble sort 40	1 / 1.00 / 1.00	1.000	0.050	bubble sort 19
sorting	0.7	40	bubble sort 36, selection sort 2, merge sort 2	3 / 1.48 / 1.23	1.000	0.125	bubble sort 33, selection sort 2, merge sort 2
sorting	1.0	40	bubble sort 27, insertion sort 5, merge sort 3, counting sort 2, selection sort 2, quicksort 1	6 / 3.04 / 2.07	0.900	0.200	bubble sort 26, insertion sort 5, merge sort 3, counting sort 2, selection sort 2, quicksort 1
sorting	1.3	40	bubble sort 18, insertion sort 7, quicksort 6, selection sort 4, merge sort 4, counting sort 1	6 / 4.49 / 3.62	0.825	0.275	bubble sort 18, insertion sort 7, quicksort 6, selection sort 4, merge sort 4, counting sort 1
sorting	1.0, top-k 10	40	bubble sort 26, insertion sort 4, merge sort 4, selection sort 3, counting sort 2, quicksort 1	distinct ideas 6 (none: 6)			bubble sort 25, insertion sort 4, merge sort 4, selection sort 3, counting sort 2, quicksort 1

Table 82. Gemma idea log-odds against 1/T (four points).

task	idea (vs mode)	slope vs 1/T [95% CI]	R^2	top-3 order at T = 0.3 / 1.0 / 1.3
open path search	Dijkstra (vs A*)	-0.83 [-1.34, -0.28]	0.637	A*, BFS, Dijkstra / Dijkstra, BFS, A* / BFS, Dijkstra, A*
open path search	greedy best-first (vs A*)	-1.87 [-2.12, -1.52]	0.667	A*, BFS, Dijkstra / Dijkstra, BFS, A* / BFS, Dijkstra, A*
open path search	BFS (vs A*)	-0.62 [-0.97, -0.23]	0.294	A*, BFS, Dijkstra / Dijkstra, BFS, A* / BFS, Dijkstra, A*
open path search	SPFA (vs A*)	-1.57 [-1.83, -1.07]	0.744	A*, BFS, Dijkstra / Dijkstra, BFS, A* / BFS, Dijkstra, A*

sorting	merge sort (vs bubble sort)	-1.08 [-1.30, -0.63]	0.965	bubble sort, selection sort, merge sort / bubble sort, merge sort, selection sort / bubble sort, selection sort, merge sort
sorting	counting sort (vs bubble sort)	-0.71 [-0.98, -0.22]	0.581	bubble sort, selection sort, merge sort / bubble sort, merge sort, selection sort / bubble sort, selection sort, merge sort
sorting	selection sort (vs bubble sort)	-1.03 [-1.26, -0.61]	0.931	bubble sort, selection sort, merge sort / bubble sort, merge sort, selection sort / bubble sort, selection sort, merge sort
sorting	insertion sort (vs bubble sort)	-1.18 [-1.37, -0.88]	0.577	bubble sort, selection sort, merge sort / bubble sort, merge sort, selection sort / bubble sort, selection sort, merge sort

The gemma transplant also fails. At layer 13, adding the ban-minus-placebo displacement at the five suffix positions to the placebo prompt recovers 6% of the sorting top-5 ban's effect [−13%, 21%] (Tables 83 and 84). It moves output away from bubble sort (13 to 4 of 30), but toward insertion sort rather than the ban's heap sort, quicksort and radix sort. Subtracting it from the ban restores nothing (−0.03). On path search gemma barely obeys the ban itself (JSD 0.11 between placebo and ban), so the recovery of 0.42 [−0.93, 0.78] is noise; one random displacement "recovers" 0.66.

Table 83. Gemma transplant summary (layer 13, n = 30 per cell).

task (ban)	JSD placebo vs ban	recovery, placebo + displacement [95% CI]	recovery, random displacements 1 / 2 / 3	restoration, ban - displacement [95% CI]	restoration, ban - random displacement 1
open path search (ban A*)	0.114	0.42 [-0.93, 0.78]	0.17 / 0.12 / 0.66	-0.05 [-1.09, 0.44]	0.19
sorting (ban top-5)	0.579	0.06 [-0.13, 0.21]	-0.11 / 0.03 / 0.01	-0.03 [-0.10, 0.05]	-0.01

Table 84. Gemma transplant cells: idea counts and pass rates.

task	cell (n = 30)	idea counts	pass rate
open path search	placebo	BFS 15, DFS 6, Dijkstra 5, A* 3, other 1	0.07

open path search	ban	BFS 19, A* 7, SPFA 2, Dijkstra 2	0.03
open path search	placebo + displacement	BFS 14, A* 5, Dijkstra 5, DFS 3, greedy best-first 2, SPFA 1	0.00
open path search	ban - displacement	BFS 19, A* 5, other 3, SPFA 2, Dijkstra 1	0.17
open path search	ban - random displacement 1	BFS 16, Dijkstra 6, A* 6, SPFA 2	0.10
open path search	placebo + random displacement 1	BFS 15, Dijkstra 4, DFS 4, A* 2, SPFA 2, greedy best-first 2, other 1	0.03
open path search	placebo + random displacement 2	BFS 15, DFS 7, A* 3, Dijkstra 2, SPFA 2, greedy best-first 1	0.10
open path search	placebo + random displacement 3	BFS 14, A* 7, Dijkstra 3, DFS 2, SPFA 2, other 2	0.10
sorting	placebo	merge sort 14, bubble sort 13, counting sort 1, selection sort 1, quicksort 1	0.97
sorting	ban	heap sort 10, quicksort 9, radix sort 7, shell sort 1, Timsort/ hybrid 1, counting sort 1, bucket sort 1	0.40
sorting	placebo + displacement	merge sort 12, insertion sort 9, bubble sort 4, counting sort 3, selection sort 1, quicksort 1	0.83
sorting	ban - displacement	quicksort 20, heap sort 4, shell sort 4, radix sort 1, other 1	0.83
sorting	ban - random displacement 1	heap sort 11, quicksort 9, unparseable 4, radix sort 3, shell sort 2, Timsort/hybrid 1	0.47
sorting	placebo + random displacement 1	bubble sort 22, merge sort 6, selection sort 2	1.00
sorting	placebo + random displacement 2	merge sort 13, bubble sort 13, shell sort 1, counting sort 1, quicksort 1, selection sort 1	0.97
sorting	placebo + random displacement 3	bubble sort 14, merge sort 12, insertion sort 1, selection sort 1, counting sort 1, quicksort 1	0.97

Sparse autoencoder features, descriptive only. Gemma Scope [19] residual SAEs (layer 13, width 16k, about 84 active latents per token) reconstruct the instruction-tuned model's prompt states poorly (fraction of variance unexplained 0.61; the SAEs were trained on the base model), so this step is marked unreliable and no verdict rests on it. With that caveat: a ban raises the latents specific to the banned idea in 10 of 16 single bans, but most of these rises are within the range of random latent sets (Table 85). Several of the latents that change most under a ban (for example latent 902, whose Neuronpedia description concerns legal or formal assessments) describe formal

or regulatory language rather than algorithms.

Table 85. Gemma Scope priming of the banned idea's top latents under each single ban (z-score; percentile against random latent sets and against other bans).

task	ban	banned idea	priming of the banned idea's latents (z)	percentile vs random latent sets	percentile vs other bans
open path search	rank-1	A*	1.00	0.905	1.00
open path search	rank-2	BFS	-0.32	0.274	0.00
open path search	rank-3	Dijkstra	-0.20	0.140	0.90
open path search	rank-4	DFS	0.40	0.841	1.00
open path search	rank-5	SPFA	0.40	0.749	0.82
open path search	rank-6	greedy best-first	1.14	0.936	0.82
open path search	rank-7	bidirectional	0.23	0.693	0.73
open path search	rank-8	IDDFS	1.05	0.962	0.44
sorting	rank-1	bubble sort	0.36	0.856	1.00
sorting	rank-2	merge sort	-0.53	0.057	0.30
sorting	rank-3	selection sort	-0.43	0.252	0.50
sorting	rank-4	insertion sort	-0.72	0.126	0.27
sorting	rank-5	counting sort	0.75	0.971	0.55
sorting	rank-6	quicksort	0.05	0.656	0.90
sorting	rank-7	heap sort	-0.09	0.327	1.00
sorting	rank-8	radix sort	0.80	0.950	0.75

2.10 What iteration 2 established, and what it did not

Established.

1. *The default answer is a mode, not an average, and it depends on the prompt's wording.* On the bare

prompt the open-task default is BFS (60%), with DFS, A* and Dijkstra as a stable minority. The menu is four ideas wide, not two. Adding "in Python." to the sorting prompt changes the mode from bubble sort to quicksort, and the influence arm's output contract had narrowed the path-search menu to two ideas.

2. *Temperature is a weak idea knob and a strong correctness knob.* It adds ideas slowly, changes the mode at $T \leq 1.3$ only once (a within-noise tie on the sorting grid), and does not follow a pure scaling law. Its most useful range for ideas is $T \leq 1.0$, where idea entropy rises by 0.3 bits on paths and 1.3 bits on sorting. Even at $T = 0.3$ the model writes 1 to 3.5 ideas per 10 programs, each in many phrasings, which is the user's case (a). Case (b) never occurs.
3. *Seed strings narrow ideas.* Prepended random strings make the model pick its mode more often (-0.34 and -0.68 bits at $T = 0.7$), although they displace the residual stream more than a ban does.
4. *Bans displace the idea distribution; they do not just remove options.* On the bare prompt bans are mostly obeyed; they lower the shares of ideas they do not name, and banning ideas the model never writes moves the whole distribution. The internal displacement at the prompt's end predicts the sign of share changes (83%) and beats text-embedding predictors, but it does not beat IIA by the required margin, and on gemma it loses to IIA.
5. *The ban is not carried by the prompt-final state.* Transplanting, subtracting or overwriting that state does nothing on either model. Blocking attention from the generated tokens to the ban sentence restores the placebo behaviour (exploratory, Qwen, four bans). The ban is applied while the program is being written.
6. *Internal directions select.* On menus of four or more ideas, the away-from-mode direction hands most of the mass to the one idea its projection predicts (A* on paths, quicksort on sorting), replicated on fresh seeds, against five random draws that do not. It never flattened. Off-menu ideas are reached by asking for them; steering toward their centroids fails except for heap sort.
7. *Off-menu correct programs.* They are rare at every temperature (at most 8%) and are reached on sorting by bans and by the heap-sort direction (up to 90%). On path search no knob tested reaches them beyond 1 in 100.

Dead ends recorded as such.

- *The temperature law as pre-registered.* Killed by Tables 41 and 45: no task meets all criteria on either draw, the path-search intercepts reject pure scaling, and the frozen law loses to fixed utilities on sorting out of sample. Linearity in $1/T$ itself survives the curvature test on most lines, and gemma's sorting lines are near-linear.

- *Seeds as shocks whose effect can be predicted from a utility gap.* Killed by Table 48 (sorting: predicted 0.39, observed 0.017) and by the failed readout gate. Seeds narrow rather than diversify.
- *The prompt-final idea state as the carrier of a ban (the transplant test).* Killed on both models by Tables 60 and 83: recovery about 0 against random displacements about 0, with a 35-layer overwrite also at about 0.
- *Internal geometry as the best predictor of substitution.* Killed as stated by Tables 55 and 80: 9.2% better than IIA on Qwen (20% required), 20% top-substitute hits (70% required), worse than IIA on the k-th-smallest task and on gemma.
- *Nearest-neighbour geometry as the predictor of where steering lands.* Wrong on both tasks (IDDFS and selection sort predicted, A* and quicksort observed).
- *Random-direction parity (iteration-1 conclusion 3).* Withdrawn by Tables 67 and 68.
- *Bans are ignored by this model (iteration-1 conclusion 5).* Withdrawn by Tables 30 to 32; true only under the contract prompt with list wording.
- *Steering toward an off-menu idea's centroid as a novelty knob.* 0% to 1% target share for three of four ideas (Table 74).

Not settled.

- Whether the commitment to an idea happens at a single token. Forking-path resampling [4] has still not been run in any iteration.
- How the ban sentence is read during decoding. The knockout is one exploratory manipulation on four bans; which heads carry it, and at which generated positions, is unmeasured.
- Why the same displacement predictor that ties IIA on Qwen loses on gemma. Gemma's weaker compliance on path search and its larger placebo shifts are candidates.
- All results are from one wording per condition, and Qwen results from one model size.

2.11 What the next iteration takes from this one

The open question is now mechanistic and narrow: where, during generation, does the model read a prohibition or a request and turn it into an idea? Iteration 2 showed that the prompt-final state does not carry it and that attention to the instruction tokens during decoding does. The next step is attention-head-level activation patching on the knockout cells, located with forking-path resampling. That would show where the idea is fixed relative to the moment the model reads the instruction. A second, cheaper step is to test the selector result on the wording variants that moved

the mode (grid wording against ban-arm wording) and on gemma, where it has not been run.

2.12 Bookkeeping

- **Artifacts.** `art_qJbC6mPQUr_5` (gen_art_experiment_4): 13,968-program ledger; new generations: 3,615 re-draw, 2,160 fresh-seed, 150 truncation continuations and 20 regenerations; about 2.5 h on one 23 GB NVIDIA data-centre GPU; judge spend $0.042. `art_e6bJAhbt41cu` (gen_art_experiment_5): 810 forced-extension and 5,930 confirmation generations under vLLM, 71 transplant cells (4,260 HF generations); about 3 h on the same GPU type; judge spend $0.015. `art_wQJ_kngaEEa6` (gen_art_experiment_6): 9,720 Qwen and gemma generations on the HF engine; about 4.5 h on one RTX 2000 Ada (16 GB); judge spend $0.020. Total paid calls this iteration: $0.077.
- **Frozen files.** Temperature law `d1dabe95…` (2026-10-16); seed predictor `1cffcec3…`; off-menu reference `783778b0…` (frozen 03:40:12 UTC); ban predictors frozen 2026-10-16 03:07:25 with git commit `9336da7a`, centroid hashes per task; Qwen selector rules and strengths (two sha256 files); gemma ban predictor `7be800bb…`.
- **Deviations.** The ban arm's top-up replicate files did not exist, so the screen noise floor uses baseline thirds. The seed readout failed its gate, so the behavioural predictor was primary. Within-idea diversity used edit-distance similarity only (the planned code-embedding variant was skipped). The selector band is layers 32 to 34, not iteration 1's 29 to 31. The third artifact disabled its code-stop rule after a check showed it changed 13% of path-search labels, and discarded 400 early generations made with it. Gemma sample sizes were cut as listed in Section 2.9.3, and gemma ran without vLLM. The ban predictor's $\beta = 19.75$ lies near the edge of its grid.
- **Independent re-derivation.** Each artifact re-derived its headline numbers with separate code and placebo checks: label reproduction 4,210/4,210, judge kappas, Arrhenius slopes (exact match), out-of-sample errors, greedy flip rates, confirmation JSDs, regularity counts with 0 violations in 20 shuffled-label repetitions, sign accuracies, transplant recoveries, selector verdicts, the entropy-against-random differences (exact match) and the gemma confirmation comparison. The gemma comparison came out at -0.88 against -0.71 in the pipeline, because only the classes present were scored, with the same direction. Random draws tested against each other and shuffled targeted/random labels give CIs that include 0.
- **Kept on the run volume only, not in the published repository.** Activation caches of the second artifact (`acts/`, about 600 MB, regenerable) and of the third (`data/acts/`, about 200 MB), and the frozen vectors, centroids and displacements (`results/*.pt`).

What we have learned so far

The run asks how an instruction-tuned model chooses the solution idea for an open design task, and whether that choice can be moved toward ideas the model does not usually produce. After two iterations the evidence comes from Qwen2.5-3B-Instruct [1] on four tasks (open path search, sorting, weighted shortest path, k-th smallest element) and from gemma-2-2b-it [20] on two. It rests on more than 50,000 generated, labelled and executed programs.

The default answer is the mode of a small menu, and the prompt's wording sets the menu. Asked to implement a path search algorithm, Qwen writes BFS in about 60% of programs, with DFS, A* and Dijkstra as a stable minority. Asked for a shortest path in a weighted graph, it writes Dijkstra in 300 of 300. The model can list many more ideas than it writes: every co-mention list names BFS, DFS, Dijkstra and A*, and most name bidirectional and greedy best-first search. Small wording changes move the menu. An output contract narrowed path search to BFS and DFS, adding "in Python." changed the sorting mode from bubble sort to quicksort, and an irrelevant prohibition sentence (the placebo) cut A* from 13% to 1%. The user's picture of a range of answers from which the model returns the middle does not fit: the distribution is multi-modal over discrete ideas, and the model returns its mode.

Temperature is not the creativity knob. Raising T adds ideas slowly and breaks code quickly. On paths, idea entropy rises by about 0.3 bits from $T = 0.3$ to 1.0 while the pass rate halves, and the modal idea never changes below $T = 1.6$ except for one within-noise tie. Even at $T = 0.3$ a set of 10 programs holds 1 to 3.5 ideas, each in several phrasings; ten ideas with one phrasing each never occurs. That answers questions (a) to (c) of the original request in favour of (a): tight temperature gives a few ideas in many phrasings, and loosening it adds phrasings first and ideas slowly. Top-k $= 10$ mainly keeps high temperatures from degenerating. The idea log-odds are roughly linear in $1/T$, but the pre-registered temperature law fails: intercepts reject pure scaling, minority ideas swap ranks, and the frozen law does not transfer across a wording change.

Seed strings make the model more predictable. Prepending `seed=` with random characters or random words lowers idea entropy at $T = 0.7$ (by 0.34 bits on paths and 0.68 on sorting) and changes the greedy idea in at most a quarter of cases.

Bans move behaviour, through the instruction tokens. On the bare prompt a short `Do not use X.` is obeyed in most programs, and the freed mass goes to particular ideas: DFS when BFS is banned, A* or Bellman-Ford when Dijkstra is banned, IDDFS when BFS, DFS and A* are banned, and counting and heap sort when the comparison sorts are banned. Bans also lower ideas they do not name, and banning an idea the model never writes moves the whole distribution (banning greedy best-first search takes A* from 3% to 58%). Read as a choice model, bans violate regularity.

The residual-stream displacement at the end of the prompt predicts the direction of these shifts (83% sign accuracy) better than text embeddings of the algorithm names. It does not predict the distribution much better than proportional renormalisation (9% against a required 20%), and on gemma it is worse. Transplanting that displacement does nothing on either model, while blocking attention to the ban sentence during decoding restores the unbanned behaviour. To the request's question of whether banning the top five sends mass to the edge or to the opposite side: on path search mass goes to the banned ideas' internal neighbours (IDDFS sits next to BFS and DFS), on sorting there is no consistent direction, and for top-5 bans the internal statistic is near zero on all three tasks.

Internal directions select; they do not spread. A difference-in-means direction pointing away from the default idea hands most of the probability to one other idea, the one its projection predicts: A* on paths (57% and 63% on two seed draws) and quicksort on sorting (65% and 68%). Five matched-norm random directions do not do this. No direction tested in either iteration flattened the idea distribution. Steering toward an off-menu idea's centroid fails for three of four ideas, while asking for the idea in the prompt succeeds in 92% to 100% of programs. Iteration 1's claim that random directions raise entropy as much as targeted ones does not survive a draw-averaged comparison.

Novelty. No condition in either iteration produced a correct program that a judge did not recognise as a known algorithm. On textbook tasks this metric is near-floor by construction. The informative quantity is the off-menu correct share. It stays at or below 8% for every temperature; on sorting it reaches 44% under a top-3 ban and 90% when steering toward heap sort, and on path search no knob tested exceeds 1 in 100. In these models, creativity as a behaviour means moving the model to another known idea. Bans, explicit requests and selective steering do that; temperature and seed strings do not.

Status of the user's questions after iteration 2:

1. *Is temperature the creativity knob, and do cases (a), (b) or (c) hold under tight temperature with top-k = 10?* Answered: no. Case (a) holds at every tested temperature on both models; ideas grow slowly with T while correctness falls; top-k = 10 caps degeneration more than it changes ideas (Sections 2.5 and 2.9.3).
2. *Does the model always design BFS for the path-search prompt?* Answered: BFS in about 60% of bare-prompt programs, with DFS 15%, A* 13% and Dijkstra 7%; the share depends on wording (Section 2.4). Gemma splits between A* and BFS.
3. *Does context seeding help?* Answered: no. Random-character and random-word seed strings of the user's form narrow ideas rather than diversify them (Section 2.6). The user's two literal

strings were not themselves tested.

4. *Is the model giving the average answer?* Answered: it gives the mode of a multi-modal distribution over discrete ideas, and context (a ban, a mention, a wording change) moves it to a different mode rather than to a different part of a continuum (Sections 2.4 and 2.8).
5. *How much do prompt instructions matter?* Answered: more than any other knob tested. A one-sentence ban or mention moves an idea's share by up to 92 points; the prompt asking for a new algorithm changes little, and long novelty prompts break the code (Sections 1.9, 2.4 and 2.8).
6. *What do the internals of BFS, DFS and Dijkstra runs look like, and can a manifold be drawn?* Partly answered. Commitment-window activations form a label-dominated map (idea label explains 77% of PC1 variance on paths) in which the shortest-path family, the unweighted searches and the heuristic searches separate (Section 2.8, Tables 64 to 66). Gemma Scope SAE features were tried and are unreliable on the instruction-tuned model; thought anchors and forking-path analysis have not been run.
7. *After banning the top ideas, does mass move to the edge or the opposite side?* Partly answered: to the banned ideas' internal neighbours on path search (IDDFS after banning BFS, DFS and A*), with no consistent direction on sorting and a near-zero statistic for top-5 bans (Section 2.8).
8. *Can we make LLMs give truly creative and novel solutions?* Answered for these models and tasks: no knob produced a correct program outside the known algorithms, and that metric is near-floor on textbook tasks. Bans, explicit requests and selective steering do reliably move the model to less typical known ideas, sometimes correctly; temperature and seeds do not.

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